



Open Chiplet Atlas Harness (OCAH)

Programmer's Guide

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WARNING

This documentation product is currently under development. More content will be added as the project matures.

Chapter 1. SMC Programming Guide

This Chapter details the bring-up process and run-time operation of the SMC for the purposes of firmware development, with references to the Production SMC ROM.

The subsequent Chapter serves as a detailed reference to the Production SMC ROM, along with documentation on configurable functionality, such as compile-time options and hookable functions.

1.1. SMC Bring-up

1.1.1. Address Map

The SMC accesses its resources and peripherals through its fixed local-alias aperture, at the RTL-fixed base address `LOCAL_BASE (0xC000_0000)`, and with programmable size `REGION_SIZE`, which defaults to 16 MiB.

External masters can access the SMC's resources through the SMC's global system address, which is assigned by the integrator. This must be matched by firmware in the `GLOBAL_BASE` register. The global aperture is also of size `REGION_SIZE`.

The integrator's assignment of global system addresses to SMCs should be a function of each chiplet's Chiplet ID, which can be read from the eFuse block. It is the integrator's responsibility to assign subsystems to global address ranges that do not overlap with each other, or with the SMC's local-alias aperture.

The following is a subset of the functional units and peripherals accessible to the SMC, relevant to early bring-up procedure. The full list can be found in the TRM SMC Memory Map section. The addresses below are expressed as `BASE + offset`, where `BASE` is either the local-alias base address `LOCAL_BASE`, or the global system base `GLOBAL_BASE`. `offset` is bounded by `REGION_SIZE`.

Table 1. Early Initialization SMC Address Map.

Component	Address Range	Instance	Description
CPU Control	BASE + 0x0001_0000 - BASE + 0x0001_1FFF	1 instance	CPU cluster management
Memory Zeroer	BASE + 0x0003_8200 - BASE + 0x0003_83FF	1 instance	Memory initialization unit
SPM ROM Memory	BASE + 0x0004_0000 - BASE + 0x0005_FFFF	128KB	Scratchpad ROM memory space
SPM Memory	BASE + 0x0006_0000 - BASE + 0x0007_FFFF	128KB	Scratchpad memory space

The `GLOBAL_BASE` and `REGION_SIZE` registers are part of the CPU Control block (`BASE + 0x0001_0000`).

The `ext_out` port is a catch-all for outbound memory requests from the SEP and SMC that do not match any programmed aperture. Unmatched inbound memory requests from external masters on `ext_in` are not passed-through to `ext_out`—only the SMC window exposed by `GLOBAL_BASE` and `REGION_SIZE` is accessible externally, otherwise a decode error is returned. Firmware **must** program `GLOBAL_BASE` and `REGION_SIZE` **before any cross-subsystem AXI traffic is issued**.

1.1.2. SMC CPU

The SMC CPU consists of either 4 Rocket RV64 cores. The cores implement the RV64GC instruction set, as well as the Zba,

Zbb, and Zbs instructions. The cores support Machine, Supervisor, and User mode privilege levels.

The CPU cold-reset vector is `0xC004_0000`, corresponding to the start of ROM memory—see the SMC’s address map above. Each core’s reset vector can be modified to a warm-reset (a reset that does not clear SRAM) vector by writing the location to the core’s `RESET_VECTOR` register, which is part of the CPU Control block.

During early bring-up, it is advised to allow only one core to perform some early initialization procedures. Each core’s ID can be read from the `mhartid` CSR. The programmer should only allow the core with `mhartid` set to 0 to perform early initialization of memory (for example, copying runtime data from ROM to SRAM), while the rest of the cores wait to be released from a loop. In the 1-core configuration, the single core has `mhartid` set to 0.

In the production SMC ROM, the `__metal_synchronize_harts` function is used by the primary core to signal initialization is complete, and for the secondary cores to wait until they are released. By default, the production SMC ROM parks the remaining secondary cores in a `wfi` loop. The programmer may redefine the `secondary_main` function in C code to provide code for the secondary cores.

The core ID read from the `mhartid` register **must** also be used to give each core a unique stack region for C code.

The production SMC ROM linker script—`hw/sys/smc/bootrom/prod/linker/smc/smc_rom.ld` places the executable `.text` and read-only data `.rodata` sections in ROM. The `.data` section and other runtime data are included in the ROM image, but have runtime addresses within SRAM at `0xC006_0000`. This data must be copied into SRAM (by the single initializing core) before it can be referenced or modified. The stack, heap, and the `.bss` section reside only in SRAM.

1.1.3. SRAM Initialization using Memory Zeroer

The SMC contains a Memory Zeroer unit at `BASE + 0x0003_8200`, which can be used to initialize SRAM before use. This can be performed like so:

1. Disable interrupts to the core.
2. Write the SRAM base address (`0xC006_0000`) to the Memory Zeroer’s `DEST_ADDR` register.
3. Write the size of SRAM (128 KiB = `0x0002_0000`) to the Memory Zeroer’s `SIZE` register.
4. Write zero to the Memory Zeroer’s `CTRL_STATUS` register, which starts the operation.
5. Polls the Memory Zeroer’s `CTRL_STATUS` register until the `STATUS` bit (bit 32) is unset, indicating the operation is complete.
6. Restore interrupt state.

See also `__metal_sram_init_early` from the Production ROM in `hw/sys/smc/bootrom/prod/boot/sram_init.S`.

The production ROM assumes that SRAM is automatically zeroed on reset. If `SRAM_AUTO_ZERO_DISABLE` or the eFuse `RESERVED_0[6]`, the Memory Zeroer is used by the production ROM.

1.1.4. Memory Built-In Self Test and Repair

The programmer may check the status of the SRAM memory built-in self-test (MBIST) and SRAM Memory Repair before usage by reading the `STATUS_SMU` register.

In the Production SMC ROM, this is performed by ``metal_mbist_early_check`` and `metal_memory_repair_check` respectively. Their operation is as follows:

`__metal_mbist_early_check` performs the following steps:

1. Check the **MEM_BIST_BYPASS** strap (register **STRAPS_HI** bit 54). If set, skip MBIST check, otherwise continue.
2. Read the DFT Control's **STATUS_SMU** register.
3. Check if the MBIST was performed (register **STATUS_SMU** bit 4). If unset, assume the MBIST was not required, and return.
4. Check if the MBIST passed (register **STATUS_SMU** bit 8). If set, return — the MBIST has passed.
5. The MBIST has failed, or aborted due to a timeout. If the eFuse **RESERVED_2[0]** is set, ignore the failure and return.
6. The MBIST has failed — park the core in a **wfi** loop.

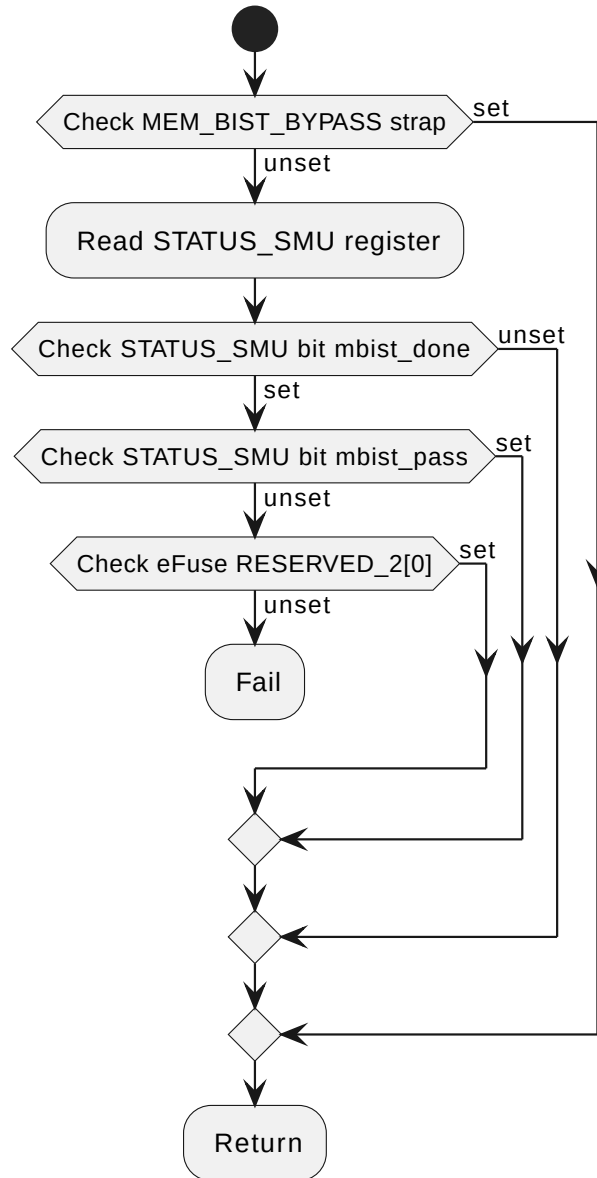


Figure 1. Production ROM **__metal_mbist_early_check**.

__metal_memory_repair_check performs the following steps:

1. Check the **MEM_REPAIR_BYPASS** strap (register **STRAPS_LO** bit 13). If set, skip Memory repair check, otherwise continue.
2. Read the DFT Control's **STATUS_SMU** register.
3. Check if Memory repair was performed (register **STATUS_SMU** bit 0). If unset, assume Memory repair was not

required, and return.

4. Check if Memory repair passed (register `STATUS_SMU` bit 1). If set, return — Memory repair has passed.
5. Memory repair has failed. If the eFuse `RESERVED_2[0]` is set, ignore the failure and return.
6. Memory repair has failed — park the core in a `wfi` loop.

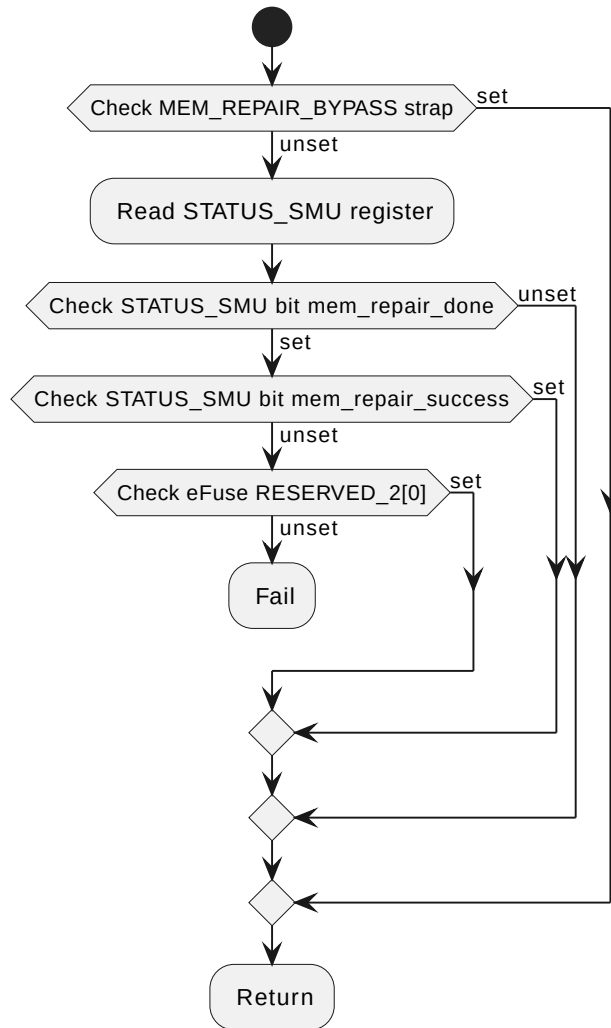


Figure 2. Production ROM `__metal_memory_repair_check`.

1.1.5. Per-Core Reset

See the *SMC CPU Reset, Boot, and Initialization* section of the Technical Reference Manual for the hardware behavior and associated risks.

WARNING Do not reset a hart while any hart is performing coherent memory accesses.

A safe startup scenario, subject to the warning above, is when three of four harts remain parked after ROM boot and must start at a different firmware address. Before they join normal coherent operation, software can program a parked hart's `RESET_VECTOR[n]` and pulse its reset to start it at that address. Avoid this flow where possible for the reasons described in the Technical Reference Manual.

1.1.6. External CPU cluster NDM reset

Read `NDMRESET_REQUEST` bit `i` as the NDM reset request from external CPU cluster `i`. Write `NDMRESET_PROCESS` bit `i` to answer that cluster. Read `NDMRESET_CLUSTER_COUNT` and mask both registers with that value.

This harness reports 4 in `NDMRESET_CLUSTER_COUNT`. Bits of `NDMRESET_REQUEST` at and above the count read as 0. Writes to those bits of `NDMRESET_PROCESS` stay in the register.

The register block is in the *Chip Config, Scratch, NDM Reset, and Misc Wrap* section of the Technical Reference Manual.

1.2. General Operation

1.2.1. Interrupt Control

There are 3 architectural interrupt sources that the cores of the SMU receive:

1. **Timer interrupts** - Interrupts generated by the system timer. In the SMC, the source of timer interrupts is the core local interruptor (CLINT).
2. **Software interrupts** - Interrupts signalled by other cores in the system. In the SMC, software-initiated interrupts are signalled through the CLINT.
3. **External interrupts** - Interrupts from all other peripherals. External interrupts are aggregated by the platform local interrupt controller (PLIC).

External interrupts have the highest priority, followed by Software interrupts, then Timer interrupts.

The `mtvec` CSR stores the trap vector which the cores transfer control to upon taking an interrupt or encountering an exception. The `mtvec` CSR contains the base address `BASE` of the trap vector, as well as a `MODE` field that determines how the trap vector address is computed:

1. If `MODE` is set to **Direct** (`0b00`), then all interrupts and exceptions cause the program counter to be set to `mtvec.BASE`.
2. If `MODE` is set to **Vectored** (`0b01`), then exceptions cause the program counter to be set to `mtvec.BASE`, and interrupts cause the program counter to be set to `mtvec.BASE + (0x4 * i)`, where `i` is the architectural interrupt type (Timer, Software, or External, per-privilege level).

Since program counter destinations in **Vectored** mode form a tightly-packed array, the entries for each interrupt should consist of a single jump instruction to a programmer-defined exception/interrupt handler function.

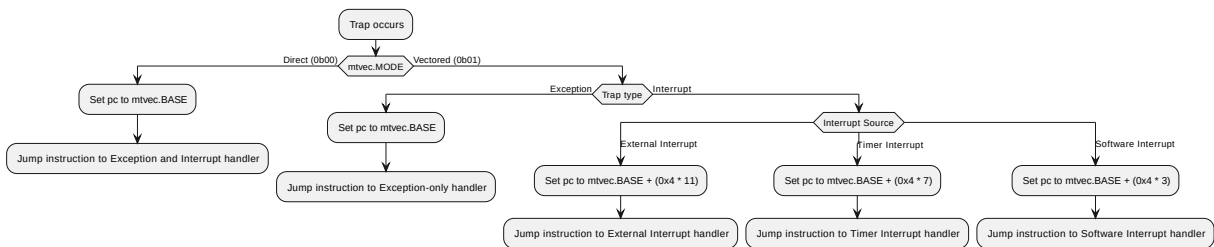


Figure 3. Exception and Interrupt Dispatch

Timer and Software Interrupt Control

The SMC uses the core local interruptor (CLINT) for Timer interrupts, as well as for software-initiated inter-processor

interrupts (IPIs). The base address for the CLINT — `CLINT_BASE` — is `0xC800_0000`.

The current value of the timer is stored in the CLINT's `MTIME` register, at `CLINT_BASE + 0xBFF8`. The CLINT stores a timer comparison value corresponding to each core of the SMC in the CLINT `MTIMECMP` register array, at `CLINT_BASE + 0x2000 + (0x8 * hartid)`. Both registers are 64-bits wide.

A timer interrupt becomes pending for a core once `MTIME` exceeds the core's corresponding `MTIMECMP` value. To dismiss the interrupt, the timer interrupt handler should set the core's corresponding `MTIMECMP` register to a new value in the future.

WARNING

There may be a delay between writing `MTIMECMP` and the timer interrupt pending bit falling, during which a spurious timer interrupt may occur if interrupts are enabled again (for example, by returning from the timer interrupt handler). For robustness against spurious timer interrupts, the timer interrupt handler should verify the interrupt condition (`MTIMECMP > MTIME`) before calling periodic functions and re-scheduling.

The following code samples should be used to read the full 64-bit `MTIME` value using 32-bit reads, guarding against overflow between the two halves, and setting the value of `MTIMECMP` without causing a spurious interrupt:

CLINT example C definitions.

```
// CLINT addresses.
#define CLINT_BASE                0xC8000000
#define CLINT_REG_MSIP(hartid)    (CLINT_BASE + (hartid * 0x4))
#define CLINT_REG_MTIMECMP_LO(hartid) (CLINT_BASE + 0x4000 + (hartid * 0x8))
#define CLINT_REG_MTIMECMP_HI(hartid) (CLINT_BASE + 0x4004 + (hartid * 0x8))
#define CLINT_REG_MTIME_LO        (CLINT_BASE + 0xbff8)
#define CLINT_REG_MTIME_HI        (CLINT_BASE + 0xbffc)
```

Reading `MTIME` example C usage.

```
uint64_t read_clint_mtime(void) {
    uint32_t hi, lo;

    // Make sure the high half has not overflowed
    // while reading the lower half.
    do {
        hi = read32(CLINT_REG_MTIME_HI);
        lo = read32(CLINT_REG_MTIME_LO);
    } while (read32(CLINT_REG_MTIME_HI) != hi);

    return ((uint64_t)hi << 32) | lo;
}
```

Writing `MTIMECMP` example C usage.

```
unsigned long read_hartid(void) {
    // Determine the current hartid by
```

```

// reading from the `mhartid` CSR.
unsigned long hartid;
__asm__ volatile(
    "csrr %[hartid], mhartid"
    : [hartid] "=r"(hartid)
    :: "memory"
);
return hartid;
}

void write_clint_mtimecmp(uint64_t value) {
    unsigned long hartid = read_hartid();

    write32(CLINT_REG_MTIMECMP_HI(hartid), 0xFFFFFFFF);
    write32(CLINT_REG_MTIMECMP_LO(hartid), (uint32_t)value);
    write32(CLINT_REG_MTIMECMP_HI(hartid), (uint32_t)(value >> 32));
}

```

Cores can send interrupts to other cores of the SMC by writing a value of **1** to the target core's corresponding **MSIP** register, at **CLINT_BASE + (0x4 * hartid)**. This writes directly to the target core's **MIE.msip** bit. Upon taking a software interrupt, a core may clear its software interrupt pending bit by writing **0** to its corresponding **MSIP** register.

Sending Software Interrupt to another core example C usage.

```

void software_interrupt_send(unsigned long target_hartid) {
    write32(CLINT_REG_MSIP(target_hartid), 1);
}

void software_interrupt_clear(unsigned long target_hartid) {
    write32(CLINT_REG_MSIP(target_hartid), 0);
}

```

External Interrupt Control

The SMC uses the standard RISC-V platform-level interrupt controller (PLIC) for managing external interrupts, see [here](#) for the RISC-V PLIC Specification and layout.

The SMC Interrupt Map can be found in the TRM (TODO: XREF). The allocation of additional interrupt IDs within the External interrupts block is defined by the integrator.

The process of determining and servicing the specific external interrupt is as follows, where **PLIC_BASE = 0xC400_0000**:

1. Claim the highest priority pending interrupt ID by reading from the PLIC **CLAIM** register for the hart at **PLIC_BASE + 0x20_0004 + (0x1000 * hartid)**.
2. If the returned interrupt ID is 0, return from the interrupt handler — any pending interrupts have been claimed by other cores on the system.
3. Dispatch to the corresponding interrupt handling routine, based on the claimed interrupt ID.

4. Complete the interrupt **at the peripheral**, clearing its interrupt pending signal **to the PLIC**.
5. Finally, complete the external interrupt **at the PLIC**, by writing the claimed interrupt ID PLIC **COMPLETE** register for the hart, also at **PLIC_BASE + 0x20_0004 + (0x1000 * hartid)**.

WARNING

If the interrupt is cleared at the PLIC before it is cleared at the interrupting peripheral, the interrupt may be raised again to the PLIC, causing a future spurious interrupt. Clearing of interrupt signals should be done from the inner-most source first (peripheral first, then PLIC).

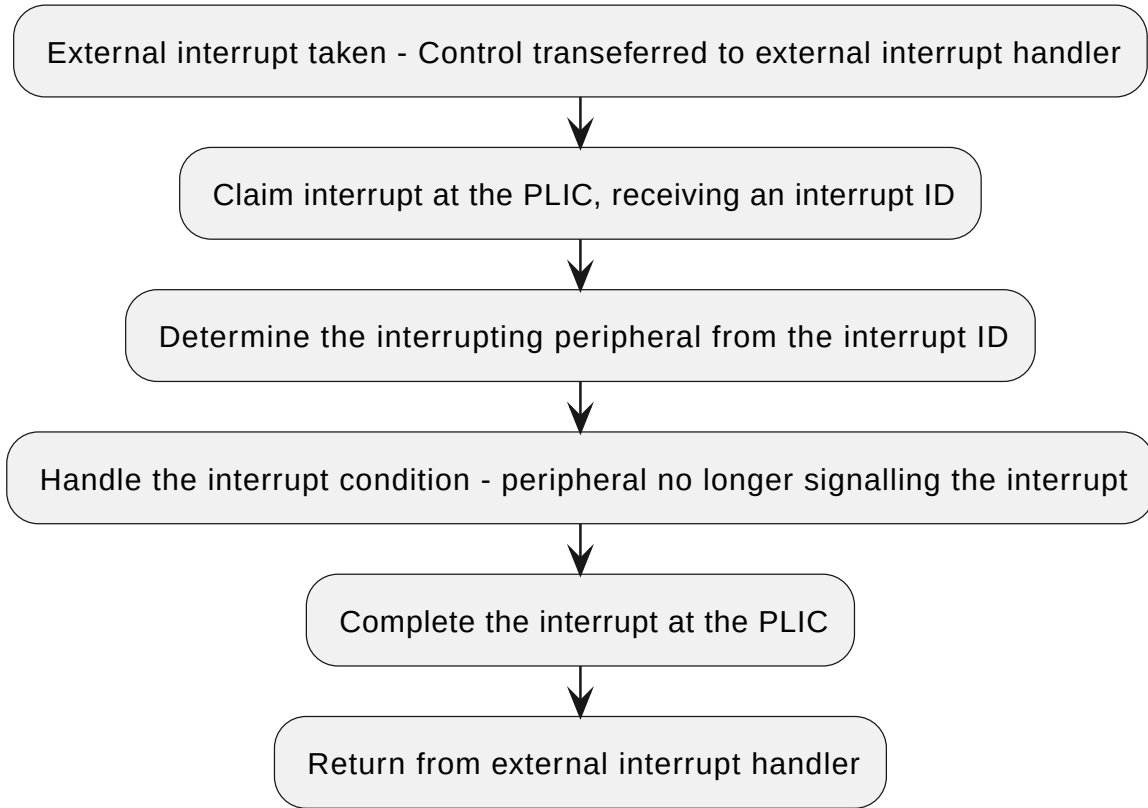


Figure 4. External Interrupt Handling Flow

1.2.2. Watchdog Timers

Each core of the SMC has a dedicated watchdog timer (WDT), that can generate an interrupt and optionally a reset of the core if not serviced within a configurable period of time. The watchdog timers are driven by the SMU clock signal **clk_smu_i**.

Table 2. WDT Address Map

Start Address	End Address	Instance
BASE + 0x0000_0000	BASE + 0x0000_0023	Core 0 WDT
BASE + 0x0000_0400	BASE + 0x0000_0423	Core 1 WDT
BASE + 0x0000_0800	BASE + 0x0000_0823	Core 2 WDT
BASE + 0x0000_0C00	BASE + 0x0000_0C23	Core 3 WDT

Table 3. WDT Per-Instance Registers

Offset	Register	Description
0x00	CTRL	Enable, scale, reset, and interrupt-pending flags. Writing requires an unlock by writing to KEY first.
0x08	COUNT	Raw 31-bit counter. Writing requires an unlock by writing to KEY first.
0x10	SCALED_COUNT	Scaled counter: <code>COUNT >> CTRL.wdogscale</code> . Read-only.
0x18	FEED	Write 0x009F00D to 'feed' the WDT, resetting COUNT to 0. Writing requires an unlock by writing to KEY first.
0x1C	KEY	Write 0x51F15E to unlock one register write. Reads return 1 if a register write is unlocked.
0x20	CMP	Comparison threshold. The WDT fires when <code>SCALED_COUNT >= CMP</code> . Writing requires an unlock by writing to KEY first.

Every register write (with the exception of KEY) requires unlocking by writing the value 0x51F15E to the KEY register. The WDT automatically locks after one register write after unlocking.

The `wdogenalways` (bit 12) and `wdogcoreawake` (bit 13) fields of CTRL control under what conditions COUNT is incremented. If `wdogcoreawake` is set, COUNT is only incremented while the core is not in reset. If `wdogenalways` is set, COUNT always increments. `wdogcoreawake` takes priority over `wdogcoreawake` if both bits are set.

If `wdogrsten` (bit 8) is enabled in CTRL and the WDT fires, the core being reset will not be able to service its own WDT interrupt. Other SMC cores **should** enable interrupts from eachother's watchdog timers to be notified when a core is reset by the watchdog, so that SMC software can robustly recover from the failure or reallocate work while the reset core is brought up again.

TODO rewrite: Key CTRL fields:

- `wdogscale` (bits 3:0) — right-shift applied to COUNT before comparison; slows the effective rate by $2^{\text{wdogscale}}$.
- `wdogzerocmp` (bit 9) — automatically reset COUNT to 0 when the compare fires.
- `wdogip0` (bit 28) — interrupt pending; write 0 to clear after servicing.

WDT example C definitions.

```
// WDT register addresses.
#define WDT_BASE          0xC000_0000
#define WDT_REG_CTRL(hartid) (WDT_BASE + (hartid * 0x400) + 0x0)
#define WDT_REG_FEED(hartid) (WDT_BASE + (hartid * 0x400) + 0x18)
#define WDT_REG_KEY(hartid)  (WDT_BASE + (hartid * 0x400) + 0x1c)
#define WDT_REG_CMP(hartid)  (WDT_BASE + (hartid * 0x400) + 0x20)

// WDT register constants.
#define WDT_KEY_UNLOCK     0x0051F15E
#define WDT_FEED_VALUE     0x0D09F00D

#define WDT_CTRL_WDOGRSTEN 0x100
#define WDT_CTRL_WDOGENALWAYS 0x1000
```

WDT example C configuration and usage.

```
unsigned long read_hartid(void) {
    // Determine the current hartid by
```

```

// reading from the `mhartid` CSR.
unsigned long hartid;
__asm__ volatile(
    "csrr %[hartid], mhartid"
    : [hartid] "=r"(hartid)
    :: "memory"
);
return hartid;
}

static void wdt_write(uintptr_t addr, uint32_t value) {
    // Helper to write to one of the core's WDT registers,
    // unlocking it before performing the actual write.
    unsigned long hartid = read_hartid();

    // Unlock.
    write32(WDT_REG_KEY(hartid), WDT_KEY_UNLOCK);

    // Write.
    write32(addr, value);
}

void wdt_configure(void) {
    unsigned long hartid = read_hartid();

    uint32_t compare = /* TODO */;
    uint32_t ctrl = /* TODO */;

    wdt_write(WDT_REG_CMP(hartid), compare);

    wdt_write(WDT_REG_CTRL(hartid), ctrl);
}

void wdt_service(void) {
    unsigned long hartid = read_hartid();

    wdt_write(WDT_REG_FEED(hartid), WDT_FEED_VALUE);
}

```

1.2.3. Logging peripherals

Software may log info and error messages by transmitting over UART. For more advanced and flexible use-cases, such as logging asynchronously and from multiple cores, the Log Engine peripheral can drive UART transmission on the CPU's behalf, with messages fed in from SMC memory.

The UARTs and Log Engines sit within the UART Controller block at address range `BASE + 0x6000 - BASE + 0x6FFF`. Within this block, there are 4 UART and Log Engine pairs (a stride of `0x400` between each).

Table 4. UART and Log Engine Address Map

Start Address	End Address	Description
BASE + 0x6100	BASE + 0x6124	UART 1
BASE + 0x6200	BASE + 0x6280	Log Engine 1
BASE + 0x6500	BASE + 0x6524	UART 2
BASE + 0x6600	BASE + 0x6680	Log Engine 2
BASE + 0x6900	BASE + 0x6924	UART 3
BASE + 0x6a00	BASE + 0x6a80	Log Engine 3
BASE + 0x6d00	BASE + 0x6d24	UART 4
BASE + 0x6e00	BASE + 0x6e80	Log Engine 4

UART

The UART baud rate is configured by writing the 16-byte clock divisor value to the **DLM** (most-significant byte) and **DLL** (least-significant byte) registers. To access these registers, the **DLAB** bit (bit 7) must be enabled in the **LCR** register. Once configured, the **DLAB** bit should be disabled to access the registers hidden by that bit.

FIFO configuration is done using the **FCR** and **ECR** registers. The FIFO is enabled by setting the least-significant bit in the **FCR** register. The **RCVR_TRIGGER_MS2B** field of **ECR** and **RCVR_TRIGGER** field of **FCR** form a 4-bit value which determines the receive FIFO trigger level. While the receive FIFO contains more than this amount of characters, the UART asserts the Received Data Ready interrupt.

Table 5. UART Receive FIFO levels

Value	ECR.RCVR_TRIGGER_MS2B	FCR.RCVR_TRIGGER	FIFO trigger level
0x0 (0b0000)	0b00	0b00	1 character.
0x1 (0b0001)	0b00	0b01	4 characters.
0x2 (0b0010)	0b00	0b10	8 characters.
0x3 (0b0011)	0b00	0b11	14 characters.
0x4 (0b0100)	0b01	0b00	32 characters.
0x5 (0b0101)	0b01	0b01	64 characters.
0x6 (0b0110)	0b01	0b10	128 characters.
0x7 (0b0111)	0b01	0b11	256 characters.
0x8 (0b1000)	0b10	0b00	512 characters.
0x9 (0b1001)	0b10	0b01	1024 characters.
0xA (0b1010)	0b10	0b10	2048 characters.
0xB (0b1011)	0b10	0b11	4096 characters.

Data is written by writing to the **THR** register. If the FIFO is enabled, the written data will be pushed onto the transmission FIFO. Data is popped from the receive FIFO by reading the **RBR** register. Note that the **THR** and **RBR** registers are accessible only when the **DLAB** bit of **LCR** is disabled).

UART example C definitions.

```
// UART Register addresses.
#define UART_BASE          0xC0006100
#define UART_REG_RBR_THR (UART_BASE + 0x0) // DLAB=0, Reads access RBR, writes access
```

```

THR.
#define UART_REG_DLL      (UART_BASE + 0x0) // DLAB=1
#define UART_REG_IER      (UART_BASE + 0x4) // DLAB=0
#define UART_REG_DLM      (UART_BASE + 0x4) // DLAB=1
#define UART_REG_IIR_FCR  (UART_BASE + 0x8) // Reads access IIR, writes access FCR.
#define UART_REG_LCR      (UART_BASE + 0xc)
#define UART_REG_LSR      (UART_BASE + 0x14)
#define UART_REG_ECR      (UART_BASE + 0x20)

// Register constants.
#define UART_IER_ALL 0x1f

#define UART_FCR_FIFO_ENABLE      0x1
#define UART_FCR_RCVR_FIFO_RESET 0x2
#define UART_FCR_XMIT_FIFO_RESET 0x4
#define UART_FCR_DMA_MODE_SELECT_0 0x0
#define UART_FCR_DMA_MODE_SELECT_1 0x8
#define UART_FCR_RCVR_TRIGGER_256C 0xc0

#define UART_LCR_PEN      0x8 // Parity enable bit.
#define UART_LCR_STB_2S 0x4 // 2 stop bits.
#define UART_LCR_STB_1S 0x0 // 1 stop bit.
#define UART_LCR_WLS_8B 0x3 // 8 bits per character.
#define UART_LCR_DLAB      0x80

#define UART_LSR_THRE 0x20

#define UART_ECR_RCVR_TRIGGER_MS2B_256C 0x1

```

UART example C configuration and usage (no FIFO).

```

void uart_configure_nofifo(void) {
    // Reset the UART.
    // Disable interrupts.
    write32(UART_REG_IER, 0);
    // A divisor of 0 disables the transmitter and receiver.
    write32(UART_REG_LCR, UART_LCR_DLAB);
    write32(UART_REG_DLL, 0);
    write32(UART_REG_DLM, 0);
    // Reset and disable FIFOs.
    write32(UART_REG_IIR_FCR,
            UART_FCR_RCVR_FIFO_RESET | UART_FCR_XMIT_FIFO_RESET
            | UART_FCR_DMA_MODE_SELECT_0);

    // UART configuration: 8 bits per character, 1 stop bit, no parity bits.
    uint32_t uart_config = UART_LCR_WLS_8B | UART_LCR_STB_1S;
}

```

```

// Write configuration to LCR, and enable DLAB to configure baud rate.
write32(UART_REG_LCR, uart_config | UART_LCR_DLAB);

// Calculate clock divisor for the intended baud rate.
uint16_t divisor = div_cceil(CLOCK_FREQUENCY, 16 * TARGET_BAUD) - 1;

// Write divisor bytes to DLL and DLM.
write32(UATR_REG_DLL, (uint32_t)((uint8_t)divisor));
write32(UART_REG_DLM, (uint32_t)((uint8_t)(divisor >> 8)));

// Write configuration again to LCR, clearing the DLAB bit.
write32(UART_REG_LCR, uart_config);

// Enable interrupts.
write32(UART_REG_IER, UART_IER_ALL);
}

void uart_write(const char *str) {
    while (*str) {
        // Poll LSR until THR is empty.
        while (read32(UART_REG_LSR) & UART_LSR_THRE);

        // Write the character to THR.
        write32(UART_REG_RBR_THR, (uint32_t)(*str));
        str++;
    }
}

```

UART example C configuration and usage (with FIFO).

```

void uart_configure_fifo(void) {
    // Reset the UART.
    // Disable interrupts.
    write32(UART_REG_IER, 0);
    // A divisor of 0 disables the transmitter and receiver.
    write32(UART_REG_LCR, UART_LCR_DLAB);
    write32(UART_REG_DLL, 0);
    write32(UART_REG_DLM, 0);
    // Reset and disable FIFOs.
    write32(UART_REG_IIR_FCR, UART_FCR_RCVR_FIFO_RESET | UART_FCR_XMIT_FIFO_RESET);

    // FIFO configuration: Enabled, 256-character trigger level.
    write32(UART_REG_ECR, UART_ECR_RCVR_TRIGGER_MS2B_256C);
    write32(UART_REG_IIR_FCR,
            UART_FCR_RCVR_TRIGGER_256C | UART_FCR_FIFO_ENABLE
            | UART_FCR_DMA_MODE_SELECT_0);
}

```

```

// UART configuration: 8 bits per character, 1 stop bit, no parity bits.
uint32_t uart_config = UART_LCR_WLS_8B | UART_LCR_STB_1S;

// Write uart configuration to LCR, and enable DLAB to configure baud rate.
write32(UART_REG_LCR, uart_config | UART_LCR_DLAB);

// Calculate clock divisor for the intended baud rate.
uint16_t divisor = div_ceil(CLOCK_FREQUENCY, 16 * TARGET_BAUD) - 1;

// Write divisor bytes to DLL and DLM.
write32(UATR_REG_DLL, (uint32_t)((uint8_t)divisor));
write32(UART_REG_DLM, (uint32_t)((uint8_t)(divisor >> 8)));

// Write configuration again to LCR, clearing the DLAB bit.
write32(UART_REG_LCR, uart_config);

// Enable interrupts.
write32(UART_REG_IER, UART_IER_ALL);
}

void uart_write_fifo(const char *str) {
    while (*str) {
        // Poll until transmit FIFO is empty.
        while (read(UART_REG_LSR) & UART_LSR_THRE);

        // Burst write up to 4096 characters at a time.
        for (size_t i = 0; i < 4096; i++) {
            if (*str == 0) return;
            write32(UART_REG_RBR_THR, (uint32_t)(*str));
            str++;
        }
    }
}

```

Log Engine Configuration

The Log Engine is a DMA-based logging peripheral that can efficiently transfer log messages from SMC memory to a UART on the CPU's behalf. The UART exposes the `txrdy_o` signal directly to the Log Engine which notifies it that more bytes can be accepted into the transmit FIFO.

Each Log Engine supports up to 16 log entries, which represent a message to be transmitted. The Log Engine uses round-robin scheduling for fairness in multi-threaded usage.

Before configuration begins, software should disable the Log Engine's operation by writing `0` to the Log Engine's `CTRL` register. After configuration is complete, the Log Engine can be enabled by writing `1` to the `CTRL` register.

The Log Engine is configured by writing the address and size of the region of SMC memory that will be used for log entries, into the `LOG_REGION_ADDR` and `LOG_REGION_SIZE` registers respectively. This memory buffer is divided evenly into

the 16 regions, each representing a separate log entry.

The Log Engine will write bytes to the UART address specified in the `LOG_WRITE_ADDR` register. This must be set to the address of the associated UART's `THR` register, which feeds the transmission FIFO.

The UART used by the Log Engine must be configured first, before it can be used by the Log Engine. The UART used must also be configured to use DMA Mode 0 by clearing the `DMA_MODE_SELECT` bit in the UART's `FCR` register. See the previous section on configuring the UART.

Table 6. Log Entry Memory Buffer

Start Address	End Address	Size	Description
<code>LOG_REGION_ADDR</code>	<code>LOG_REGION_ADDR + (1 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_SIZE / 16</code>	Log Entry 1/16
<code>LOG_REGION_ADDR + (1 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_ADDR + (2 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_SIZE / 16</code>	Log Entry 2/16
<code>LOG_REGION_ADDR + (2 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_ADDR + (3 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_SIZE / 16</code>	Log Entry 3/16
...	...	...	...
<code>LOG_REGION_ADDR + (15 * (LOG_REGION_SIZE / 16))</code>	<code>LOG_REGION_ADDR + LOG_REGION_SIZE</code>	<code>LOG_REGION_SIZE / 16</code>	Log Entry 16/16

The size of memory region written to `LOG_REGION_SIZE` must be a multiple of 128-bytes, meaning each log entry is a multiple of 8-bytes in size.

Log Engine and UART example C definitions.

```
// Log Engine Register addresses.
#define LOG_ENGINE_BASE          0xC0006200
#define LOG_ENGINE_REG_CTRL      (LOG_ENGINE_BASE + 0x0)
#define LOG_ENGINE_REG_LOG_REGION_SIZE (LOG_ENGINE_BASE + 0x4)
#define LOG_ENGINE_REG_LOG_REGION_ADDR_LO (LOG_ENGINE_BASE + 0x8)
#define LOG_ENGINE_REG_LOG_REGION_ADDR_HI (LOG_ENGINE_BASE + 0xc)
#define LOG_ENGINE_REG_LOG_WRITE_ADDR (LOG_ENGINE_BASE + 0x10)
#define LOG_ENGINE_REG_INTR_ENABLE (LOG_ENGINE_BASE + 0x14)
#define LOG_ENGINE_REG_LOG_CTRL(num) (LOG_ENGINE_BASE + 0x40 + (num * 0x4))

// UART Register addresses.
#define UART_BASE          0xC0006100
#define UART_REG_RBR_THR (UART_BASE + 0x0)

// Register constants.
#define LOG_ENGINE_CTRL_ENABLE      1
#define LOG_ENGINE_CTRL_DISABLE    0
#define LOG_ENGINE_INTR_LOG_FETCH_ERR 0x1
#define LOG_ENGINE_INTR_LOG_WRITE_ERR 0x10

// Sample log region.
#define LOG_ENTRY_SIZE  1024
#define LOG_REGION_SIZE (LOG_ENTRY_SIZE * 16)
char log_region[LOG_REGION_SIZE];
```

Log Engine example C configuration.

```
// Configure the Log Engine to read logs from `log_region`.
void configure(char *log_region, uint32_t log_region_size) {
    // Disable operation.
    write32(LOG_ENGINE_REG_CTRL, LOG_ENGINE_CTRL_DISABLE);

    // Write address of log region.
    write32(LOG_ENGINE_REG_LOG_REGION_ADDR_LO, (uint32_t)log_region);
    write32(LOG_ENGINE_REG_LOG_REGION_ADDR_HI, (uint32_t)(log_region >> 32));

    // Write size of log region.
    write32(LOG_ENGINE_REG_LOG_REGION_SIZE, log_region_size);

    // Write the address of the UART THR register.
    write32(LOG_ENGINE_REG_LOG_WRITE_ADDR, UART_REG_RBR_THR);

    // Enable error interrupts.
    write32(LOG_ENGINE_REG_INTR_ENABLE,
            LOG_ENGINE_INTR_LOG_FETCH_ERR | LOG_ENGINE_INTR_LOG_WRITE_ERR);

    // Enable operation.
    write32(LOG_ENGINE_REG_CTRL, LOG_ENGINE_CTRL_ENABLE);
}
```

Log Engine Operation

To engage the Log Engine in transmitting a log entry, software must:

1. Choose a vacant log entry index.
2. Copy or write the message to be sent into the chosen log entry index's corresponding region in the memory buffer.
3. Write the length of the to be sent message into the log entry's corresponding **LOG_CTRL** register in the Log Engine.

Reading from the **LOG_CTRL** register array returns the number of remaining bytes for each log entry. A log entry has finished transmission once reading the corresponding **LOG_CTRL** register returns 0.

Log Engine example C usage.

```
// Print formatted output to UART, using the Log Engine.
void log_engine_printf(const char *format, ...) {
    // Determine and pick a vacant log entry index.
    // In this example, the arbitrary index `5` suffices.
    int index = 5;

    va_list args;
    va_start(args, format);
```

```

// Write the message into the vacant log entry region,
// writing at most `LOG_ENTRY_SIZE` bytes.
int len = snprintf(&log_region[index * LOG_ENTRY_SIZE],
                  LOG_ENTRY_SIZE, format, args);

va_end(ap);

if (len < 0) {
    // Printing failed.
    return;
}

// Write the length of the message. This queues up the transmission.
write32(LOG_ENGINE_REG_LOG_CTRL(index), len);

// Poll for completion.
while (read32(LOG_ENGINE_REG_LOG_CTRL(index)) != 0);
}

```

A more performant implementation may choose to skip polling for completion, and only initially block while all log entries are in-use.

As the log engine can only feed the UART transmit FIFO, message reception and draining the receive FIFO remains the responsibility of the cores.

1.3. GPIO

The SMC exposes 65 GPIO pins, each controlled through its own register instance within the GPIO block at **BASE + 0x0000_3000**. Per-pin registers start at **GPIO_BASE + (pin * 0x10)**.

Table 7. GPIO Address Map

Start Address	End Address	Description
BASE + 0x0000_3000	BASE + 0x0000_3FFF	GPIO block

Table 8. GPIO Per-Pin Registers

Offset	Register	Description
+0x0	DATA_CTRL	Direction, data, interrupt, and LSIO (Low-Speed I/O hardware peripheral) control
+0x4	DATA_CTRL_ENABLE	Per-field LSIO override enables
+0x8	ACCESS_FILTER	AXI protection filtering

The per-pin **DATA_CTRL** register contains the pin's configuration and value.

The **pad2core** field (bit 31) contains the current sampled value of the pin. The **core2pad** field (bit 0) contains the current value driven to the pad.

The **enable_rx_tx** field determines the direction of pin: .GPIO **enable_rx_tx** values.

Value	Description
0b01	Output / TX
0b10	Input / RX

The pin may also raise edge-sensitive or level-sensitive interrupts. The interrupt condition is configured in the `interrupt_type` field:

Table 9. GPIO `interrupt_type` values.

Value	Description
0b00	Level interrupt, Active-high
0b01	Level interrupt, Active-low
0b10	Edge interrupt, Rising-edge
0b11	Edge interrupt, Falling-edge

To enable interrupts for the specific pin, `interrupt_enable` (bit 18) must be set.

The per-pin `DATA_CTRL_ENABLE` register provides per-field LSIO overrides. When an LSIO peripheral controls the pad (`DATA_CTRL.interface_enable` is not set), setting a bit here forces the corresponding `DATA_CTRL` field to take effect for that field only.

1. `use_reg_core2pad` (bit 0) forces the register-controlled `DATA_CTRL.core2pad` onto the pad, regardless of LSIO output.
2. `use_reg_tx` (bit 4) forces the register-controlled `DATA_CTRL.enable_rx_tx[0]` (TX enable) value, regardless of the LSIO output.
3. `use_reg_rx` (bit 5) forces the register-controlled `DATA_CTRL.enable_rx_tx[1]` (RX enable) value, regardless of the LSIO output.

`DATA_CTRL_ENABLE` is only meaningful for pins where an LSIO peripheral is used. For general-purpose register-driven pin usage by the cores, `DATA_CTRL_ENABLE` should be left at 0.

GPIO example C definitions.

```
#define GPIO_BASE                0xC0003000
#define GPIO_PIN_BASE(pin)      (GPIO_BASE + ((pin) * 0x10))
#define GPIO_REG_DATA_CTRL(pin) (GPIO_PIN_BASE(pin) + 0x0)
#define GPIO_REG_DATA_CTRL_ENABLE(pin) (GPIO_PIN_BASE(pin) + 0x4)
#define GPIO_REG_ACCESS_FILTER(pin) (GPIO_PIN_BASE(pin) + 0x8)

#define GPIO_DATA_CTRL_CORE2PAD    0x1
#define GPIO_DATA_CTRL_ENABLE_RX_TX_TX 0x10
#define GPIO_DATA_CTRL_ENABLE_RX_TX_RX 0x20
#define GPIO_DATA_CTRL_INTERFACE_ENABLE 0x10000
#define GPIO_DATA_CTRL_INTERRUPT_ENABLE 0x40000
#define GPIO_DATA_CTRL_INTR_RISING_EDGE 0x200000
#define GPIO_DATA_CTRL_PAD2CORE    0x80000000
```

```
void gpio_configure_output(size_t pin, bool value) {
    // Enable the GPIO and set the direction to TX.
    uint32_t gpio_ctrl = GPIO_DATA_CTRL_INTERFACE_ENABLE |
GPIO_DATA_CTRL_ENABLE_RX_TX_TX;
    // If 'value' is set, drive the GPIO.
    if (value) {
        gpio_ctrl |= GPIO_DATA_CTRL_CORE2PAD;
    }
    write32(GPIO_REG_DATA_CTRL(pin), gpio_ctrl);
}

void gpio_set(size_t pin, bool value) {
    // Read the GPIO settings, then drive the value based on 'value'.
    uint32_t gpio_ctrl = read32(GPIO_REG_DATA_CTRL(pin));
    if (value) {
        gpio_ctrl |= GPIO_DATA_CTRL_CORE2PAD;
    } else {
        gpio_ctrl &= ~GPIO_DATA_CTRL_CORE2PAD;
    }
    write32(GPIO_REG_DATA_CTRL(pin), gpio_ctrl);
}

bool gpio_read(size_t pin) {
    // Configure as input first if not already done.
    write32(GPIO_REG_DATA_CTRL(pin),
        GPIO_DATA_CTRL_INTERFACE_ENABLE | GPIO_DATA_CTRL_ENABLE_RX_TX_RX);
    return (read32(GPIO_REG_DATA_CTRL(pin)) & GPIO_DATA_CTRL_PAD2CORE) ? true : false;
}

void gpio_configure_interrupt(size_t pin) {
    write32(GPIO_REG_DATA_CTRL(pin),
        GPIO_DATA_CTRL_INTERFACE_ENABLE | GPIO_DATA_CTRL_ENABLE_RX_TX_RX
        | GPIO_DATA_CTRL_INTERRUPT_ENABLE | GPIO_DATA_CTRL_INTR_RISING_EDGE);
}
```

Chapter 2. SMC Production ROM Reference

2.1. Overview

The SMC production ROM establishes a usable SMC execution environment and then services Open Chiplet Configuration Protocol (OCCP) requests over target-mode I2C and I3C interfaces. It performs early SRAM handling and memory-health checks before using a stack, initializes status and coordination state, applies the ROM's lifecycle-based access policy, and exposes commands for status, memory transfer, and boot handoff.

The ROM follows one common flow regardless of the `PRIMARY_CHIPLET` strap. The strap value is reported for observability, but the ROM does not implement separate primary and secondary boot algorithms.

The implementation is divided into:

- assembly startup in `boot/`, including SRAM initialization, MBIST, memory-repair checking, trap setup, and C-runtime entry;
- the Phase 0 through Phase 5 C flow in `src/main.c`;
- I2C, I3C, strap, eFuse, and optional PLL drivers in `drivers/`;
- OCCP, security, status, scratch-register coordination, and ring-buffer support in `lib/`; and
- the ROM/SRAM image layout in `linker/smc/smc_rom.ld`.

Selected adopter-facing services are weak-symbol ROM FW integration hooks. The open definitions allow the image to build without adopter-owned PLL or I3C implementations; a final adopter ROM FW image can replace them with strong definitions before the ROM FW is frozen. The detailed sections identify the hooks where they are used. For the build-time override method and the SMC override inventory, see [Immutable ROM FW Integration](#) in the Integrator Guide.

The ROM has two terminal handoff paths. `ExecuteImage` sends a success response and transfers control directly in non-secure lifecycle states. `AuthenticateImage` sends a success response, publishes a manifest offset to the coordination scratch registers, and parks the SMC in `wfi`.

2.2. OCCP protocol

OCCP is a request-response protocol carried as an ordered little-endian byte stream. A controller writes one request frame to an SMC target and reads the resulting response frame. The ROM implements version 1.0.0 of the Base and Boot applications.

2.2.1. Packet format

Every frame begins with the same eight-byte packet header. The body immediately follows the header, followed by a body CRC only when the `Body CRC present` bit is set. The 11-bit `Length` field is the body length in bytes and does not include either CRC.

OCCP frame

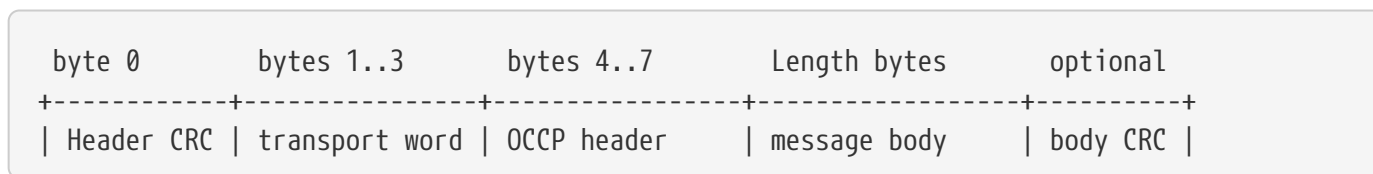


Table 10. Packet-header fields

Frame bits	Field	Meaning
7:0	Header CRC	CRC-8 over frame bytes 1 through 7.
8	Body CRC present	A CRC follows the body when set.
31:9	I3C flags	Reserved by this implementation; transmitted as zero in responses.
39:32	Application ID	0x00 for Base or 0x01 for Boot.
47:40	Message ID	Command within the selected application.
51:48	Flags	Request flags. Responses echo these bits.
52	Error	Set in an error response.
63:53	Length	Body size from 0 through 2047 bytes.

Multi-byte integers in command bodies and responses are little-endian. Addresses are 64-bit values.

2.2.2. Integrity checks

Header CRC uses an initial value of **0xFF**, polynomial **0xD3**, most-significant-bit-first processing, and no final XOR. It always covers the seven bytes after the CRC byte.

When a body CRC is present, bodies of up to 14 bytes use the same CRC-8 algorithm. Bodies longer than 14 bytes use CRC-32 with initial value **0xFFFFFFFF**, polynomial **0x992C1A4C**, most-significant-bit-first processing, and final XOR **0xFFFFFFFF**. The CRC covers only the body.

Requests with a clear **Body CRC present** bit are accepted without body integrity validation. Responses that contain a body set the bit and append the applicable CRC. Successful responses with no body clear it.

2.2.3. Applications and commands

Table 11. Implemented commands

AppID	MsgID	Command	Behavior
0x00	0x00	GetVersion	Returns Base application version 1.0.0.
0x00	0x01	GetStatus	Returns one OCCP status field or one entry from a status ring buffer.
0x00	0x02	WriteData	Writes bytes to an allowed, 32-bit-aligned address.
0x00	0x03	ReadData	Reads bytes from an allowed, 32-bit-aligned address.
0x01	0x00	GetVersion	Returns Boot application version 1.0.0.
0x01	0x01	ExecuteImage	Responds and transfers control to an allowed address in a non-secure lifecycle state.
0x01	0x02	AuthenticateImage	Responds, publishes a manifest address to the coordination registers, and parks in wfi .
0x01	0x03	TrainD2D	Not implemented; returns error code 0x07 .

GetVersion

The request body is empty. Both **GetVersion** handlers also flush the selected receive interface before responding. If bytes remain after the nominal zero-length request, the ROM FW returns **Overflow_msg** instead of the version. The four-byte

response body is:

Table 12. GetVersion response body

Offset	Size	Field
0	1	Major version
1	1	Minor version
2	2	Patch version

Both applications currently return **1, 0, 0**.

GetStatus

The request body is exactly two bytes containing a 16-bit status ID. The response body is one 32-bit status value.

Table 13. Supported status IDs

Status ID	Returned value
0x0000	OCCP boot-status nibble, encoded as an SMC BL0 status message.
0x0001	OCCP interface-status nibble, encoded as an SMC BL0 status message.
0x0002	OCCP command count in bits 7:0, encoded as an SMC BL0 status message. The current implementation also copies the internal-error byte into bits 15:8.
0x0003	Last internal implementation error, encoded as an SMC BL0 error message.
0x8000	Next unread SEP ring-buffer entry.
0x8001	Next unread SMC ring-buffer entry.

Other status IDs return **Unsupported_StatusID**. If the **STATUS_RPT_DISABLE** strap is asserted, every **GetStatus** request returns that error. An empty SEP or SMC ring buffer returns a zero value.

The internal error field returned for status ID **0x0003** is distinct from the protocol error codes carried by error-response bodies:

Table 14. Internal implementation errors

Value	Meaning
0x00	No error
0x01	Invalid command
0x02	Access violation
0x03	Alignment error
0x04	Interface error
0x05	Security violation
0x06	Buffer overflow
0x07	Timeout
0x08	CRC failure
0x09	Incomplete transport frame
0x0A	Oversized transport frame
0xFF	General error

These are the values defined by the software OCCP status interface; not every value is emitted by the current handlers.

In particular, misaligned read, write, execute, and authenticate addresses record access violation (0x02); alignment error (0x03) is defined but not emitted.

For a packet-header read that is incomplete or oversized, the wire response uses `Incomplete_msg` or `Oversize_msg`, but the internal error field remains timeout (0x07) because the command-loop return value has not yet been replaced. Equivalent failures returned from a command-body handler record incomplete (0x09) or overflow (0x0A).

WriteData

The request body contains a 12-byte command prefix followed by the write data.

Table 15. WriteData request body

Offset	Size	Field
0	8	Destination address
8	2	Bits 10:0: write size; bits 15:11: address attributes
10	2	Reserved
12	Write size	Data

The header body-length field must be from 12 through 2047 bytes. The encoded write size must be from 1 through 2035 bytes. The current handler validates these two values independently; it does not require the body-length field to equal 12 plus the encoded write size. The address attributes and reserved field are parsed but do not affect behavior. The destination must be 32-bit aligned and pass the target's memory-access policy. For a four-byte transfer, the ROM FW performs one 32-bit access. For an eight-byte transfer to an eight-byte-aligned address, it performs one 64-bit access. All other transfer sizes use byte accesses in increasing address order. A successful response has an empty body.

ReadData

The request body is exactly 12 bytes.

Table 16. ReadData request body

Offset	Size	Field
0	8	Source address
8	2	Bits 10:0: read size; bits 15:11: address attributes
10	2	Reserved

The read size must be from 1 through 2047 bytes. The address attributes and reserved field do not affect behavior. The source must be 32-bit aligned and pass the target's memory-access policy. For a four-byte transfer, the ROM FW performs one 32-bit access. For an eight-byte transfer from an eight-byte-aligned address, it performs one 64-bit access. All other transfer sizes use byte accesses in increasing address order. The response body is the requested data and always carries a body CRC.

ExecuteImage and AuthenticateImage

Both commands use the same ten-byte body:

Table 17. Boot command request body

Offset	Size	Field
0	8	Execution or manifest address

Offset	Size	Field
8	1	CPU ID
9	1	Bits 2:0: reserved; bits 7:3: address attributes

The CPU ID, reserved bits, and address attributes do not affect ROM behavior. The address must be nonzero, 32-bit aligned, and pass the target's memory-access policy. The SMC ROM applies that policy to an eight-byte range beginning at the supplied address.

ExecuteImage is accepted only in a non-secure lifecycle state. In a secure lifecycle state it returns **Invalid_Msgid**. When accepted, the ROM sends an empty success response, records boot completion, and jumps to the address.

AuthenticateImage is accepted in every valid lifecycle state. After the empty success response, the ROM converts an in-range SRAM address to an offset in scratch register 8, sets **MANIFEST_READY** in scratch register 9, and remains in **wfi**. An address outside the SMC SRAM range produces the scratch-register marker **0xFFFFFFFF**; the current handler still sets **MANIFEST_READY**.

2.2.4. Error responses

An error response sets the header Error bit and has a four-byte body containing the numeric error code. It always includes a CRC-8 body CRC. The response normally echoes the request AppID, MsgID, and flags. For a corrupt header CRC, AppID and MsgID are both **0xFF** because the request identifiers are not trusted.

Table 18. Protocol error codes

Code	Name	Condition
0x01	Invalid_Appid	The application is not implemented.
0x02	Invalid_Msgid	The message is not implemented for the application, or ExecuteImage was requested in a secure lifecycle state.
0x03	Invalid_header	A command body length or encoded transfer size is invalid.
0x04	Corrupt_header	Header CRC validation failed.
0x05	Corrupt_Data	Body CRC validation failed.
0x06	Invalid_Address	An address is null, misaligned, outside an allowed range, or overlaps protected memory.
0x07	Unsupported_StatusID	A status ID is unsupported, status reporting is disabled, or Train02D was requested.
0x08	Invalid_Request_Length	Defined by the implementation but not emitted by current handlers.
0x09	Invalid_Request	Defined by the implementation but not emitted by current handlers.

The transport binding also uses **0x00010000** for an incomplete frame, **0x00010001** for an oversized frame, and **0x00010002** for a transport CRC failure. The I2C/I3C binding does not generate the transport-CRC error. When the eight-byte header itself is incomplete or oversized, the error response also uses AppID and MsgID **0xFF** because no valid header is available.

2.2.5. I2C and I3C transport binding

The controller sends the packet header, body, and optional body CRC in one target-write transaction. The ROM first consumes the eight-byte header, then consumes the declared body and CRC from the same transaction. A transaction ending before the expected byte count produces **Incomplete_msg**; bytes beyond the expected end produce **Oversize_msg**. On protocol or framing errors, the ROM discards residual bytes belonging to the failed frame before sending the error response.

The controller obtains the response with a target-read transaction. For the open HCI I3C target implementation, response data is queued before the transmit descriptor advertises its length. Reads made before a response is armed are not accepted by the target hardware; the controller retries until the response descriptor is available. Large responses are streamed while the hardware drains the transmit queue. The I3C receive path retains frame-length accounting across the separate header and body reads, allowing it to distinguish incomplete, exact, and oversized writes.

I2C uses the target driver's transaction boundaries for the same exact-length checks. The transport timeout comes from eFuse `RESERVED_1` in system-clock polling iterations; a zero value selects 10,000.

2.3. Boot flow

The assembly entry path runs before any stack or writable global data is used. After the C runtime is ready, hart 0 runs Phase 0 through Phase 5. Other harts remain in the runtime's `wfi` loop.

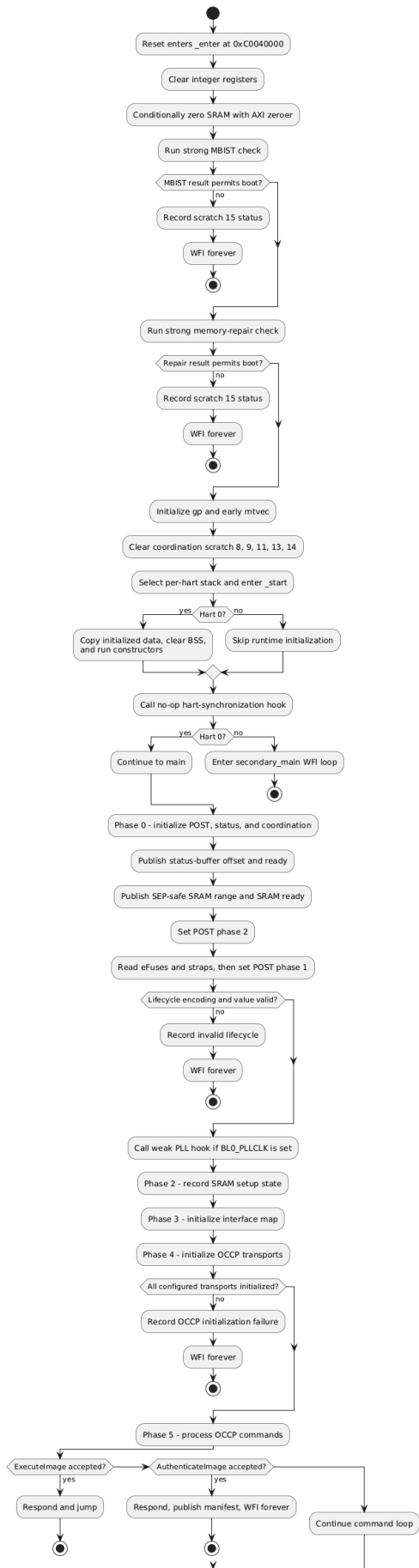


Figure 5. Detailed boot flow

2.3.1. Assembly early boot

`_enter` performs these operations in order:

1. Clear general-purpose registers `x1` through `x31`.
2. Call `__metal_sram_init_early`.
3. Call the strong `__metal_mbist_early_check` implementation.
4. Call the strong `__metal_memory_repair_check` implementation.
5. Establish `gp`.
6. Install `early_trap_vector` in `mtvec`.
7. Clear coordination scratch registers 8, 9, 11, 13, and 14 while preserving scratch register 15.
8. Establish a 16-byte-aligned stack for the current hart and call `_start`.

On hart 0, the C runtime copies initialized data from ROM, clears BSS, and initializes TLS and constructors. Other harts skip that initialization. Every hart calls `__metal_synchronize_harts`, whose implementation linked into the production ROM FW is a no-op, and then disables its interrupt sources. Hart 0 calls `main`; the other harts enter the `secondary_main wfi` loop.

Unused ROM is filled with the 32-bit `EBREAK` instruction. Once the early trap vector is installed, a breakpoint whose `mepc` lies in the linker-defined padding range writes `0xBADF00D0` to scratch register 15 and loops. Other early traps write `0xFFFFFFFF` to scratch register 0 and loop.

2.3.2. C boot phases

Phase 0 — initialization

Initializes POST tracking, the software OCCP status word, both status ring buffers, and scratch coordination. Status initialization publishes the SEP ring-buffer offset, sets the status-buffer-ready bit, and emits its initial ring-buffer marker. The main flow then publishes the SEP-safe SRAM range and sets the SRAM-initialized bit. After publishing that state, the implementation writes POST phase 2 before it starts the phase 1 strap and eFuse work.

Phase 1 — straps, eFuses, and lifecycle

Caches the eFuse and strap fields used by the ROM. The ROM validates the differential lifecycle encoding and halts on an invalid encoding or unsupported lifecycle value. If `BL0_PLLCLK` is set, it calls `smc_pll_init`; the open implementation is a weak hook that returns success without programming a PLL.

Phase 2 — SRAM state

Records the SRAM setup phase again after phase 1 completes. The actual conditional zeroing and memory-health checks have already completed in assembly.

Phase 3 — interface map

Marks I2C0, I2C1, I3C0, I3C1, and I3C3 enabled in the software map. The current ROM uses one common map for all strap roles. OCCP initialization currently configures that same fixed set directly rather than querying the software map.

Phase 4 — OCCP ready

Initializes the fixed target-interface set, updates the OCCP interface status, and drives status GPIO 58 high on success or low on failure. Any initialization error is fatal to the C boot flow.

Phase 5 — OCCP processing

Enters the non-returning command loop described in [OCCP command processing](#). The ROM FW emits the `SMC_STATUS_BOOT_COMPLETE` ring-buffer value when the loop is entered, but it does not set POST boot phase 6 at that point. POST phase 6 is set only before an accepted `ExecuteImage` transfers control. Direct execution or authenticated handoff terminates command processing as described in [OCCP protocol](#).

2.4. Hardware initialization

2.4.1. Straps and eFuses

Early assembly reads strap registers at `0xC0405800` and `0xC0405804`. The C strap driver later caches the fields used for reporting or policy.

Table 19. ROM strap use

Strap	Bit	ROM use
<code>MEM_REPAIR_BYPASS</code>	13	Skips the assembly memory-repair result check and records the bypass status.
<code>TEST_EN</code>	14	Cached by the strap driver; no later ROM behavior depends on it.
<code>BOOT_STALL</code>	17	Reported in the POST word. CPU stalling occurs before ROM execution in hardware.
<code>BOOT_I2C</code>	18	Reported in the POST word; it does not alter the interface map.
<code>BOOT_RECOVERY</code>	19	Reported as a warning and in the POST word; it does not select a different ROM flow.
<code>BL0_PLLCLK</code>	20	Controls whether the ROM calls the optional <code>smc_pll_init</code> hook.
<code>STATUS_RPT_DISABLE</code>	21	Causes <code>GetStatus</code> to return <code>Unsupported_StatusID</code> .
<code>SPI_USE_FUSED_CONFIG</code>	22	Cached by the strap driver; no later ROM behavior depends on it.
<code>PRIMARY_CHIPLET</code>	25	Reported in status and POST state; it does not select a different boot algorithm.
<code>SRAM_AUTO_ZERO_DISABLE</code>	26	Requests explicit SRAM clearing with the AXI zeroer.
<code>MEM_BIST_BYPASS</code>	54	Skips the assembly MBIST result check and records the bypass status.
<code>ROTATE_UPDATE</code>	58	Cached by the strap driver; no later ROM behavior depends on it.

The four chip-ID strap bits are read from `STRAPS_LO` bits 11, 12, 15, and 23. They contribute to the I2C/I3C identifier fallback described in [Address selection](#).

The eFuse driver reads I2C/I3C identifier slots 0, 1, 3, 6, and 7 and the `RESERVED_1` transport timeout. Early assembly additionally reads:

- `RESERVED_0[6]`, which requests explicit SRAM clearing; and
- `RESERVED_2[0]`, which permits boot to continue after an MBIST or memory-repair failure while preserving the failure code in scratch register 15.

2.4.2. SRAM initialization

Hardware auto-zero is assumed unless either `SRAM_AUTO_ZERO_DISABLE` or eFuse `RESERVED_0[6]` is set. When explicit clearing is requested, `__metal_sram_init_early`:

1. disables machine interrupts while configuring the operation;
2. writes SRAM base `0xC0060000` to zeroer destination register `0xC0038200`;
3. writes size `0x00100000` to zeroer size register `0xC0038208`;

4. writes zero to control/status register `0xC0038210` to start the operation;
5. waits while busy bit 32 is set; and
6. restores the previous `mstatus`.

The ROM does not perform a DMA-reset step.

2.4.3. MBIST and memory repair

The production ROM FW image links strong overrides for both early checks, so the weak no-op definitions in `mbist_check.S` are not used. Both checks read the SMU DFX status register at `0xC000B800`. They run after SRAM initialization and before stack setup.

The MBIST check uses:

- bit 4, `mbist_done`;
- bit 8, `mbist_pass`; and
- bit 12, `mbist_abort`.

If MBIST was not run, the ROM continues. If it passed, the ROM continues. An abort records `0xDEADC0DE` and halts. A failure records `0xDEADBEEF` and halts unless eFuse `RESERVED_2[0]` permits continuation. Scratch register 15 contains `0x12345678` while the MBIST result check is in progress. `MEM_BIST_BYPASS` records `0x12340002` and skips the check.

The memory-repair check uses:

- bit 0, `mem_repair_done`; and
- bit 1, `mem_repair_success`.

If repair was not run or succeeded, the final status is normally `0x600DCAFE`. A failure records `0xBADC0FFE` and halts unless eFuse `RESERVED_2[0]` permits continuation. `MEM_REPAIR_BYPASS` records `0x12340001` and skips the check. On its normal completion path, the repair check preserves a preceding MBIST failure, timeout, or bypass status. If both bypass straps are asserted, the memory-repair check runs second and replaces the MBIST-bypass value with `0x12340001`.

2.4.4. Clock hook

The C flow calls `smc_pll_init` only when `BL0_PLLCLK` is asserted. The open implementation is weak and its internal configuration function performs no register writes, so the default behavior is a successful no-op. An adopter ROM FW build may provide a strong implementation through the build's additional driver-source hook. If that implementation returns any value other than `SMC_PLL_SUCCESS`, the ROM records an interface error and remains in `wfi`.

2.5. Memory and security

2.5.1. Image and SRAM layout

Table 20. ROM-visible memory regions

Range	Size	Use
<code>0xC0040000--0xC005FFFF</code>	128 KiB	Executable ROM image. Unused bytes are filled with <code>EBREAK</code> .
<code>0xC0060000--0xC015FFFF</code>	1 MiB	SMC SRAM used for initialized data, TLS, BSS, stacks, heap, status buffers, and externally loaded data.

The linker places `.data`, TLS, `.bss`, stacks, and heap upward from the SRAM base. The global OCCP request buffer and the large read-response workspace are ordinary `.bss` objects; they do not occupy a separately fixed high-address window.

Two `smc_ring_buffer_t` objects are addressed downward from the exclusive SRAM end, `0xC0160000`. The SEP ring buffer is the last object in SRAM, and the SMC ring buffer immediately precedes it:

```
SMC_STATUS_BUFFER_ADDR = 0xC0160000 - 2 * sizeof(smc_ring_buffer_t)
SEP_STATUS_BUFFER_ADDR = 0xC0160000 -      sizeof(smc_ring_buffer_t)
```

Each structure contains three 32-bit metadata fields and 512 32-bit entries. The code also defines `0xC0066400` through `0xC015AFFF` as the SEP-safe region published in scratch registers 13 and 14.

2.5.2. OCCP memory-access policy

All read, write, execute, and authenticate handlers reject a zero-length policy check, arithmetic wraparound, or an access overlapping `0xC0060000` through `0xC00663FF`. That fixed 25 KiB range protects ROM-owned writable state regardless of lifecycle. Read, write, execute, and authenticate addresses must also be 32-bit aligned; execute and authenticate addresses must be nonzero.

In a secure lifecycle state, the complete access must lie in `0xC0066400` through `0xC015FFFF`. In another valid lifecycle state, the policy function allows addresses outside SRAM as long as they do not overlap the protected ROM-owned SRAM range.

`ExecuteImage` adds a lifecycle check and is rejected in secure states. `AuthenticateImage` uses the common policy check, then converts the address to an SRAM-relative offset for scratch register 8. Both boot-command handlers validate an eight-byte range beginning at the supplied address, so the complete range must satisfy the policy. The offset helper accepts only `0xC0066400` through `0xC015FFFF`; if a non-secure request supplies another otherwise permitted address, it writes the invalid marker `0xFFFFFFFF`. The current command handler still sets `MANIFEST_READY` after that marker is written.

The access policy does not reserve the status ring buffers from OCCP memory commands.

2.5.3. Lifecycle handling

The lifecycle register is `CHIP_CONFIG.LC_STATE` at `0xC000290C`. The ROM reads all eight bits as a differential encoding:

```
raw[7:4] == ~raw[3:0]
```

An encoding mismatch is invalid. For a valid encoding, the low nibble selects the policy:

Table 21. Lifecycle values used by the ROM

Low nibble	Classification	ROM policy
<code>0x0</code>	Test/development	Non-secure
<code>0x1</code>	Production	Secure
<code>0x2--0x3</code>	RMA SoP	Non-secure
<code>0x4--0x7</code>	RMA chiplet	Non-secure
<code>0x8</code>	Production end	Secure

Low nibble	Classification	ROM policy
0x9--0xF	Invalid	Record an error and remain in <code>wfi</code>

This classification affects OCCP address bounds and `ExecuteImage`. It does not select different interface maps or role-specific boot flows.

2.6. Communication interfaces

The ROM enables the same target-interface set for every strap role:

- I3C controllers 0, 1, and 3; and
- I2C controllers 0 and 1.

I3C2 remains disabled in the software interface map. The `BOOT_I2C`, `BOOT_RECOVERY`, and `PRIMARY_CHIPLET` straps are observable state but do not change this map.

2.6.1. Address selection

I3C controllers 0, 1, and 3 use eFuse identifier slots 0, 1, and 3. The eFuse driver reads each slot as two 32-bit words but caches it in a 32-bit field, so the current ROM passes only the low 32 bits onward. I2C controllers 0 and 1 take the low seven bits of slots 6 and 7. An I2C value outside `0x08` through `0x77` is replaced with `0x55`.

If a selected eFuse slot is zero, the ROM forms a fallback identifier by ORing the four-bit chip ID from straps with `CHIP_CONFIG.CHIP_ID << 5`. The same fallback function is used for I2C and I3C slots.

I3C channels are started in subordinate mode. I2C channels are initialized in target mode.

2.6.2. Driver selection

The default I3C target symbols are weak. Their initialization functions succeed so the ROM remains linkable without an active I3C implementation, while their data methods report a hardware error and their receive-availability method reports no data. An adopter ROM FW build can replace those symbols with a strong driver.

Setting `I3C_CORE=swap` at build time enables the open MIPI HCI/I3CCSR implementation in `i3c_hci_driver.c`. That implementation configures the target transaction interface, static address, dynamic-address-assignment acceptance, queue thresholds, and open-drain timing.

The HCI path reaches the pads through the LSIO selection and therefore leaves the GPIO hardware-function override clear. Every non-HCI build enables the GPIO hardware-function override for the I3C pad set before driver initialization, whether it uses the weak default or a strong platform driver. The ROM does not program GPIO drive strength from eFuses.

2.6.3. I3C response flow control

The HCI target implementation advertises a response only after response data is available. For responses that fit in the transmit queue, it writes all data and then writes the transmit descriptor. For larger responses, it fills the queue, writes the descriptor when the queue first becomes full, and continues filling as hardware drains data. The target hardware does not accept a private read until its response-start threshold is met.

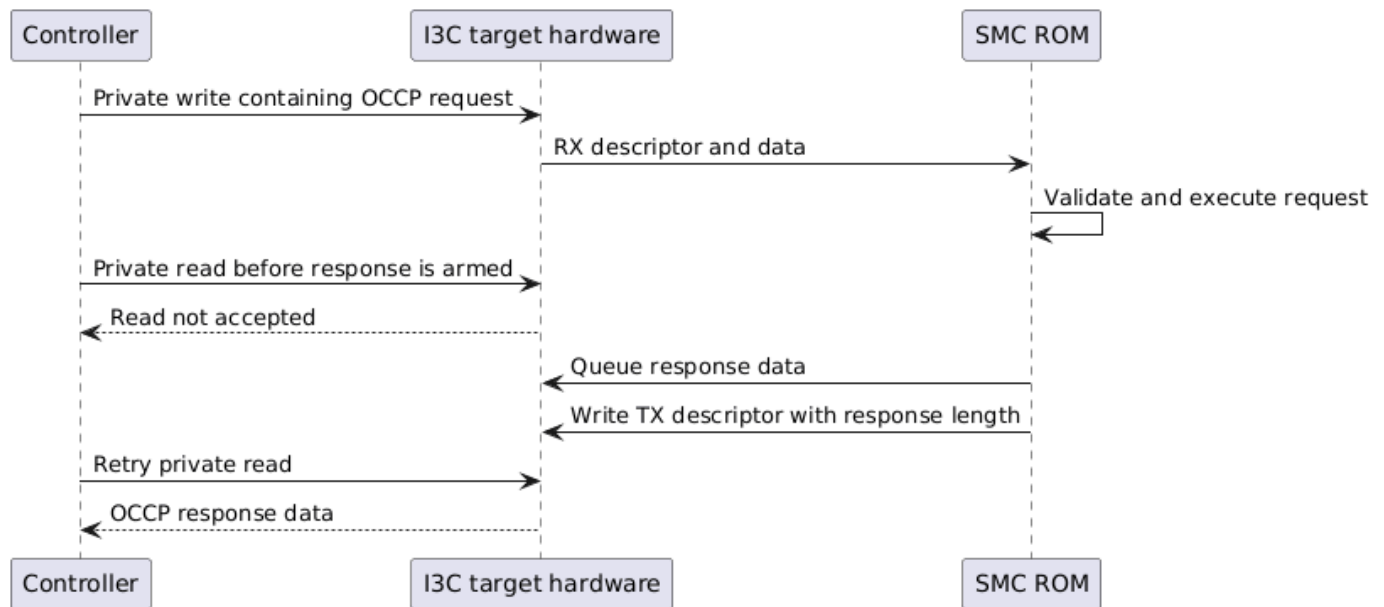


Figure 6. HCI target response flow

The HCI receive implementation separately tracks each inbound frame's descriptor length and unconsumed bytes. That accounting persists across the OCCP layer's header and body reads and supports exact incomplete/oversized-frame reporting. See [OCCP protocol](#) for the packet binding and controller-visible errors.

2.6.4. GPIO side effects

During OCCP initialization, the ROM enables the hardware-function override for CAT_THERM on GPIO 52. After transport initialization, GPIO 58 is configured as an output and driven high when all requested channel initializations succeeded or low when any failed. No status-sequence protocol is implemented on that GPIO.

2.7. Status and coordination

2.7.1. Scratch registers

The CPU-control block begins at `0xC0039000`. Scratch register 0 is at `0xC0039080`; consecutive scratch registers are eight bytes apart.

Table 22. Scratch registers used by the ROM

Index	Address	Use
0	<code>0xC0039080</code>	Simulation pass/fail and generic early-trap indication.
1	<code>0xC0039088</code>	Structured POST word.
2	<code>0xC0039090</code>	Virtual console output in debug-enabled builds.
8	<code>0xC00390C0</code>	Manifest offset from SRAM base, or <code>0xFFFFFFFF</code> when invalid.
9	<code>0xC00390C8</code>	SMC-to-SEP coordination bits.
10	<code>0xC00390D0</code>	Available through the MBIST-failure helper; the assembly checks use register 15.
11	<code>0xC00390D8</code>	SEP status-buffer offset from SRAM base.
13	<code>0xC00390E8</code>	SEP-safe SRAM start offset from SRAM base.
14	<code>0xC00390F0</code>	SEP-safe SRAM size in bytes.

Index	Address	Use
15	0xC00390F8	MBIST, memory-repair, or ROM-padding-trap status.

Assembly clears registers 8, 9, 11, 13, and 14 after installing the early trap vector. It deliberately preserves register 15.

Table 23. Scratch register 9

Bit	Name	Set when
0	SRAM initialized	Status buffers and the safe-region description have been initialized.
1	Manifest ready	<code>AuthenticateImage</code> has published scratch register 8.
2	Status buffer ready	Scratch register 11 contains the SEP ring-buffer offset.
3	SRAM protected	Defined by the interface but not set by the current ROM.

The published SRAM addresses are offsets from 0xC0060000. The normal values in registers 13 and 14 describe 0xC0066400 through 0xC015AFFF.

2.7.2. Ring buffers

The ROM initializes separate SMC and SEP ring buffers at the end of SRAM. Each holds 512 32-bit entries plus `head`, `tail`, and `num_entries` fields. A write to a full buffer advances the tail and overwrites the oldest entry. A read returns the oldest unread entry and advances the tail.

Status messages use this format:

Table 24. Status message

Bits	Field	Encoding
31:24	Message type	0x01 status, 0x08 warning, 0x0F error
23:16	Firmware ID	0x01 SEP BLO, 0x02 SEP BL1, 0x03 SMC BLO, 0x04 SMC BL1
15:0	Value	Boot or command-specific status value

2.7.3. Ring-buffer status values

The SMC ROM FW writes the following boot values to the SMC ring buffer. Primary, secondary, and recovery values report sampled straps; they do not select different ROM FW algorithms.

Table 25. Boot status values

Value	Name	Type	Current emission
0x001	SMC_STATUS_ROM_STARTED	Status	<code>smc_status_init</code> emits this value as its initialization marker, and <code>main</code> emits it again explicitly after publishing the initial coordination state.
0x010	SMC_STATUS_BOOT_START	Status	After strap, eFuse, lifecycle, and optional PLL processing.
0x020	SMC_STATUS_RECOVERY_MODE	Warning	When the recovery strap is asserted.
0x021	SMC_STATUS_PRIMARY_MODE	Status	When the primary-chiplet strap is asserted.
0x022	SMC_STATUS_SECONDARY_MODE	Status	When the primary-chiplet strap is deasserted.
0x025	SMC_STATUS_INVALID_SEC_MODE	Error	Before halting on an invalid lifecycle encoding or value.
0x030	SMC_STATUS_OCCP_INIT_FAILED	Error	Before halting when any configured OCCP transport fails initialization.
0x031	SMC_STATUS_OCCP_READY	Status	After all configured OCCP transports initialize successfully.
0x040	SMC_STATUS_COORDINATION_ACTIVE	Status	After the SEP-safe SRAM range and SRAM-ready indication are published.

Value	Name	Type	Current emission
0x050	SMC_STATUS_BOOT_COMPLETE	Status	When the ROM FW enters the OCCP command loop, and again before an accepted <code>ExecuteImage</code> transfers control.
0x0FF	SMC_STATUS_UNEXPECTED_EXIT	Error	Only if the normally non-returning OCCP command loop returns.

OCCP processing uses the following value families. Values described as “base OR data” use `SMC_OCCP_ERROR_WITH_DATA` to OR the base with the low eight bits of the associated command or error value. Access-denied details use `SMC_OCCP_ERROR_WITH_NIBBLE`, which computes `base | ((detail & 0xF) << 8)`.

Table 26. OCCP ring-buffer status values

Base value	Name	Current use
0x100	SMC_OCCP_ERROR_CMD_READ	Packet-header transport read failure.
0x101 OR data	SMC_OCCP_ERROR_CMD_UNKNOWN	Unknown application or message ID; data identifies the rejected value.
0x110 OR data	SMC_OCCP_ERROR_CMD_FAILED	Header validation, command execution, or interface-unlatch detail. Unlatch events use warning type; command failures use error type.
0x120	SMC_OCCP_ERROR_READ_OVERFLOW	Invalid or excessive <code>ReadData</code> size, or an invalid <code>ReadData</code> , <code>ExecuteImage</code> , or <code>AuthenticateImage</code> body length.
0x121, optionally with shifted detail	SMC_OCCP_ERROR_READ_ACCESS_DENIED	Misaligned or policy-rejected read address.
0x130	SMC_OCCP_ERROR_WRITE_OVERFLOW	Invalid or excessive write-command length.
0x131, optionally with shifted detail	SMC_OCCP_ERROR_WRITE_ACCESS_DENIED	Misaligned or policy-rejected write address.
0x140	SMC_OCCP_ERROR_VALIDATE_SECURITY	Defined for authenticate security failures but not emitted by the current implementation.
0x141	SMC_OCCP_ERROR_VALIDATE_ADDRESS_FAILED	Authenticate transport, address, or policy failure.
0x200 OR address data	SMC_OCCP_STATUS_JUMP_EXECUTED	Accepted <code>ExecuteImage</code> ; data is address bits 23:16.
0x201	SMC_OCCP_ERROR_JUMP_SECURITY	<code>ExecuteImage</code> rejected in a secure lifecycle state.
0x202	SMC_OCCP_ERROR_JUMP_READ_FAILED	<code>ExecuteImage</code> body, transport, address, or policy failure.

Handler-specific error values do not replace the command loop’s common error report. After most handlers return an error, the loop emits a second `SMC_OCCP_ERROR_CMD_FAILED` value with the returned internal error code.

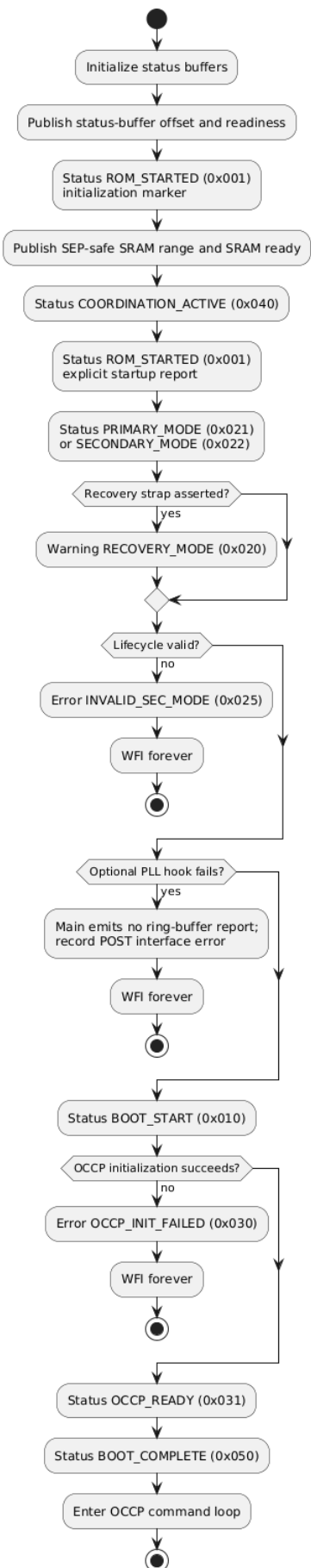


Figure 7. SMC ROM FW status-reporting flow

The **STATUS_RPT_DISABLE** strap does not disable ring-buffer initialization or writes. It only rejects OCCP **GetStatus**.

2.7.4. Software OCCP status word

The ROM maintains an in-memory 32-bit status word:

Table 27. OCCP status word

Bits	Field	Current update behavior
3:0	Boot status	Initialized to zero. The current main flow does not advance this field.
7:4	Interface status	Set to ready after successful transport initialization or error after failure.
15:8	Command count	Incremented when a successful packet-header transport read reaches the common end of the command iteration, including header-validation and dispatch errors; wraps from 255 to 0. Header-read transport failures and successful terminal handoffs are not counted.
23:16	Last internal error	Updated after command failures.
31:24	Reserved	Zero.

GetStatus IDs **0x0000** through **0x0003** expose individual fields in the status-message envelope described in [OCCP protocol](#).

2.7.5. POST word

Scratch register 1 carries a structured development-observability word.

Table 28. POST fields

Bits	Field	Values used by the ROM
31:28	Boot phase	0 initialization, 1 straps/eFuses, 2 SRAM setup, 3 interface setup, 4 OCCP ready, 5 OCCP processing, 6 boot complete, 7 error
27:24	OCCP state	0 idle, 1 command received, 2 processing, 3 response ready, 4 complete, 5 error
23:20	Latched interface	0 none, 1 I3C0, 2 I3C1, 4 I3C3, 5 I2C0, 6 I2C1
19:16	Error	0 none, 1 SRAM, 2 interface, 3 command, 4 access, 5 invalid security mode
15:12	Security decisions	Bits indicate secure, production, and RMA SoP classifications.
11:8	Boot observations	Bits indicate primary strap, recovery strap, I2C boot strap, and SRAM-auto-zero-disable strap.
7:4	Strap observations	Bits indicate boot stall, status-report disable, and memory-repair bypass.
3:0	Coordination	Bits indicate SRAM ready, manifest ready, status buffer ready, and coordination active.

POST state reports decisions made by the current ROM; it is not a separate boot-control interface. During startup, the implementation writes boot-phase values in the sequence 0, 2, 1, 2, 3, 4, and 5. The first phase-2 write follows publication of the SRAM coordination state; phase 1 then records strap and eFuse processing, and the second phase-2 write records the explicit C-flow SRAM phase. The **SMC_STATUS_BOOT_COMPLETE** ring-buffer value emitted on entry to the OCCP loop does not set POST boot phase 6. The ROM FW sets phase 6 only immediately before an accepted **ExecuteImage** transfers control.

2.8. OCCP command processing

This section describes the ROM command-loop implementation. Packet formats, command semantics, CRCs, transport binding, and controller-visible errors are defined in [OCCP protocol](#).

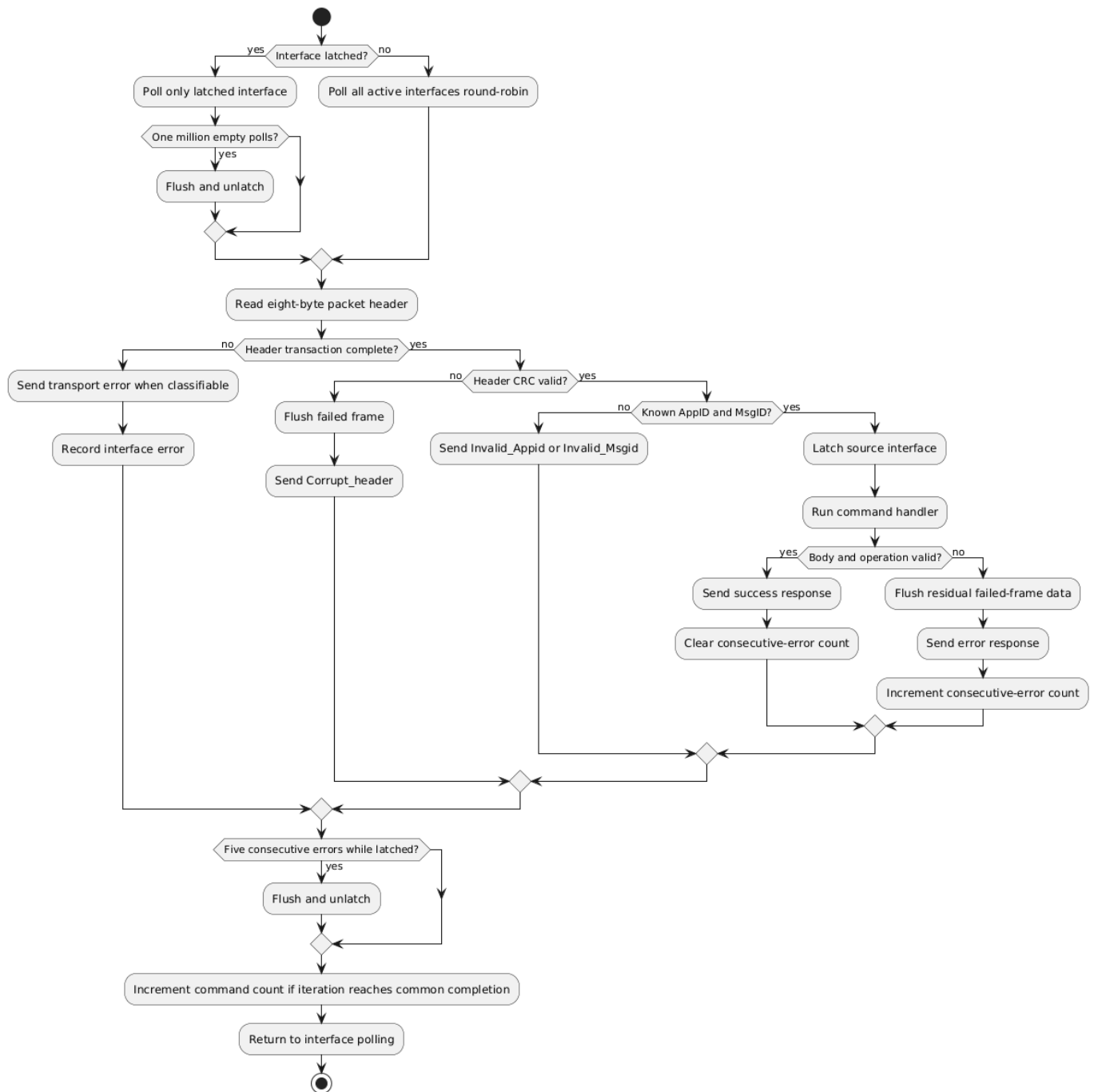


Figure 8. OCCP command-processing flow

2.8.1. Polling and latching

Before latching, the loop polls each active interface in array order: I3C0, I3C1, I3C3, I2C0, and I2C1. After an unlatch, round-robin polling begins with the interface after the one just released.

The ROM latches when it receives a header with a valid CRC and a numerically recognized AppID/MsgID. Latching occurs before command-body length, CRC, address, and execution checks, so a recognized command that later fails validation still selects the interface. While latched, traffic on other interfaces is ignored.

One million polling iterations without receive data causes the current interface to be flushed and unlatched. While an interface is latched, five consecutive header-read or command-execution errors also cause an unlatch. Errors received

before any interface is latched do not increment that threshold counter. A successfully completed handler clears the consecutive-error count. `smc_occp_force_unlatch` exposes the same unlatch operation to linked ROM FW code.

2.8.2. Receive and dispatch

The loop reads the eight-byte packet header first. For a valid header, it dispatches directly on AppID and MsgID. Each handler then checks its required body length, reads the body plus optional CRC, validates body integrity, performs command-specific policy checks, and sends a response.

The POST OCCP-state field progresses through command received, processing, response ready, and complete or error. The software OCCP command count increments when a packet-header transport read succeeds and processing reaches the common end of the command iteration. This includes header-validation failures, invalid application or message IDs, and handlers that return success or error. Successful `ExecuteImage` and `AuthenticateImage` handlers do not return and therefore do not increment the count. Initial header-read failures return to polling before that increment.

2.8.3. Failed-frame cleanup

Failed-frame cleanup occurs in these contexts:

- header-validation errors flush residual body data before transmitting the error response;
- the `GetVersion` handlers flush unexpected residual bytes and report `Oversize_msg` when any are found;
- a command failure flushes residual data unless its error-response path already performed the flush; and
- an unlatch flushes the selected interface before returning to all-interface polling.

The I2C flush is bounded by the configured transport timeout and a maximum of 2,079 bytes. The HCI I3C flush is frame-aware: it discards bytes owed by the abandoned frame and complete stale frames already represented by descriptors, then stops when that frame ledger is clean. It does not use an idle-time window to consume newly arriving commands.

2.8.4. Terminal commands

`ExecuteImage` sends its success response before transferring control and therefore never returns to this loop. `AuthenticateImage` sends its success response before publishing the manifest coordination state and parking in `wfi`. No command-processing or status-only service loop remains active after `AuthenticateImage`.

Chapter 3. SEP Programming Guide

This Chapter details the bring-up process and run-time operation of the SEP for the purposes of firmware development, with reference to the Production SEP ROM.

The subsequent Chapter serves as a detailed reference to the Production SEP ROM implementation.

3.1. Bring-Up

3.1.1. Address Map

Table 29. SEP Address Map.

Base Address	End Address	Size	Unit	Description
0x0000_0000	0x0FFF_FFFF	256 MB	External	External to chiplet, but not in SMU (SMC, DTP, etc).
0x1000_0000	0x1FFF_FFFF	256 MB	SEP Local	SEP Local resources (DMA, Crypto, etc)
0x2000_0000	0x3FFF_FFFF	512 MB	SEP External	Reserved for adopter extension IP
0x4000_0000	0x7FFF_FFFF	1 GB	SMC	SMC Resources
0x8000_0000	0xBFFF_FFFF	1 GB	SMU	SMU Resources (DTP, etc)
0xC000_0000	0xCFFF_FFFF	256 MB	SEP CPU Resources	EL2 Veer Resources (DCCM, ICCM, PIC)
0xD000_0000	0xFFFF_FFFF	768 MB	SEP Local Alias	Corresponds to 0x1000_0000 - 0x3FFF_FFFF

The following table lists the Memory regions accessible to the SEP. Scratch RAM and BL0 ROM are part of the SEP Local address range. ICCM and DCCM are part of the SEP CPU Resources address range.

Table 30. SEP Memory Regions.

Base Address	End Address	Size	Unit	Description
0x1000_0000	0x1000_FFFF	64 kB	SRAM	Scratch SRAM
0x1001_0000	0x1003_FFFF	192 kB	RSV	Reserved Scratch SRAM expansion
0x1004_0000	0x1004_FFFF	64 kB	ROM	BL0 immutable instruction memory
0xC000_0000	0xC003_FFFF	256 kB	ICCM	Tightly Coupled Instruction SRAM
0xC004_0000	0xC005_FFFF	128 kB	DCCM	Tightly Coupled Data SRAM

3.1.2. Early Initialization

The SEP CPU cold-reset vector is `0x1004_0000`, corresponding to the start of BL0 ROM.

The production BL0 ROM linker script — `hw/sys/sep/bootrom/prod/link/rom.ld` places the executable `.text` and read-only data `.rodata` sections in ROM. The `.data` section and other runtime data are included in the ROM image, but have runtime addresses within DCCM at `0xC004_0000`. This data is copied by BL0 ROM into DCCM by early assembly code in `vector.S`. The stack and the `.bss` section reside only in DCCM. Unlike the SMC SRAM, ICCM and DCCM do not require

initialization.

3.1.3. NMI Vector Programming

The Security Processor (SEP) runs a VeeR-EL2 core whose Non-Maskable Interrupt is driven by the watchdog timer bark (`sep.sv` wires `nmi_int` to `intr_wdog_timer_bark`). When the NMI asserts, the core jumps to the address held in the `SEP_NMI_VEC` register of the `sep_cpu_ctrl` block (offset `0x180`).

Software reads and writes `SEP_NMI_VEC` as a byte address. The register's `nmi_vec` field occupies bits 31:1, so bit 0 is reserved and always reads back as zero; the hardware imposes no stricter alignment than that. The reference DV firmware nevertheless aligns its NMI trampoline to 256 bytes, which the linker script enforces.

WARNING

The `SEP_NMI_VEC` reset value (`0xC000_0100`) is a placeholder. It does not point at a valid handler, and no part of the boot flow makes it one. Firmware must complete the following sequence before enabling the watchdog timer or any other NMI source:

1. Place an NMI handler in ICCM.
2. Write the handler's address to `SEP_NMI_VEC`.
3. Set `SEP_NMI_VEC_LOCK` (offset `0x188`) to freeze the vector until the next reset.
4. Only then enable the WDT.

If a bark arrives while the reset value is still in place, the core jumps to `0xC000_0100` and executes whatever occupies that location: arbitrary instructions if the loaded image covers it, or an uncorrectable ICCM ECC error if it does not. The ECC case re-raises NMI, so the core makes no forward progress.

Because `SEP_NMI_VEC_LOCK` is write-one-to-set and sticky until reset, the vector cannot be repointed once locked. Firmware that needs to install a different handler after lock must do so through its own indirection — the reference firmware keeps a runtime-writable handler pointer that the trampoline dispatches through, so the locked vector always targets the trampoline.

If the watchdog bite threshold is also enabled, an unserviced bark still escalates to the watchdog reset path and resets the SEP. A mis-programmed vector therefore degrades to a watchdog reset rather than a silent hang, but it does so only after the bite interval and without running any handler.

3.2. OCAH TRNG Bring-Up

SEP has three entropy streams: Key Manager, the AES/KMAC/OTBN group, and the fabric entropy pool. `SEP_CPU_CTRL.EXT_TRNG_SRC_SEL` selects the source of each stream independently:

- bit 0 selects the Key Manager stream;
- bit 1 selects the AES/KMAC/OTBN stream; and
- bit 2 selects the fabric entropy-pool stream.

A bit value of 0 selects the internal ESRC → CSRNG → EDN chain. A bit value of 1 selects the corresponding external TRNG stream and bypasses the internal chain for that stream.

WARNING

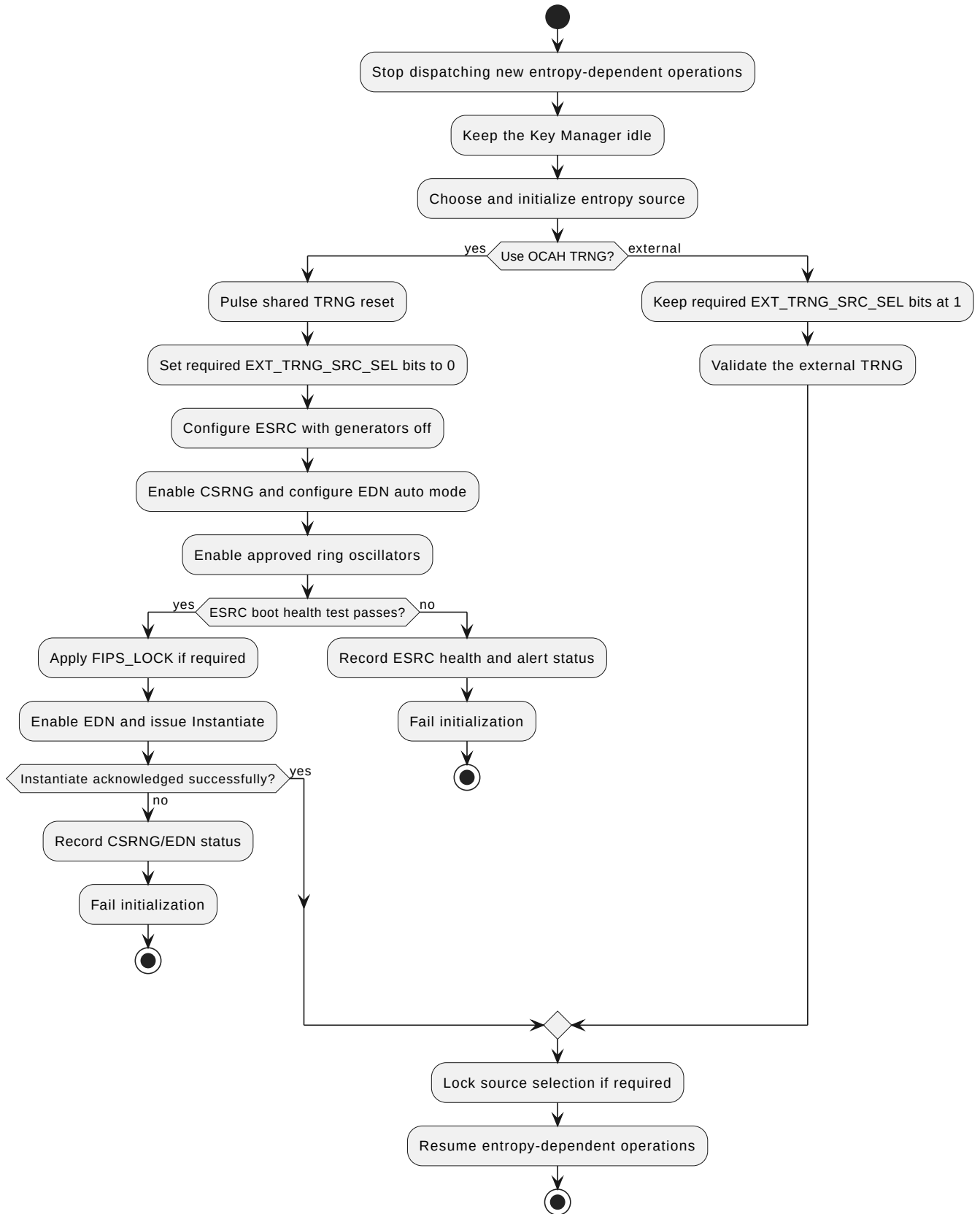
The OCAH internal entropy source has not been certified. Consequently, `EXT_TRNG_SRC_SEL` resets to `0x7`, selecting the external TRNG for all three streams. Production firmware should retain that

default and use an appropriately qualified external source unless the product security authority explicitly approves the OCAH internal path.

The CSRNG and EDN blocks are derived from OpenTitan. OpenTitan describes CSRNG as compatible with NIST SP 800-90A and AIS31 and implements the approved CTR_DRBG construction, but this is not a certification of the OCAH integration. NIST CAVP validation applies to a specific implementation and operational environment. No CAVP or FIPS 140 validation is claimed for the OCAH entropy chain. See the [OpenTitan CSRNG specification](#) and [NIST CAVP](#).

`EXT_TRNG_SRC_SEL_LOCK.LOCK` makes the selection read-only until SEP reset. Production firmware should lock the selection after it has verified that the chosen source is operational. An external-TRNG integration must initialize and test that device through its platform-specific control interface; the OCAH sequence below applies only to streams firmware explicitly changes to 0.

3.2.1. Bring-up sequence at a glance



3.2.2. Internal entropy-chain initialization

Use generated register definitions and named fields rather than copying complete register literals. Several control fields use MuBi4 encodings, for which true is `0x6` and false is `0x9`.

1. Prevent new entropy-dependent operations during initialization. On first boot, initialize entropy before accepting crypto work; the Key Manager is held in software reset, which keeps it idle. A runtime recovery must wait until KM is idle, or hold it in software reset, before pulsing the TRNG reset. AES, KMAC, and OTBN may stall until fresh entropy is available, or, after each is idle, firmware may hold them in software reset to abort their operations.
2. Reset the internal entropy chain by clearing and then setting `SEP_RESET_CTRL.SW_RESET_N.trng_sw_rst_n` with read-modify-write operations that preserve the other reset bits. This reset covers ESRC, CSRNG, EDN, their control-plane adapters, buffered internal entropy, and the fabric entropy pool. `ESRC.CTRL[0]` is reserved and is not a reset control.
3. Select the internal source by clearing each required bit in `EXT_TRNG_SRC_SEL`. This mux programming is a required part of OCAH TRNG initialization; the reset value deliberately continues to select the external source.
4. Disable `ESRC.RING_OSC_ENABLE.ENABLE` while programming the entropy-source policy. Program the approved sampling, health-test, FIFO, conditioning, and module-enable fields. In particular, the intended standards-aligned profile requires `CTRL.MODULE_ENABLE=1`, the approved health-test thresholds and window, entropy-compressor bypass disabled, and SHA-256 whitening enabled.
5. Enable the CSRNG with `CSRNG.CTRL.ENABLE`. Leave `SW_APP_ENABLE`, `READ_INT_STATE`, and `FIPS_FORCE_ENABLE` false unless the security policy explicitly requires those diagnostic interfaces.
6. Configure EDN continuous-operation mode while EDN is disabled:
 - program `RESEED_CMD` and `GENERATE_CMD` with commands that request entropy from CSRNG;
 - program `MAX_NUM_REQS_BETWEEN_RESEEDS`; and
 - set `BOOT_REQ_MODE` false and `AUTO_REQ_MODE` true.
7. Enable the approved ring oscillators and sample clocks in `ESRC.RING_OSC_ENABLE`.
8. Poll `ESRC.MAIN_SM_STATUS.BOOT_PHASE_DONE` with a bounded timeout. While waiting, fail initialization if `MAIN_SM_STATUS.ALERT`, `MAIN_SM_STATUS.ERR`, or `INTR_STATUS.PERSISTENT_FAILURE` asserts. `BOOT_PHASE_DONE` proves that the startup health-test gate opened; it does not by itself prove that CSRNG accepted a complete seed.
9. If required by the security policy, set `ESRC.FIPS_LOCK.LOCK` after startup passes and before exposing entropy to consumers. The lock freezes the certified ESRC configuration until the internal entropy chain is reset. `DEBUG_CTRL` intentionally remains outside this lock; the platform must separately restrict any externally routed debug-observation signal.
10. Enable EDN with `EDN_ENABLE=true`, `AUTO_REQ_MODE=true`, `BOOT_REQ_MODE=false`, and `CMD_FIFO_RST=false`.
11. Wait for both `EDN.SW_CMD_STS.CMD_RDY` and `CMD_REG_RDY`, then write an Instantiate command with no additional data to `EDN.SW_CMD_REQ`. Poll `CMD_ACK` or `INTR_STATE.EDN_CMD_REQ_DONE` with a bounded timeout and require `CMD_STS` to report success.
12. Lock `EXT_TRNG_SRC_SEL` if required, then release or initialize the entropy consumers.

The EDN Instantiate acknowledgement is the software-visible readiness event: CSRNG cannot acknowledge that command until it has accepted a complete ESRC seed. Firmware therefore does not need a delay or a hierarchical `drbg_seed_valid` probe. `BOOT_PHASE_DONE` remains useful for separating an ESRC startup-health failure from a downstream CSRNG or EDN failure.

WARNING

Do not set EDN boot-request and auto-request mode true at the same time. Boot-request mode takes precedence and remains in its completed state until firmware clears it; merely setting both modes does not transition EDN to continuous auto-request operation.

`SEP_CPU_CTRL.CLOCK_GATE_CTRL` currently has no entropy clock-gate field. Firmware must not write an undocumented bit 10 as part of entropy bring-up.

3.2.3. Failure handling and recovery

Treat entropy initialization as a fallible operation. Every poll must have a platform-defined timeout, and firmware should preserve enough status to distinguish:

- ESRC startup-health failure (`MAIN_SM_STATUS`, `HEALTH_TEST_STATUS`, and `INTR_STATUS`);
- CSRNG command, exception, recoverable-alert, or fatal-alert failure; and
- EDN command, recoverable-alert, or fatal-alert failure.

The boot policy must define whether any such failure stops secure boot or enters a platform recovery flow. Continuing into an entropy-dependent operation is not recovery: the consumer can wait indefinitely for entropy.

For a runtime restart of the internal chain:

1. Stop dispatching new entropy-dependent operations. The Key Manager must be idle before the TRNG reset is asserted: wait until it is not consuming entropy, or hold it in software reset. If policy requires aborting AES, KMAC, or OTBN in-flight work, wait for each to become idle, save `SW_RESET_N`, and hold those consumers in reset.
2. Clear `trng_sw_rst_n`; hardware drains accepted control transactions, isolates the three internal TRNG register ports, and clears buffered internal and post-mux entropy before asserting reset.
3. Set `trng_sw_rst_n`.
4. Repeat the complete internal initialization sequence, including the bounded ESRC startup poll and acknowledged EDN Instantiate command.
5. Restore any consumer-reset state saved in the first step after initialization succeeds.

The native EDN boundary uses request/acknowledge, not valid/ready. During TRNG reset the post-mux adapters suppress acknowledgements and discard buffered entropy. A request already asserted by AES, KMAC, or OTBN remains asserted and is serviced with fresh entropy after the adapter and internal chain restart. Resetting those consumers is an operation-abort policy, not a prerequisite for the coordinated TRNG reset.

The Key Manager AXI-Stream has no post-mux adapter, so TRNG reset does not scrub it. A reset while the DRBG sampler is requesting or assembling a word can mix leftover bytes into the next DATA read or raise a stream error. The Key Manager must be idle before the SEP host asserts the TRNG reset.

Do not try to recover only ESRC with `CTRL[0]`. A shared TRNG reset invalidates the entire internal chain.

Chapter 4. SEP Production ROM

WARNING

The production ROM firmware and this documentation are works in progress and are not ready for production use.

4.1. Overview

The SEP production ROM is the immutable BL0 Root-of-Trust firmware. It starts at the SEP CPU reset vector, establishes a usable machine-mode environment, selects a manifest source, validates that manifest and its payload, and transfers control to SEP BL1 in ICCM.

The firmware has three cold-boot modes. A primary chiplet in normal mode reads the primary or backup manifest from SPI flash. A primary chiplet in recovery mode and every secondary chiplet wait for the SMC to stage a manifest in SMC SRAM. A warm reset takes a separate assembly path and may return directly to a BL1 handler in ICCM.

The implementation is divided into:

- reset, warm-dispatch, memory-health, PMP, and C-runtime setup in `src/vector.S`;
- boot orchestration in `src/rom_main.c`;
- manifest and payload processing in `src/oca_boot.c`, which drives the OCA validator library vendored at `tools/tt-oca-manifest/`, with this ROM's platform callbacks (key authorization, revocation, anti-rollback, signature verification, payload decryption) in `src/oca_platform.c`;
- SPI, DMA, cryptographic, lifecycle, and SMC-interface support in `src/` and `include/`; and
- ROM and DCCM placement in `link/rom.ld`.

These implementation files are the source of truth for firmware ordering and policy. SEP RTL, generated register collateral, the SEP memory-map chapter, and `link/rom.ld` define hardware addresses and image placement. The production Makefile and `configs/` define build and packaging behavior. DV tests provide observable execution evidence but do not define the architecture.

Successful BL0 execution does not return and does not report a BL0 pass. `rom_handoff_bl1()` transfers control to BL1, which owns the eventual success verdict. A terminal BL0 error records its status and parks the hart in `wfi` until reset.

4.2. Boot flow

The assembly entry path handles every operation that must run before a stack or trustworthy DCCM exists. The C path starts only after the memory-health gate, DCCM scrub, and runtime initialization complete.

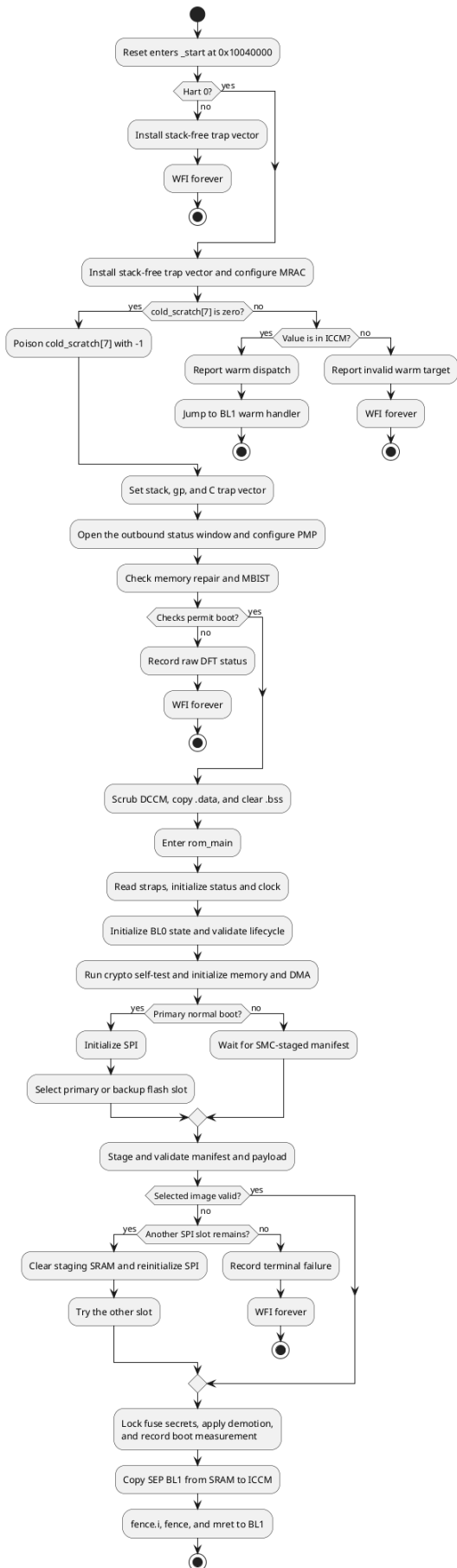


Figure 9. Cold and warm boot flow

4.2.1. Assembly entry

`_start` parks every hart except hart 0. Hart 0 installs a trap vector that does not use the stack, configures the CPU memory attributes, and examines `cold_scratch[7]` before touching DCCM. This ordering preserves the BL1 runtime state on a watchdog reset.

The cold path establishes the stack and global pointer, installs the C-capable trap vector, opens the outbound filter window used by later firmware, and programs PMP. It then evaluates the SMC memory-repair and MBIST status. A failed check stops the boot unless the corresponding strap or `SKIP_MEM_CHECK` fuse policy permits it. Only after this gate passes does the ROM scrub DCCM, copy initialized data from ROM, clear BSS, and call `rom_main()`.

4.2.2. C boot sequence

`rom_main()` performs these operations in order:

1. Verify that `.data` and `.bss` were initialized.
2. Probe the SMC scratch round-trip and report the ROM version and hash.
3. Read the captured straps and initialize status reporting and the selected clock.
4. Initialize `bl0_state`, check that its DCCM reserve does not overlap the stack, validate the lifecycle state, and read the chip ID.
5. Run the peripheral/bus reset sequence and the SHA-256 self-test, apply the configured SRAM and ICCM initialization, read `SBOOT_DIS`, install the stack canary, and initialize the DMA.
6. Select SPI, recovery, or secondary mode.
7. Probe the SMC coordination state and check the stack canary.
8. Load and validate a manifest and payload.
9. Decide demotion, lock sensitive fuse reads, verify the lock, and record the boot measurement.
10. Copy BL1 to ICCM and transfer control.

Errors detected by C code converge on `rom_err_fail()`. The function records the error in `bl0_state` when available, publishes machine-readable status, writes the failure verdict, and enters a non-returning `wfi` loop.

A trap takes a separate path. `trap_handler_c()` writes the failure verdict alone, so neither `bl0_state` nor the machine-readable status word records a trap-terminated boot. [Status and error reporting](#) describes both trap vectors.

4.3. Hardware initialization

4.3.1. Captured straps

The ROM reads the captured GPIO straps through the SMC supplementary window. `include/sep_smc_interface.h` defines the SEP-visible offsets and bit positions.

Strap	Register bit	ROM field	Effect
Primary chiplet	<code>STRAPS_LO[25]</code>	<code>primary_chiplet</code>	Selects the SPI path when recovery is clear; otherwise selects the secondary SMC-SRAM path.
Boot recovery	<code>STRAPS_LO[19]</code>	<code>boot_recovery</code>	Makes a primary chiplet wait for an SMC-staged manifest.
BL0 PLL clock	<code>STRAPS_LO[20]</code>	<code>bl0_pll_clk</code>	Selects the PLL initialization hook instead of the reference clock.

Strap	Register bit	ROM field	Effect
Disable status reporting	<code>STRAPS_LO[21]</code>	<code>status_report_disable</code>	Skips the SMC-SRAM status ring; scratch-register status remains available.
Bypass SRAM repair	<code>STRAPS_LO[13]</code>	assembly only	Skips the memory-repair result check because repair did not run.
Bypass MBIST result	<code>STRAPS_HI[22]</code>	assembly only	Ignores the MBIST result.
Rotate update	<code>STRAPS_HI[26]</code>	<code>rotate_update</code>	Swaps the order of the primary and backup SPI manifest slots.

4.3.2. Early CPU and fabric setup

The reset path programs the VeeR EL2 memory-region attributes before its first scratch-register access. It then configures the outbound filter window used by later firmware status traffic and programs PMP regions for the Boot ROM, DCCM, status mailbox, SEP SRAM, SEP peripherals, and ICCM.

The current `vector.S` labels PMP entry 6 as SMC MMIO, but programs `0xC000_0000` through `0xCFFF_FFFF`. SEP actually accesses SMC through `0x4000_0000`, so this entry does not protect the SEP-visible SMC aperture. The PMP table must be corrected before it can be treated as the final SMC access policy.

`PMP_ENABLE` controls whether these entries are programmed. `PMP_LOCK` controls their lock bits. The production Makefile defaults to PMP enabled and unlocked; an integrator must select the final lock policy for the immutable image.

4.3.3. Memory-health gate

Before writing DCCM, the assembly path reads the SMC DFX status. It checks that memory repair succeeded unless repair was bypassed, waits for MBIST completion, and checks that MBIST passed unless its result was bypassed. A failed, aborted, or timed-out check records the raw status in SMC scratch register 10.

The current code uses a fixed `MBIST_DONE_WAIT_ITERS` bound. This value is an implementation limit rather than a calibrated silicon duration. If `STATUS_RPT.SKIP_MEM_CHECK` is clear, failure is terminal; if it is set, the boot continues. That bit is not allocated by the current SEP eFuse RDL and must not be treated as a production fuse contract without a corresponding register specification.

4.3.4. Runtime memory initialization

The cold path scrubs `DCCM_SCRUB_BYTES`, then copies `.data` from its ROM load address and clears `.bss`. The default is the full 128 KiB DCCM, including the reserved `bl0_state` area, so each word receives valid ECC before C uses it.

The C path can clear part or all of SEP SRAM according to `SRAM_SCRUB_BYTES`. The open build defaults this value to zero because the programmed simulation flow may preload data in that memory. The manifest load overwrites its staging region, but this simulation choice does not define a production sanitization policy.

ICCM is initialized through the secure DMA because the CPU load/store path cannot write it. `ROM_ICCM_CLEAR_ENABLE` defaults to zero to control simulation cost. A release configuration should enable it so all ICCM fetch words have valid ECC, including words adjacent to the end of BL1, but the current Makefile does not define or enforce a release mode. `ROM_ICCM_CLEAR_ENABLE` is also absent from its build-flags stamp, so changing the variable requires a clean build to ensure `rom_main.c` is recompiled.

4.3.5. C initialization

The C path validates its runtime state, initializes the SMC-backed status channel when enabled, and selects the reference or PLL-derived clock. It then initializes `bl0_state`, validates the lifecycle state, reports the SMC chip ID, and runs SHA-

256 over the NIST "abc" vector as a crypto-engine self-test.

The peripheral/bus reset function currently reports that the sequence is not implemented. DMA setup and the selected SPI-controller initialization occur later, immediately before boot-source processing.

4.4. Memory and security

4.4.1. Image and runtime layout

Region	Address	Size	Use
SEP SRAM	0x1000_0000	256 KiB	Manifest and payload staging.
Boot ROM	0x1004_0000	64 KiB	BL0 code, metadata, read-only data, and the load image of <code>.data</code> .
SPI XIP	0x3000_0000	256 MiB aperture	Memory-mapped flash source for the default controller path.
SMC SRAM	SEP view 0x4006_0000	1 MiB	Recovery/secondary manifest source and status ring storage.
ICCM	0xC000_0000	256 KiB	Validated BL1 execution image.
DCCM	0xC004_0000	128 KiB	BL0 writable data, BSS, stack, and BL0-to-BL1 state.

The linker keeps `.text`, `.metadata`, and `.rodata` in ROM. `.data` executes from DCCM but has a ROM load image. `.bss` and the stack occupy only DCCM. A 256-byte reserve at the top of DCCM holds `struct bl0_state` (`BL0_STATE_RESERVE_BYTES` in `include/bl0_state.h`, which `link/rom.ld` must match); runtime and compile-time checks keep the structure from overlapping the downward-growing stack.

`bl0_state` gives BL1 the validated manifest address, boot mode, secure-boot decision, lifecycle and feature-control values, last error, boot measurement, deferred BL2 demotion decision, and the SRAM address of the BL1 image.

4.4.2. Security ordering

The following ordering is part of the ROM security contract:

1. Memory repair and MBIST are evaluated before the ROM writes or trusts DCCM.
2. The lifecycle state is validated before it can select secure-boot policy.
3. Manifest bounds and structure are checked before payload addresses are used.
4. For an encrypted image, the payload hash covers ciphertext and is checked before in-place decryption; the TOC is parsed only after decryption.
5. Image ranges are checked before image data is hashed.
6. Sensitive fuse reads are locked before the demotion register is written.
7. The stack canary is checked before the non-returning manifest and handoff path.
8. BL1 is copied only after the manifest, payload, TOC, and BL1 entry have all passed validation.

Hash and public-key digest comparisons accumulate differences over every byte instead of returning on the first mismatch.

4.4.3. Lifecycle and secure-boot policy

The lifecycle controller accepts TEST_DEV, PROD, PROD_END, and the defined RMA encodings. An invalid value causes BL0 to hold the SMC cores in reset, report a terminal error, and stop.

SB00T_DIS is an eFuse override that disables secure boot. When it is clear, PROD and PROD_END enforce secure boot regardless of the manifest request. TEST_DEV and RMA enable secure boot when the manifest requests it. An encrypted payload is rejected when secure boot is not active.

4.4.4. Manifest authentication

The ROM always checks the SHA-256 manifest hash over the to-be-signed region. When secure boot is active it additionally:

1. compares the manifest security version with the thermometer-encoded **BL1_VERSION** fuse;
2. selects a ROM-embedded or eFuse public-key digest;
3. checks the corresponding revocation bit;
4. binds the manifest RSA modulus to the selected digest; and
5. verifies the RSA-3072 PKCS#1 v1.5 signature with OTBN.

The public key in the manifest is therefore not trusted by itself. The Key Manager does not participate in this verification path.

4.4.5. Payload protection

The payload SHA-256 is checked on both secure and non-secure boots. Secure encrypted images derive an AES-128 key from the **CLASS_KEY** fuse with KBKDF-HMAC-SHA256 and decrypt the payload in place with AES-CBC. Temporary key, salt, and derivation buffers are explicitly cleared.

After decryption, the ROM validates the TOC and every image extent, requires a usable SEP BL1 entry, verifies each image SHA-256, and clears bytes in gaps between described images and after the final image.

4.4.6. Secret locking, demotion, and measurement

Before handoff, BL0 read-locks **CLASS_KEY**, both RMA token digests, and the chiplet, SiP, and system UIDs. Failure to read back all requested lock bits is terminal.

PROD_END prevents demotion and locks both demotion registers. Other lifecycle states apply the BL1 decision or preserve a deferred BL2 decision according to the manifest. BL0 records a SHA-256 boot measurement over the manifest hash, lifecycle, demotion decision, secure-boot state, and **SB00T_DIS**. This digest reports the selected boot state to BL1; it does not itself bind a key to that state.

4.5. SPI and manifest processing

4.5.1. Flash transports

include/boot_flash.h presents one flash-offset interface to the manifest loader. **BOOT_SPI_CONTROLLER_OT** selects the implementation at build time:

- The default path uses a memory-mapped SPI XIP aperture and the secure DMA. The open `src/sep_spi.c` implementation is a weak unavailable-device stub so an adopter can supply the controller integration when producing the final image.
- The OpenTitan SPI host issues command/FIFO transfers. Its receive FIFO can drain through the secure DMA (`BOOT_OT_SPI_USE_PIO=0`) or CPU programmed I/O (`BOOT_OT_SPI_USE_PIO=1`). These are separate ROM build variants over the same packed flash bytes.

Controller initialization receives the clock selected earlier in the boot. Failure is not immediately terminal because the loader can proceed to another configured slot.

4.5.2. Boot source and slot selection

The SPI path uses fixed manifest offsets:

Slot	Manifest offset	Selection
Primary	<code>0x0000_1000</code>	First unless <code>rotate_update</code> is set.
Backup	<code>0x0004_1000</code>	Second unless <code>rotate_update</code> is set.

On recovery or secondary boot, the ROM does not use either SPI offset. It waits for `SMC_SEP_STATUS_MANIFEST_READY`, reads the manifest offset from SMC scratch register 8, and adds that offset to the SEP-visible SMC SRAM base.

4.5.3. Manifest layout

The manifest is an OCA classic manifest with a 4096-byte body. Bytes `[0, 3172)` are the signed region — everything from the magic through the four-byte `0xCA` trailer at offset 3168 — which both the RSA signature and the manifest hash cover. Bytes `[3172, 4096)` are the unsigned tail holding the signature, the manifest hash and the payload offset. Because the payload offset is in that tail it is attacker-controlled even on a validly signed manifest, so the ROM bounds it against the region the slot may reach rather than trusting it. The payload begins with a `PTOC` header followed by the image entries and their image data.

The format and its numeric constants come from the OCA specification and its layout header, `tools/tt-oca-manifest/validators/oca/lib/oca_layout_classic.h`, vendored as a submodule. The YAML files under `configs/` are test-image producer inputs; they do not supersede the format the library consumes on BL0's behalf.

4.5.4. Per-slot validation

For each selected slot, BL0 performs this sequence:

1. Read the fixed manifest header into SEP SRAM.
2. Validate the identifier, major version, manifest length, alignment, payload offset and length, SRAM bounds, and non-overlap.
3. Verify the manifest hash.
4. Decide whether secure boot is active and apply selected lifecycle, chiplet ID, and package ID usage constraints.
5. Read any permitted manifest extension and stage the payload.
6. Under secure boot, check rollback, authorize and check the root key, verify RSA-3072, verify the payload hash, and decrypt an encrypted payload. Without secure boot, still verify the payload hash.
7. Validate the TOC, image count, image order and bounds, per-image hashes, and the presence and placement of SEP BL1.

An encrypted payload is authenticated as ciphertext before decryption. The TOC inside it is interpreted only after decryption.

4.5.5. Retry behavior

The SPI path can attempt both slots. `rotate_update` exchanges their order. A failed attempt records its error; before another attempt BL0 clears the SEP SRAM staging region and reinitializes the flash controller. A failed reinitialization is terminal because the remaining slot cannot be read.

The SMC-SRAM path has one source and no alternate slot. Exhausting the available attempts reports `SEP_MSG_MANIFEST_LOAD_FAILED` and enters the common failure path.

4.5.6. Verification evidence

`hw/sys/sep/dv/cocotb/tests/rom_fw/sep_rom_ot_secure_boot_test.py` exercises the signed OpenTitan SPI-host DMA path. It requires evidence that the ROM took the SPI source, read the primary manifest at `0x1000`, started RSA verification, accepted the signature, and completed crypto validation. It also forbids `SBOOT_OFF`, so a silent downgrade cannot satisfy the test.

That testcase supplies OTBN entropy through the `+sep_crypto_edn_force` DV shortcut because the current BL0 does not bring up the entropy-source, CSRNG, and EDN chain. The test therefore demonstrates the manifest, transport, OTBN RSA, and handoff path; it does not demonstrate ROM entropy-chain initialization.

The DMA and PIO receive paths are covered separately because they are separate datapaths. `sep_rom_ot_pio_boot_test` runs the `build_ot_pio` image, which does not link `ot_spi_flash_read_dma` at all, so passing one of the two proves nothing about the other. `hw/sys/sep/dv/testlists/rom_fw.toml` lists the whole ROM regression; the cases named here are the ones that bear directly on manifest and transport processing, not the extent of ROM coverage.

4.6. SMC coordination

SEP reaches SMC resources through its outbound AXI path at `SEP_SMC_GLOBAL_BASE`, `0x4000_0000`. The offsets and status bits in `include/sep_smc_interface.h` form the SEP side of the interface contract. They are SEP-visible offsets and must not be replaced with SMC-local absolute addresses.

4.6.1. Shared scratch registers

SMC CPU-control scratch registers have an eight-byte stride. The table covers every low 32-bit entry SEP BL0 consumes or writes, together with the entries the shared interface reserves for SMC alone. `include/sep_smc_interface.h` names all of them except index 7, which BL0 addresses by literal index:

Index	Name	BL0 role	Use
1	POST code	None	SMC boot-status publication.
2	Virtual console	None	SMC debug text channel.
7	Scratch round-trip probe	Writes	BL0 writes four test patterns, reads each back, and clears the entry to zero before it selects a boot source. A read-back mismatch is terminal. SMC must not keep its own state here.
8	Manifest address	Consumes	Byte offset of an SMC-staged manifest from the SMC SRAM base.
9	Status to SEP	Consumes	SMC-to-SEP readiness flags.

Index	Name	BL0 role	Use
10	MBIST failure	Writes	Raw DFX status recorded by the SEP pre-C memory-health gate.
11	Status-buffer address	Consumes	Byte offset of the SEP status ring from the SMC SRAM base.
13, 14	SEP-safe SRAM start and size	None	SMC-published range available to SEP.
15	Memory-repair status	None	SMC memory-repair result.

Scratch register 9 defines four readiness flags: SRAM initialized, manifest ready, status buffer ready, and SRAM protected. BL0 reads the flags relevant to the operation it is about to perform rather than treating the word as one global ready indication.

4.6.2. Manifest handoff

Recovery and secondary boot wait until the manifest-ready bit is set. BL0 then reads the offset from scratch register 8 and forms the source address as:

```
SEP_SMC_GLOBAL_BASE + SMC_SRAM_OFFSET + manifest_offset
```

The resulting manifest and payload must remain inside the one-MiB SMC SRAM window. The ROM copies them to SEP SRAM before validation and execution.

The two halves of that window requirement are enforced differently, and an integrator has to account for both. The ROM checks the payload source range itself: a payload that starts below the SMC SRAM base or ends above it is rejected with `OCA_FAIL_PAYLOAD_LOCATION` (console `PAYLOAD_LOC_FAIL`), as is an address computation that overflows. The manifest header read is not checked. BL0 adds the offset from scratch register 8 to the SMC SRAM base and issues the header transfer without a range or overflow test, so keeping that offset inside the window is SMC's obligation. A header read from outside the window is caught only afterwards, by the manifest identifier, version, length, and hash checks on the bytes it returned.

The normal primary-chiplet SPI path does not consume the SMC manifest offset, but it still uses SMC for captured straps, DFX status, chip identity, and status reporting.

4.6.3. Status-buffer handoff

When status reporting is enabled, BL0 waits for the status-buffer-ready bit in scratch register 9. It reads the ring-buffer offset from scratch register 11, checks that the ring header lies within SMC SRAM, and then publishes non-debug status through that ring. SEP owns the ring head and SMC owns the tail.

An invalid offset leaves the scratch-register status channel operational and prevents BL0 from writing through an untrusted ring pointer.

4.6.4. Probe behavior

Before loading a manifest, `rom_smc_coordination_probe()` reports the status word, manifest offset, and status-buffer offset for diagnosis. The probe does not replace the blocking checks in manifest and status initialization; it records whether the corresponding SMC state was already ready at that point.

4.7. Status and error reporting

BL0 uses SEP cold scratch registers for channels that must be reachable before the outbound fabric and C runtime are available.

Register	Channel	Contract
<code>cold_scratch[0]</code>	Terminal verdict	BL0 writes the failure value only. A successful boot transfers to BL1, which owns the pass verdict.
<code>cold_scratch[1]</code>	Machine-readable status	Carries encoded status and error values throughout assembly and C execution.
<code>cold_scratch[2]</code>	Virtual console	Carries packed debug characters when debug output is compiled in.
<code>cold_scratch[7]</code>	Warm-reset handler	Carries the BL1 ICCM re-entry address across a watchdog reset.

4.7.1. Status encoding

`include/errors.h` encodes a status as an eight-bit type, an eight-bit firmware stage ID, and a 16-bit value. BL0 uses information, warning, error, debug, and extended-information types. `include/status_values.h` is the authoritative registry for `SEP_MSG_*` values; consumers should use its names rather than copying numeric values from this document.

Every status is written to `cold_scratch[1]`. Debug values stop there. Other types are also offered to the SMC-SRAM ring when ring reporting is enabled and valid.

4.7.2. Status ring

The status ring is lock-free: SEP advances `head`, and SMC advances `tail`. Before an entry write, BL0 validates the entry count, head, tail, remaining capacity, and destination address. An invalid state reports `SEP_MSG_STATUS_REPORTING_INVALID` through `cold_scratch[1]` and does not write the entry.

Capacity is reserved by severity. Informational status preserves space for a warning and an error; warning status preserves space for an error. At the reservation boundary BL0 inserts the corresponding overflow marker. An error can use the final free entry.

4.7.3. Trap handling

The reset path first installs a stack-free trap vector. It records a general exception status followed by `mcause`, `mtval`, and `mepc`, then restores the decodable status word and writes the failure verdict. It uses only SEP scratch registers, so an early exception does not depend on external address decode.

After the stack is valid, BL0 installs the C-capable trap vector. It reports the same CSRs through the debug console, writes the terminal failure verdict, and stops.

4.7.4. Error convergence

C error paths converge on `rom_err_fail()`. When `bl0_state` is initialized, the function stores the full error code there. It also publishes an encoded error in `cold_scratch[1]`, writes the failure verdict in `cold_scratch[0]`, and enters a `wfi` loop.

Manifest failures use the `MANIFEST_ERR_*` namespace, formed as `OCA_BOOT_ERR_BASE | oca_result_t`—the OCA library's verdict carried in the low byte. The enumeration is `tools/tt-oca-manifest/validators/oca/lib/oca_validator.h`, and `status_for_result()` in `src/oca_boot.c` is the only bridge between it and the `SEP_MSG_*` codes the status ring carries. Other

terminal firmware errors are defined next to their owning implementation or in `include/errors.h`. These registries, rather than this document, are the source of numeric values.

Memory-repair and MBIST failure can occur before the C runtime and `bl0_state` exist. That path records the raw DFX word in SMC scratch register 10, leaves a terminal status in `cold_scratch[1]`, and stops without relying on the outbound mailbox.

4.8. BL1 handoff

The payload TOC may describe SEP and SMC images. BL0 scans it for the first `IMAGE_TYPE_SEP_BL1` entry and validates that the complete image lies inside ICCM, that its length is nonzero and no larger than ICCM, and that its entry offset lies inside the image. These checks run during payload validation and are repeated at handoff.

4.8.1. ICCM copy

The VeeR EL2 instruction-fetch unit can execute from ICCM, but the CPU load/store unit cannot read or write it. ICCM and DCCM occupy the same VeeR region, and load/store address checking treats ICCM addresses outside the DCCM offset as unmapped.

BL0 therefore uses the secure DMA to copy the validated BL1 bytes from the payload in SEP SRAM to the entry's ICCM load address. A CPU `memcpy` cannot replace this transfer.

BL1's binary contains executable code and the load image of its own read-only and writable data. Because BL1 also cannot read its ICCM copy, BL0 stores the payload source address in `bl0_state.bl1_image_src_addr`. BL1 startup uses that SRAM source when initializing DCCM.

4.8.2. Transfer of control

The entry address is the ICCM load address plus the TOC entry offset. After the DMA succeeds, BL0 writes the entry to `mepc`, executes `fence.i` and `fence`, and uses `mret` to transfer control. This makes the newly written instructions visible to the fetch unit and completes prior memory activity before BL1 runs.

The handoff is non-returning. Reaching its caller again is an error. BL0 does not publish a success verdict before the transfer; BL1 proves successful execution through its own status path.

4.8.3. Shared state

BL1 locates `struct bl0_state` at the top of DCCM and validates its leading and trailing magic words and size. The structure communicates:

- the validated manifest and BL1 source addresses;
- selected boot mode and secure-boot state;
- lifecycle and feature-control values;
- the boot measurement and deferred BL2 demotion decision; and
- the last BL0 error, when execution did not reach handoff.

BL0 and BL1 must use the same `include/bl0_state.h` layout.

4.8.4. Warm-reset return

BL1 may place an ICCM handler address in SEP `cold_scratch[7]` before enabling the watchdog. This register bank survives the warm watchdog reset. On reset, BL0 reads the slot before setting a stack or touching DCCM:

- zero selects a cold boot, after which BL0 poisons the slot with `-1`;
- an address inside ICCM transfers directly to the handler and remains in the slot for subsequent watchdog resets; and
- any other nonzero value reports an invalid target and stops.

The direct warm path preserves BL1 DCCM and ICCM state. The handler must restore any CPU state reset by the watchdog before resuming normal execution.

Chapter 5. AoU Software operation guide

This section gives information about operation guide for the AOU_CORE.

5.1. SFR setting

Register	Offset	Bit field name	Bit field	Type	Reset value
IP_VERSION	0x0000				0x00010000
		MAJOR_VERSION	[31:16]	RO	0x0001
		MINOR_VERSION	[15:0]	RO	0x0000
AOU_CON0	0x0004				0x00000000
		Rsvd	[31:28]	RO	0x0
		RP3_ERROR_INFO_ACCESS_EN	[27]	RW	0x0
		RP2_ERROR_INFO_ACCESS_EN	[26]	RW	0x0
		RP1_ERROR_INFO_ACCESS_EN	[25]	RW	0x0
		RP0_ERROR_INFO_ACCESS_EN	[24]	RW	0x0
		RP3_AXI_AGGREGATOR_EN	[23]	RW	0x0
		RP2_AXI_AGGREGATOR_EN	[22]	RW	0x0
		RP1_AXI_AGGREGATOR_EN	[21]	RW	0x0
		RP0_AXI_AGGREGATOR_EN	[20]	RW	0x0
		TX_LP_MODE_THRESHOLD	[19:12]	RW	0x4
		TX_LP_MODE	[11]	RW	0x0
		Rsvd	[10:5]	RO	0x0
		AOU_SW_RESET	[4]	RW	0x0
		CREDIT_MANAGE	[3]	RW	0x0
		AXI_SPLIT_TR_EN	[2]	RW	0x0
		Rsvd	[1:0]	RO	0x0
AOU_INIT	0x0008				0x00004608
		Rsvd	[31:11]	RO	0x0
		MST_TR_COMPLETE	[10]	RO	0x1
		SLV_TR_COMPLETE	[9]	RO	0x1
		INT_ACTIVATE_START	[8]	W1C	0x0
		INT_DEACTIVATE_START	[7]	W1C	0x0
		DEACTIVATE_TIME_OUT_VALUE	[6:4]	RW	0x0
		ACTIVATE_STATE_DISABLED	[3]	RO	0x1
		ACTIVATE_STATE_ENABLED	[2]	RO	0x0
		DEACTIVATE_START	[1]	RW	0x0
		ACTIVATE_START	[0]	RW	0x0
AOU_INTERRUPT_MASK	0x000C				
		Rsvd	[31:9]	RO	0x0

Register	Offset	Bit field name	Bit field	Type	Reset value
		INT_REQ_LINKRESET_ACT_ACK_MASK	[8]	RW	0x0
		INT_REQ_LINKRESET_DEACT_ACK_MASK	[7]	RW	0x0
		INT_REQ_LINKRESET_INVALID_ACTMSG_MASK	[6]	RW	0x0
		INT_REQ_LINKRESET_MSGCREDIT_TIMEOUT_MASK	[5]	RW	0x0
		INT_EARLY_RESP_MASK	[4]	RW	0x0
		INT_MI0_ID_MISMATCH_MASK	[3]	RW	0x0
		INT_SI0_ID_MISMATCH_MASK	[2]	RW	0x0
		Rsvd	[1]	RO	0x0
		Rsvd	[0]	RO	0x0
LP_LINKRESET	0x0010				
		Rsvd	[31:14]	RO	0x0
		ACK_TIME_OUT_VALUE	[13:11]	RW	0x4
		MSGCREDIT_TIME_OUT_VALUE	[10:8]	RW	0x4
		ACT_ACK_ERR	[7]	W1C	0x0
		DEACT_ACK_ERR	[6]	W1C	0x0
		INVALID_ACTMSG_INFO	[5:2]	RO	0x0
		INVALID_ACTMSG_ERR	[1]	W1C	0x0
		MSGCREDIT_ERR	[0]	W1C	0x0
DEST_RP	0x0014				0x00000000
		Rsvd	[31:14]	RO	0x0
		RP3_DEST	[13:12]	RW	0x3
		Rsvd	[11:10]	RO	0x0
		RP2_DEST	[9:8]	RW	0x2
		Rsvd	[7:6]	RO	0x0
		RP1_DEST	[5:4]	RW	0x1
		Rsvd	[3:2]	RO	0x0
		RP0_DEST	[1:0]	RW	0x0
PRIOR_RP_AXI	0x0018				
		Rsvd	[31:28]	RO	0x0
		AXI_QOS_TO_NP	[27:24]	RW	0xA
		AXI_QOS_TO_HP	[23:20]	RW	0x5
		Rsvd	[19:18]	RO	0x0
		RP3_PRIOR	[17:16]	RW	0x3
		Rsvd	[15:14]	RO	0x0
		RP2_PRIOR	[13:12]	RW	0x2
		Rsvd	[11:10]	RO	0x0
		RP1_PRIOR	[9:8]	RW	0x1
		Rsvd	[7:6]	RO	0x0

Register	Offset	Bit field name	Bit field	Type	Reset value
		RP0_PRIOR	[5:4]	RW	0x0
		Rsvd	[3:2]	RO	0x0
		ARB_MODE	[1:0]	RW	0x0
PRIOR_TIMER	0x001C				0x00000000
		TIMER_RESOLUTION	[31:16]	RW	0x0
		TIMER_THRESHOLD	[15:0]	RW	0x0
AXI_SPLIT_TR_RP0	0x0020				0x00000F0F
		Rsvd	[31:16]	RO	0x0
		MAX_AWBURSTLEN	[15:8]	RW	0xF
		MAX_ARBURSTLEN	[7:0]	RW	0xF
ERROR_INFO_RP0	0x0024				0x0
		Rsvd	[31:22]	RO	0x0
		SPLIT_BID_MISMATCH_INFO	[21:12]	RO	0x0
		SPLIT_RID_MISMATCH_INFO	[11:2]	RO	0x0
		SPLIT_BID_MISMATCH_ERR	[1]	W1C	0x0
		SPLIT_RID_MISMATCH_ERR	[0]	W1C	0x0
WRITE_EARLY_RESPONSE_RP0	0x0028				0x0
		Rsvd	[31:15]	RO	0x0
		WRITE_RESP_DONE	[14]	RO	0x0
		WRITE_RESP_ERR	[13]	W1C	0x0
		WRITE_RESP_ERR_TYPE_INFO	[12:11]	RO	0x0
		WRITE_RESP_ERR_ID_INFO	[10:1]	RO	0x0
		EARLY_BRESP_EN	[0]	RW	0x0
AXI_ERROR_INFO0_RP0	0x002C				0x0
		DEBUG_UPPER_ADDR	[31:0]	RW	0x0
AXI_ERROR_INFO1_RP0	0x0030				0x0
		DEBUG_LOWER_ADDR	[31:0]	RW	0x0
AXI_SLV_ID_MISMATCH_ERR_RP0	0x0034				
		Rsvd	[31:22]	RO	0x0
		AXI_SLV_BID_MISMATCH_INFO	[21:12]	RO	0x0
		AXI_SLV_RID_MISMATCH_INFO	[11:2]	RO	0x0
		AXI_SLV_BID_MISMATCH_ERR	[1]	W1C	0x0
		AXI_SLV_RID_MISMATCH_ERR	[0]	W1C	0x0
AXI_SPLIT_TR_RP1	0x0038				0x00000F0F
		Rsvd	[31:16]	RO	0x0
		MAX_AWBURSTLEN	[15:8]	RW	0xF
		MAX_ARBURSTLEN	[7:0]	RW	0xF
ERROR_INFO_RP1	0x003C				0x0

Register	Offset	Bit field name	Bit field	Type	Reset value
		Rsvd	[31:22]	RO	0x0
		SPLIT_BID_MISMATCH_INFO	[21:12]	RO	0x0
		SPLIT_RID_MISMATCH_INFO	[11:2]	RO	0x0
		SPLIT_BID_MISMATCH_ERR	[1]	W1C	0x0
		SPLIT_RID_MISMATCH_ERR	[0]	W1C	0x0
WRITE_EARLY_RESPONSE_RP1	0x0040				0x0
		Rsvd	[31:15]	RO	0x0
		WRITE_RESP_DONE	[14]	RO	0x0
		WRITE_RESP_ERR	[13]	W1C	0x0
		WRITE_RESP_ERR_TYPE_INFO	[12:11]	RO	0x0
		WRITE_RESP_ERR_ID_INFO	[10:1]	RO	0x0
		EARLY_BRESP_EN	[0]	RW	0x0
AXI_ERROR_INFO0_RP1	0x0044				0x0
		DEBUG_UPPER_ADDR	[31:0]	RW	0x0
AXI_ERROR_INFO1_RP1	0x0048				0x0
		DEBUG_LOWER_ADDR	[31:0]	RW	0x0
AXI_SLV_ID_MISMATCH_ERR_RP1	0x004C				
		Rsvd	[31:22]	RO	0x0
		AXI_SLV_BID_MISMATCH_INFO	[21:12]	RO	0x0
		AXI_SLV_RID_MISMATCH_INFO	[11:2]	RO	0x0
		AXI_SLV_BID_MISMATCH_ERR	[1]	W1C	0x0
		AXI_SLV_RID_MISMATCH_ERR	[0]	W1C	0x0
AXI_SPLIT_TR_RP2	0x0050				0x00000F0F
		Rsvd	[31:16]	RO	0x0
		MAX_AWBURSTLEN	[15:8]	RW	0xF
		MAX_ARBURSTLEN	[7:0]	RW	0xF
ERROR_INFO_RP2	0x0054				0x0
		Rsvd	[31:22]	RO	0x0
		SPLIT_BID_MISMATCH_INFO	[21:12]	RO	0x0
		SPLIT_RID_MISMATCH_INFO	[11:2]	RO	0x0
		SPLIT_BID_MISMATCH_ERR	[1]	W1C	0x0
		SPLIT_RID_MISMATCH_ERR	[0]	W1C	0x0
WRITE_EARLY_RESPONSE_RP2	0x0058				0x0
		Rsvd	[31:15]	RO	0x0
		WRITE_RESP_DONE	[14]	RO	0x0
		WRITE_RESP_ERR	[13]	W1C	0x0
		WRITE_RESP_ERR_TYPE_INFO	[12:11]	RO	0x0
		WRITE_RESP_ERR_ID_INFO	[10:1]	RO	0x0

Register	Offset	Bit field name	Bit field	Type	Reset value
		EARLY_BRESP_EN	[0]	RW	0x0
AXI_ERROR_INFO0_RP2	0x005C				0x0
		DEBUG_UPPER_ADDR	[31:0]	RW	0x0
AXI_ERROR_INFO1_RP2	0x0060				0x0
		DEBUG_LOWER_ADDR	[31:0]	RW	0x0
AXI_SLV_ID_MISMATCH_ERR_RP2	0x0064				
		Rsvd	[31:22]	RO	0x0
		AXI_SLV_BID_MISMATCH_INFO	[21:12]	RO	0x0
		AXI_SLV_RID_MISMATCH_INFO	[11:2]	RO	0x0
		AXI_SLV_BID_MISMATCH_ERR	[1]	W1C	0x0
		AXI_SLV_RID_MISMATCH_ERR	[0]	W1C	0x0
AXI_SPLIT_TR_RP3q	0x0068				0x00000F0F
		Rsvd	[31:16]	RO	0x0
		MAX_AWBURSTLEN	[15:8]	RW	0xF
		MAX_ARBURSTLEN	[7:0]	RW	0xF
ERROR_INFO_RP3	0x006C				0x0
		Rsvd	[31:22]	RO	0x0
		SPLIT_BID_MISMATCH_INFO	[21:12]	RO	0x0
		SPLIT_RID_MISMATCH_INFO	[11:2]	RO	0x0
		SPLIT_BID_MISMATCH_ERR	[1]	W1C	0x0
		SPLIT_RID_MISMATCH_ERR	[0]	W1C	0x0
WRITE_EARLY_RESPONSE_RP3	0x0070				0x0
		Rsvd	[31:15]	RO	0x0
		WRITE_RESP_DONE	[14]	RO	0x0
		WRITE_RESP_ERR	[13]	W1C	0x0
		WRITE_RESP_ERR_TYPE_INFO	[12:11]	RO	0x0
		WRITE_RESP_ERR_ID_INFO	[10:1]	RO	0x0
		EARLY_BRESP_EN	[0]	RW	0x0
AXI_ERROR_INFO0_RP3	0x0074				0x0
		DEBUG_UPPER_ADDR	[31:0]	RW	0x0
AXI_ERROR_INFO1_RP3	0x0078				0x0
		DEBUG_LOWER_ADDR	[31:0]	RW	0x0
AXI_SLV_ID_MISMATCH_ERR_RP3	0x007C				
		Rsvd	[31:22]	RO	0x0
		AXI_SLV_BID_MISMATCH_INFO	[21:12]	RO	0x0
		AXI_SLV_RID_MISMATCH_INFO	[11:2]	RO	0x0
		AXI_SLV_BID_MISMATCH_ERR	[1]	W1C	0x0

Register	Offset	Bit field name	Bit field	Type	Reset value
		AXI_SLV_RID_MISMATCH_ERR	[0]	W1C	0x0

Table 7. Register Map

ERR SET is restricted to change the value while the system is running

For details on the ACTIVATE and DEACTIVATE SFRs, please refer to [Activation & Deactivation Flow](#).

AOU_INIT.ACTIVATE_START

- After link-up is completed, AOU_CORE can send or receive an ActivateReq message.
- AoU Activation Request can only be triggered by AOU_INIT.ACTIVATE_START, and it is allowed to be set before and after the completion of the UCle Link-up.

AOU_INIT.DEACTIVATE_START

- AOU_CORE must exchange both DeactivateReq and DeactivateAck messages before the UCle Core transitions to the Link Down state.
- The DEACTIVATE_START field in AOU_INIT initiates deactivation and ensures that a DeactivateReq is issued only after both conditions below are satisfied:
 1. All outstanding responses to remote die requests have been completed.
 2. No additional messages remain to be transmitted within a window defined by **TIME_OUT_VALUE**.
- **AOU_INIT.DEACTIVATE_TIME_OUT_VALUE** configures the timeout value used for this condition. This field is 3 bits wide and encodes a timeout of $2^{(\text{DEACTIVATE_TIME_OUT_VALUE} + 3)}$ cycles.

MAX BURSTLEN

- The **MAX_AWBURSTLEN** and **MAX_ARBURSTLEN** register fields define the maximum burst length that the splitter may issue to the local bus.
- The splitter inside **AOU_CORE** issues AXI transfers with burst lengths less than or equal to the configured values (**MAX_AWBURSTLEN**, **MAX_ARBURSTLEN**).
- If **AOU_CORE** receives a larger burst length from the remote die, the splitter divides it into smaller bursts based on **MAX_AxBURSTLEN**.
- Since Splitter only supports maximum burst lengths in the form of 2^n , the field must be configured with one of the following: 1, 3, 5, 7, 15, 31, 63, 127 or 255.

ACTIVATE_STATE_ENABLED/DISABLED

- This signal indicates the activation status of the AOU_ACTIVATION_CTRL.
- The ENABLE signal is asserted to indicate that activation has been completed.

The DISABLED signal is asserted to indicate that deactivation has been completed.

MST_TR_COMPLETE/SLV_TR_COMPLETE

- This signal indicates that there are no pending transactions in both local die and remote die.

ACK_TIME_OUT_VALUE/MSG_CREDIT_TIME_OUT_VALUE

- After sending an AoU Activation/Deactivation Request message, if an Ack message is not received within the specified ACK_TIME_OUT_VALUE, a LinkReset interrupt is sent to the SW.
- After receiving the Activation Ack message from the remote die, if a creditgrant message is not received within the specified MSGCREDIT_TIME_OUT_VALUE, a LinkReset interrupt is sent to the SW.
- These fields are 3 bits wide and encode a timeout of $2^{(TIME_OUT_VALUE + 18)}$ core clock cycles.

WRITE_EARLY_RESPONSE

- For the EARLY_RESPONSE_EN SFR setting to enable or disable early response, there must be no pending write requests.
- EARLY_RESPONSE_EN: If this field is set(1), It performs the early response function.
- WRITE_RESP_DONE: It indicates that there are no pending write requests and all write responses have been received from the remote die.
- WRITE_RESP_ERR: It indicated that although the early response feature was enabled and an early response has already been delivered to the local die, an error occurred later in the write response received from the remote die. The BRESP value of the error is stored in WRITE_RESP_ERR_TYPE_INFO, and the BID of the transaction that caused the error is stored in WRITE_RESP_ERR_ID_INFO SFR. When an error occurs, an interrupt is generated. Afterward, writing 1 to the WRITE_RESP_ERR SFR clears ERR_TYPE_INFO, ERR_ID_INFO, and RESP_ERR.

Low Power mode

- For lowering power, AOU_CORE uses low power mode. By setting AOU_CON0.TX_LP_MODE = 1, AOU_CORE only sends payload to FDI when there are remaining AXI messages to send. Since AOU_CORE need to send Credit when there are no remaining AXI messages, by setting AOU_CON0.TX_LP_MODE_THRESHOLD value, SW can configure the message transmission frequency.
- AOU_CON0.TX_LP_MODE_THRESHOLD = 0 selects a fully-gated low-power mode: the periodic heartbeat is disabled and the TX only launches an FDI flit when there is real work to send (AXI traffic, CrdtGrant credit-return messages, or pending MsgCredit-header credits). Once a flit is started it always finishes; any granule slots that were not filled are sent as zeros and the corresponding MsgStart bits are 0.

[Figure 22. Bubble inserted when Transmitter de-assert LP_VALID] | *image21.png*

Figure 22. Bubble inserted when Transmitter de-assert LP_VALID

- Default value is 0. AOU_CORE always sends messages to FDI for lowering latency

5.2. Debugging Features

5.2.1. Overview

AOU_CORE provides several error-detection and debug features, each with a dedicated interrupt output and a per-feature mask bit in the AOU_INTERRUPT_MASK register (0x000C). The table below summarises the mask bits and corresponding interrupt signals.

Mask Bit	Feature	Interrupt Signal
[8]	Activate-ack timeout	INT_REQ_LINKRESET
[7]	Deactivate-ack timeout	INT_REQ_LINKRESET
[6]	Invalid ACTMSG	INT_REQ_LINKRESET

Mask Bit	Feature	Interrupt Signal
[5]	MsgCredit timeout	INT_REQ_LINKRESET
[4]	Early write-response error	INT_EARLY_RESP_ERR
[3]	Manager ID mismatch	INT_MIO_ID_MISMATCH
[2]	Subordinate ID mismatch	INT_SIO_ID_MISMATCH

Setting a mask bit to 1 masks (disables) the corresponding interrupt.

5.2.2. AXI ID Mismatch Error

AOU_CORE can detect AXI ID mismatches on both the manager and subordinate interfaces. A mismatch occurs when a B or R response arrives with an AXI ID that does not correspond to any outstanding request.

Manager side. The split/aggregator path (AOU_AXI_SPLIT_TR / AOU_AXI4MUX_3X1_TOP) maintains pending-ID tables for both read and write transactions (AOU_SPLIT_RD_PENDING_INFO and AOU_SPLIT_WR_PENDING_INFO). When a B response arrives with an ID that does not match any pending write entry, SPLIT_BID_MISMATCH_ERR is asserted and the offending ID is captured in SPLIT_BID_MISMATCH_INFO. The same logic applies to R responses via SPLIT_RID_MISMATCH_ERR and SPLIT_RID_MISMATCH_INFO. These fields are in the per-RP ERROR_INFO_RPn register (RP0 at 0x0024, RP1 at 0x003C, RP2 at 0x0054, RP3 at 0x006C). SW clears each error by writing 1 to the corresponding W1C bit. The aggregate interrupt is INT_MIO_ID_MISMATCH.

Table depth and backpressure. Each pending-ID table is sized by M_RD_MO_CNT / M_WR_MO_CNT (default 32), matching the manager interface outstanding transaction limit. An entry is allocated on the split-AR or split-AW handshake and freed when the corresponding R (with RLAST) or B response completes. When all entries are occupied, the O_AR_Slot_Available_Flag / O_AW_Slot_Available_Flag signal goes low, which suppresses new split transactions and effectively deasserts ARREADY / AWREADY toward the upstream path. This is backpressure, not an error – no entries are overwritten or dropped. Because the table depth equals the AXI outstanding count, the table should never overflow during correct operation.

No disable mechanism. There is no RTL parameter or SFR to disable the pending-ID tables or the mismatch check. The interrupt can be masked via AOU_INTERRUPT_MASK.INT_MIO_ID_MISMATCH_MASK [3], but this only suppresses the interrupt signal. The tables continue to operate, backpressure still applies when full, and the sticky error bits in ERROR_INFO_RPn are still set on a mismatch.

Subordinate side. An analogous pending-ID table exists in AOU_SLV_AXI_INFO, with SFR fields in AXI_SLV_ID_MISMATCH_ERR_RPn (RP0 at 0x0034, RP1 at 0x004C, RP2 at 0x0064, RP3 at 0x007C). However, the subordinate-side mismatch outputs are currently tied to 0 in RTL; the SFR fields are allocated but will always read 0. The aggregate interrupt INT_SIO_ID_MISMATCH will therefore never fire in the current implementation.

5.2.3. LinkReset Error Reporting

When AOU_CORE detects a protocol-level error during activation or deactivation, it asserts O_AOU_REQ_LINKRESET (to FDI) and INT_REQ_LINKRESET (to SW). Both signals share the same net and are the logical OR of the four sub-conditions below. After handling the interrupt, SW must perform a LinkReset sequence.

All LinkReset-related SFR fields reside in LP_LINKRESET (0x0010).

Activate/Deactivate Ack Timeout. After sending an ActivateReq or DeactivateReq message, a counter begins. If the remote die does not return an Acknowledge within $2^{(ACK_TIME_OUT_VALUE + 18)}$ core clock cycles, the timeout fires. ACK_TIME_OUT_VALUE [13:11] is 3 bits wide with a default of 4, giving a default timeout of $2^{22} = \sim 4\text{M}$ cycles. The sticky error bits are ACT_ACK_ERR [7] and DEACT_ACK_ERR [6], both W1C.

Invalid ACTMSG. The activation FSM (AOU_ACTIVATION_CTRL) checks each received activation opcode against the current state. If the opcode is not permitted in the current state, INVALID_ACTMSG_ERR [1] is set and the offending 4-bit opcode is captured in INVALID_ACTMSG_INFO [5:2]. Clear via W1C.

MsgCredit Timeout. After the local die receives ActivateAck, it expects a CreditGrant message from the remote die indicating the number of resource planes. If no CreditGrant arrives within $2^{(MSGCREDIT_TIME_OUT_VALUE + 18)}$ core clock cycles, MSGCREDIT_ERR [0] is set. MSGCREDIT_TIME_OUT_VALUE [10:8] defaults to 4. Clear via W1C.

5.2.4. R/B Response Error Debug FIFO

When the manager interface receives an AXI B or R response with a non-OKAY status (SLVERR or DECERR), AOU_CORE captures the error details in a dedicated per-RP FIFO so that the remote die can retrieve them via AXI read.

FIFO structure. The AOU_ERROR_INFO module (one instance per RP) implements a 4-entry circular FIFO. Each entry stores:

Field	Width	Description
AXI ID	AXI_ID_WD	ID of the failing transaction
AXI ADDR	AXI_ADDR_WD	Address of the failing transaction
RESP	2 bits	BRESP or RRESP value

Both BRESP and RRESP errors can be pushed simultaneously (two entries in one cycle). If the FIFO is full, new errors are silently dropped.

Enabling access. Before the remote die can read error information, SW must:

1. Set AOU_CON0.RPn_ERROR_INFO_ACCESS_EN [27:24] to 1 for the target RP.
2. Configure the dedicated address window by writing the upper 32 bits to AXI_ERROR_INFO0_RPn.DEBUG_UPPER_ADDR and the lower 32 bits to AXI_ERROR_INFO1_RPn.DEBUG_LOWER_ADDR.

The AOU_AXIMUX_1XN_SS address decoder routes any AXI transaction whose address matches {DEBUG_UPPER_ADDR, DEBUG_LOWER_ADDR} to the AOU_ERROR_INFO module instead of the normal datapath.

Read protocol. The remote die issues an AXI read (AR) to the configured debug address. The module returns the FIFO head entry on RDATA (containing the concatenated ID, address, and response fields) with RLAST = 1.

Pop protocol. To advance to the next entry, the remote die issues a dummy AXI write (AW + W) to the same debug address. The simultaneous AW and W handshake advances the FIFO read pointer, discarding the head entry. A B response (OKAY) is returned.

The typical access sequence is: read the head entry, then pop it, and repeat until no more entries remain.

[Figure 23. R/B Response Error Debug FIFO Access Sequence] | *error-debug-fifo-access.svg*

Figure 23. R/B Response Error Debug FIFO Access Sequence

5.2.5. Write Early Response Error

When the early write-response feature is enabled (EARLY_BRESP_EN = 1) and the actual BRESP from the remote die carries an error for a transaction that already received an early OKAY, the error is reported via the INT_EARLY_RESP_ERR interrupt. The failing transaction's AXI ID and BRESP type are captured in

WRITE_EARLY_RESPONSE_RPn. SW clears the error by writing 1 to WRITE_RESP_ERR (W1C).

+For a detailed description of the early response error flow, see +ifndef::backend-pdf[] +[Transaction Flow -- Error Case](#) +and +[SFR Fields](#). +endif::backend-pdf[] +ifndef::backend-html5[] +[Transaction Flow -- Error Case](#) +and +[SFR Fields](#). +endif::backend-html5[]

5.3. Activation & Deactivation Flow

The Activation of AOU_CORE begins after the UCIE Link-up process has been successfully completed. The completion of UCIE link-up is indicated by the **I_INT_FSM_IN_ACTIVE** signal of the **AOU_CORE_TOP**.

There is no dependency between the activation of the local die and the remote die. Each die may initiate the activation sequence independently, which means the timing of sending **ActivateReq** can differ. Both dies must exchange **ActivateReq** and **ActivateAck** messages to transition to the **ENABLED** state, at which point credited messages can be transmitted.

[Figure 24. Independent Activation Flow] | *image23.png*

Figure 24. Independent Activation Flow

Regardless of its own activation state, a die must send an **ActivateAck** in response to an **ActivateReq** from the other die, to acknowledge receipt of the request.

5.3.1. Credit Management Type

There are two types of Credit Management Type. Since current AOU_SPEC cannot resolve pending AXI response without Activate again. When Credit Management type is set to 1, During Deactivate state, AOU_CORE can resolve pending AXI response itself. It can be configured through AOU_CON0.CREDIT_MANAGE, and the default value is 0.

- 0 (default) Based on AoU v0.5
 - After deactivation, if a new request is received from the remote die, no response message can be sent.
 - To deliver the corresponding response, the system must go through activation again after deactivation.
 - Credit management and transmission availability for both Request-related messages and Response-related message are controlled together.
 - Credited messages must not be sent, after the DeactivateReq message is sent.
 - Credits must not be sent after the DeactivateAck message is sent.
 - The Activate Interrupt and Deactivate Interrupt that occur in the process of resuming the exchange of response messages impose mandatory requirement to set Activate and Deactivate SFR.
- 1 Proposal by BOS.
 - When the AoU Activity state is DEACTIVATE, manage Request-related messages(WERQ, RREQ, WDATA) and Response-related messages(RDATA, WRESP) separately.
 - RDATA, WRESP Credited messages can be sent, after the DeactivateReq message is sent.
 - Credits for RDATA, WRESP messages must be sent after the DeactivateAck message is sent.
 - The Deactivate interrupt only provides a hint indicating that the remote die has started activate/deactivate.

5.3.2. Activation start

Activation can be initiated by setting `ACTIVATE_START` SFR: In the current implementation, credits are granted based on the depth of the RX FIFO, so it is required to ensure that all messages in the RX FIFO have been popped before sending the `ActivateReq` message.

Activation START(via SFR `AOU_INIT.ACTIVATE_START` SFR)

- Activation can be triggered by setting `ACTIVATE_START`.
- Activation does not proceed until `I_INT_FSM_IN_ACTIVE` is asserted, since no flits can be transmitted beforehand. Therefore, it is allowed to set the `ACTIVATE_START` SFR before `I_INT_FSM_IN_ACTIVE` is asserted. If it is set before the assertion, the activation process will automatically proceed after `I_INT_FSM_IN_ACTIVE` becomes asserted.
- `AOU_INIT.ACTIVATE_START` SFR is automatically cleared to 0 once the activation is completed and `AOU_ACTIVATE_STATE` transitions to `ENABLED`.

5.3.3. Deactivation start

Deactivation is initiated by setting the `AOU_INIT.DEACTIVATE_START` register. Setting this register does not immediately trigger sending a `DeactivateReq` message. The detailed conditions and sequence for deactivation can be found in the **Deactivation Sequence Flow** section. This section will only describe `AOU_INIT.DEACTIVATE_TIME_OUT_VALUE` SFR.

- When software initiates a `DeactivateReq` by setting the SFR, there may still be outstanding requests that have not yet reached out `AOU_CORE`.
- To handle this safely, a timeout mechanism is implemented to ensure that no valid packets remain in the `AOU_TX_CORE`.
- If software guarantees that all responses to its issued requests have been received before setting the deactivation start SFR, the deactivate `TIME_OUT_VALUE(AOU_INIT.DEACTIVATE_TIME_OUT_VALUE` SFR) can be safely configured to a shorter duration.

The `ACTIVATION_OP` field encodes deactivation messages as follows:

- 2 = `DeactivateReq`
- 3 = `DeactivateAck`

[Figure 25. Deactivation sequence with `DEACTIVATE_START`] | *image24.png*

Figure 25. Deactivation sequence with `DEACTIVATE_START`

5.3.4. Deactivation Sequence Flow

Once the local die issues a `DeactivateReq`, it can no longer provide responses to transactions initiated by the remote die. If the remote die continues to send requests or waits for responses without being notified, it may enter a hang state. To manage this safely, the remote die must take explicit action upon receiving a `DeactivateReq`: 1. Immediately generate an interrupt to the CPU to inform the system software that a `DeactivateReq` has been received and that it must set the `DEACTIVATE_START` SFR.

Although deactivation of the local die and the remote die operate independently, it is essential at the system level to communicate the deactivation & activation state through interrupts. This ensures that system software is explicitly informed of deactivation events and prevents the remote die from continuing to expect response that will never arrive, thereby avoiding hang conditions. This approach guarantees that once deactivation is initiated, both dies can coordinate

the transition into safe and consistent DISABLED state.

If the software on both dies can explicitly coordinate to guarantee that all outstanding transactions have been completed and that no new transactions will be issued, then such an complicated implementation would not be necessary.

Figure 25 illustrates the current activate/deactivate implementation of AOU_CORE.

[Figure 26. CREDIT_MANAGE 1 Deactivate sequence flow] | *image25.png*

Figure 26. CREDIT_MANAGE 1 Deactivate sequence flow

[Figure 27. CREDIT_MANAGE 0 Deactivate sequence flow] | *image26.png*

Figure 27. CREDIT_MANAGE 0 Deactivate sequence flow

This system provides a bus cleany mechanism:

- Subordinate bus cleanly indicates whether the local die has received all responses to the requests it sent to the remote die.
- Manager bus cleanly indicates whether the remote die has received all responses to the requests it sent to the local die.

After the local die sends a DeactivateReq, if the remote die issues a new request and expects a response, the system must re-enter the activation sequence before any response can be provided.

Before sending a new ActivateReq, the Rx FIFO must be completely emptied - that is, all messages must be popped. After deactivation, the credit count is reset, and during the subsequent activation process, credits are advertised based on the RX FIFO depth. Therefore, before sending an ActivateReq, the local die must confirm, as described in the local die's sequence 10.1, that all messages in the RX FIFO have been popped.

To meet satisfy the condition in the local die's sequence 10.1, no backpressure must occur on the local die's MI AW/AR/W channels.

5.3.5. Activation/Deactivation Interrupt

INT_ACTIVATE_START

- When this interrupt is detected, AOU_INIT.ACTIVATE_START SFR must be set to 1.
- When AOU_INIT.ACTIVATE_STATE_ENABLED becomes 1, write 1 to clear the AOU_INIT.INT_ACTIVATE_START W1C SFR.
- An interrupt occurs when AOU_INIT.ACTIVATE_START SFR is not set and any of the following conditions are met:
 1. There is a message to send
 - When AOU_CON0.CREDIT_MANAGE = 0, if a new request arrives from the remote die after the local die has sent a DeactivateReq, this interrupt is asserted because the local die must return a response.
 - When AOU_CON0.CREDIT_MANAGE = 1, this interrupt can also be asserted if a new request arrives after the remote die has sent a DeactivateReq.
 2. There is a response to be received (I_SLV_BUS_CLEANY_COMPLETE == 0)
 - When AOU_CON0.CREDIT_MANAGE = 0, this interrupt can be asserted if the local die issues new AXI requests after the remote die has sent a DeactivateReq.

3. An ActivateReq message is received from the remote die.

INT_DEACTIVATE_START

- When AOU_CON0.CREDIT_MANAGE = 0: if the interrupt is detected, the manager IP must stop sending new request messages and AOU_INIT.DEACTIVATE_START SFR must be set to 1.
- When AOU_CON0.CREDIT_MANAGE = 1: if the interrupt is detected, this serves only as a hint that the remote die intends to deactivate. The local die can continue sending request messages. Once all messages have been sent, AOU_INIT.DEACTIVATE_START SFR must be set. Otherwise, the remote die may end up in a state where it can never send requests again.
- When AOU_INIT.ACTIVATE_STATE_DISABLED becomes 1, write 1 to clear the AOU_INIT.INT_DEACTIVATE_START W1C SFR.
- An interrupt is asserted when AOU_INIT.DEACTIVATE_START SFR is not set and the local die receives a DeactivateReq message from the remote die.

5.3.6. PM entry / LinkReset / LinkDisabled sequence

For PM entry / LinkReset / LinkDisable entry sequence, UCIE_CORE check AOU_CORE state and try to change state. For this sequence, resolving pending AXI transactions is necessary.

Since current AOU SPEC has no way to send AXI response after sending DeactiveReq. CREDIT_MANAGE = 0 (Type 0) is matched with current AOU_SPEC. For this case, SW need to check whether there is pending AXI transaction.

• PM entry SW sequence

1. Write AOU_INIT.DEACTIVATE_START to 1.
2. Polling AOU_INIT.ACTIVATE_STATE_DISABLED and AOU_INIT.SLV_TR_COMPLETE & AOU_INIT.MST_TR_COMPLETE.
 - a. Although AOU_CORE state becomes DISABLED, there may have pending AXI transaction.
 - b. AOU_CORE issue Interrupt for resolving pending AXI transactions.
 - c. Activate AOU_CORE and resolve pending AXI transactions.
 - d. Deactivate AOU_CORE
3. Polling AOU_INIT.ACTIVATE_STATE_DISABLED and AOU_INIT.SLV_TR_COMPLETE & AOU_INIT.MST_TR_COMPLETE.
4. Do PM Entry Sequence on UCIE_CORE.

[Figure 28. PM Entry Sequence] | *image27.png*

Figure 28. PM Entry Sequence

• LinkDisable SW sequence

Same as PM entry, before doing UCIE state transition, AOU_CORE need to Disabled properly.

1. Write AOU_INIT.DEACTIVATE_START to 1.
2. Polling AOU_INIT.ACTIVATE_STATE_DISABLED and AOU_INIT.SLV_TR_COMPLETE & AOU_INIT.MST_TR_COMPLETE.
 - a. Although AOU_CORE state becomes DISABLED, there may have pending AXI transaction.
 - b. AOU_CORE issue Interrupt for resolving pending AXI transactions.

- c. Activate AOU_CORE and resolve pending AXI transactions.
 - d. Deactivate AOU_CORE
3. Polling AOU_INIT.ACTIVATE_STATE_DISABLED and AOU_INIT.SLV_TR_COMPLETE & AOU_INIT.MST_TR_COMPLETE.
 4. Do LinkReset/LinkDisabled Sequence on UCIE_CORE.
 - **LinkReset sequence**

If AOU_CORE faces uncorrectable error (ex. AOU SPEC violation), AOU_CORE sends LinkReset to CPU and D2D adapter. LinkReset indicates that an error has occurred which requires the Link to go down.

While handling LinkReset, SW needs to do AOU_CORE SW reset by setting AOU_CON0.AOU_SW_RESET.

Chapter 6. Key Manager Firmware Reference

This section is the developer reference for the KM ROM firmware that runs on the PicoRV32 and services SEP requests. It documents the SEP-facing command/response API in detail, along with the firmware-internal subsystems that implement it. Use this document when writing SEP firmware that drives the KM mailbox.

The KM firmware is responsible for bringing the subsystem to a secure operational state, servicing SEP mailbox commands, managing key material in the KPV, sideloading keys into approved crypto engines, and handling recoverable and unrecoverable fault conditions. In this revision, the ROM firmware includes the following major functional areas:

- Boot and secure initialization
- Mailbox command/response protocol
- Message receive/transmit buffering
- Command parsing and validation
- Interrupt and fault handling
- DRBG and firmware PRNG support
- KPV management
- Crypto-engine sideload drivers for HMAC, KMAC, AES, OTBN, and Adams Bridge
- CRC support functions
- SP 800-108r1 counter-mode key derivation
- Key registry and key lifecycle management
- Mutable firmware handover (load, lock, scramble, jump)
- Main event loop and runtime state management

The firmware supports the SEP-facing command set for version and status queries, random key generation, key loading, key transfer, key revocation, engine shredding, recoverable-fault acknowledgement, and mutable firmware handover. In this ROM revision, `CMD_SRAM_VER` returns failure unless mutable (SRAM-loaded) firmware is present and executing; because the ROM hands control to the mutable firmware before returning, `CMD_SRAM_VER` is only meaningful when sent to running mutable firmware.

6.1. Firmware Global Parameters

`SHRED_ITER` indicates the number of extra overwrite passes used during shred operations. Any shred sequence therefore performs `SHRED_ITER + 1` total passes.

6.2. Command Message Reference

The SEP communicates with the KM through a framed command/response protocol carried over the mailbox FIFOs. Every direct SEP command elicits a `RESP_CMD` response from the KM. The KM may also emit unsolicited responses, including `RESP_KM_READY`, `RESP_RECOVERABLE_FAULT`, and `RESP_UNRECOVERABLE_FAULT`.

All reserved fields read as 0 and must be written as 0. The maximum payload length for a single message is 255 32-bit words (1020 bytes). The message header CRC and payload CRC are validated independently before command execution. Header validation, sequence checking, command decoding, payload-length checking, and payload CRC checking all

occur before a command handler is allowed to act on the payload.

Packet diagrams show each 32-bit word from bit 31 (most significant) on the left to bit 0 (least significant) on the right. Field widths are proportional to their bit counts; reserved fields are shaded grey. Word offsets start at zero within the message or payload shown. In the message layouts, *n* is `PAYLOAD_LEN`.

All command messages use the following container format.

[Key Manager command-message format](#) shows the command header, payload words, and checksum placement.

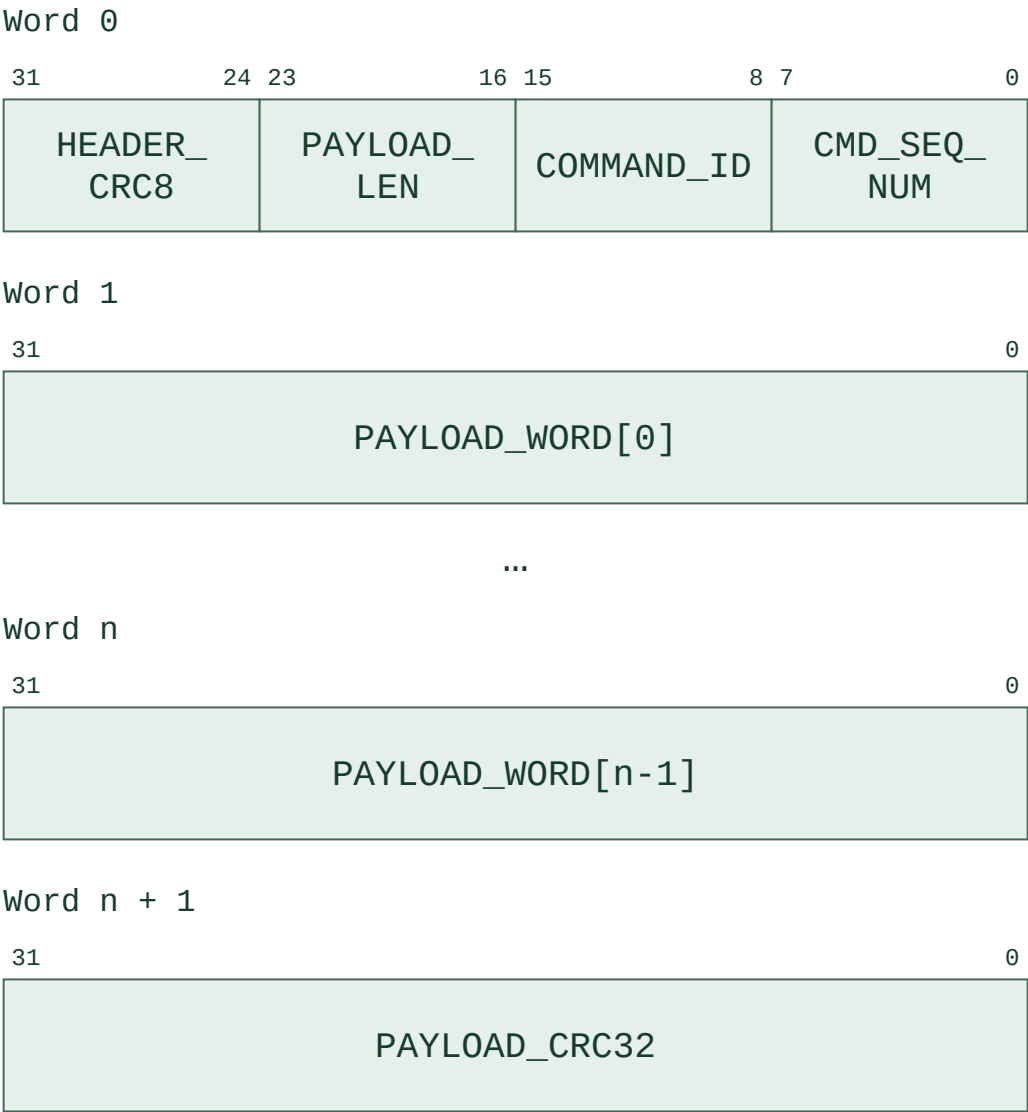


Figure 10. Key Manager command-message format

[Key Manager command-message fields](#) defines the command container fields.

Table 31. Key Manager command-message fields

Field	Description
<code>CMD_SEQ_NUM</code>	The command message sequence number. It is reset to zero during initialization or during any message buffer flush. This value must increment with each message. It must roll over to 0 after reaching the maximum value.
<code>COMMAND_ID</code>	The ID of the command for the KM to execute. This determines how the <code>PAYLOAD_WORD</code> s are interpreted.

Field	Description
PAYLOAD_LEN	The length of the message payload in number of 32-bit words, <i>not</i> including the PAYLOAD_CRC32. A payload length of 0 indicates the message has no payload.
HEADER_CRC8	A CRC-8/ROHC computed over the 24-bit message header {PAYLOAD_LEN, COMMAND_ID, CMD_SEQ_NUM} in little endian order, starting with CMD_SEQ_NUM.
PAYLOAD_WORD[n]	A 32-bit data word for the command message payload, where <i>n</i> is always less than PAYLOAD_LEN. PAYLOAD_LEN equal to 0 implies no PAYLOAD_WORD s are present. The contents of the PAYLOAD_WORD s are command-specific.
PAYLOAD_CRC32	A CRC-32C (Castagnoli) computed from the command payload in little endian order from PAYLOAD_WORD[0] to PAYLOAD_WORD[n-1]. PAYLOAD_LEN equal to 0 implies no PAYLOAD_CRC32 is present.

The rest of this section describes all valid commands and their payloads.

[Key Manager command summary](#) lists the command IDs and their functions.

Table 32. Key Manager command summary

ID	Command	Purpose
0x00	CMD_HW_VER	Request KM hardware version.
0x01	CMD_ROM_VER	Request KM ROM firmware version.
0x02	CMD_SRAM_VER	Request KM SRAM firmware version (handled by mutable firmware once loaded).
0x03	CMD_STAT	Request KM status information.
0x04	CMD_RECOV_ACK	Acknowledge a recoverable fault.
0x10	CMD_EXEC_ROM	Commit KM to ROM execution; disable all subsequent handover commands.
0x11	CMD_SRAM_LOAD_EXEC	Stream a firmware image from the mailbox, load to SRAM, and execute.
0x12	CMD_SRAM_EXEC	Jump to previously-loaded mutable firmware in SRAM.
0x22	CMD_KEY_GENERATE	Generate a random key.
0x23	CMD_KEY_REVOKE	Revoke an existing key.
0x24	CMD_KEY_TRANSFER	Transfer a key to crypto engines.
0x25	CMD_ENGINE_SHRED	Purge crypto engine key registers.
0x26	CMD_KEY_LOAD	Load host-provided key material into the KPV.
0x27	CMD_ABR_SK_TRANSFER	Capture the Adams Bridge ML-KEM shared key into the KPV.
0x28	CMD_OTP_READ_LOCK_COLD	Set cold-reset-domain OTP field read-lock bits (survive warm reset).

6.2.1. 0x00 - CMD_HW_VER

Description

Requests KM hardware version information.

Payload

This command has no payload.

Response

This command has a RESP_CMD response with a return code of *success*, and a return argument containing the version of the hardware.

[Key Manager hardware-version response](#) shows the hardware-version fields returned on success.

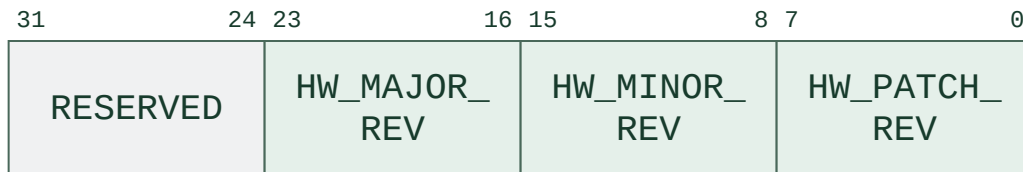


Figure 11. Key Manager hardware-version response

6.2.2. 0x01 - CMD_ROM_VER

Description

Requests KM ROM firmware version information.

Payload

This command has no payload.

Response

This command has a **RESP_CMD** response with a return code of *success*, and a return argument containing the version of the ROM firmware.

[Key Manager ROM-version response](#) shows the ROM-version fields returned on success.

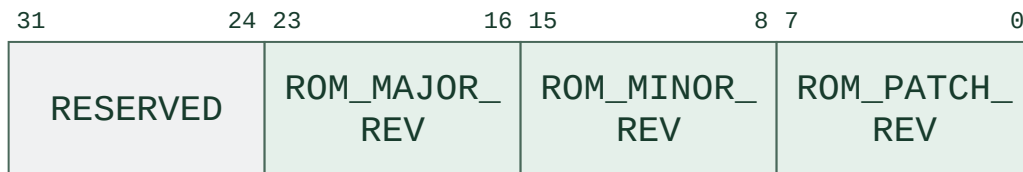


Figure 12. Key Manager ROM-version response

6.2.3. 0x02 - CMD_SRAM_VER

Description

Requests KM SRAM firmware version information (if any).

Mutable SRAM firmware may now be loaded via **CMD_SRAM_LOAD_EXEC**. If the KM has jumped to mutable firmware (via either **CMD_SRAM_LOAD_EXEC** or **CMD_SRAM_EXEC**), the mutable firmware is responsible for handling this command and returning its own version information. While still in ROM, the ROM itself handles this command and returns *failure* with no return argument, indicating no SRAM firmware is currently active.

Payload

This command has no payload.

Response

If no SRAM firmware is loaded, this command has a **RESP_CMD** response with a return code of *failure*, and no return argument.

If SRAM firmware is loaded and executing, this command has a **RESP_CMD** response with a return code of *success*, and a return argument containing the version of the SRAM firmware.

[Key Manager SRAM-version response](#) shows the SRAM firmware-version fields returned on success.



Figure 13. Key Manager SRAM-version response

6.2.4. 0x03 - CMD_STAT

Description

Requests KM status information.

Payload

This command has no payload.

Response

This command has a **RESP_CMD** response with a return code of *success*, and a return argument containing the KM status information.

[Key Manager status response](#) shows the status word returned by CMD_STAT.

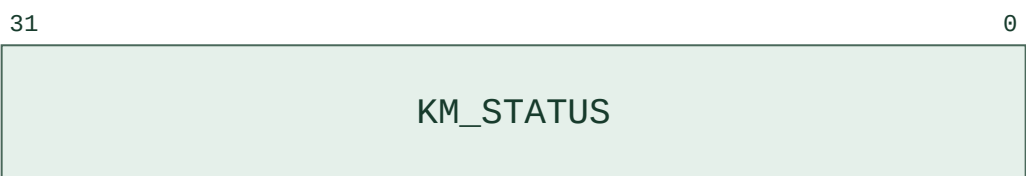


Figure 14. Key Manager status response

[Key Manager status fields](#) defines the recoverable-fault flag and reserved status bits.

Table 33. Key Manager status fields

Bit	Name	Description
0	recov_fault	The value of the RECOVERABLE_ERR register bit. Set to 1 if the KM is waiting for the SEP to acknowledge a recoverable fault.
[31:1]	reserved	All other fields reserved.

6.2.5. 0x04 - CMD_RECOV_ACK

Description

Acknowledges a recoverable error event in the KM. This command clears the `RECOVERABLE_ERR` bit in the KMCSR.

Payload

This command has no payload.

Response

This command has a `RESP_CMD` response with a return code of *success*, and no return argument.

6.2.6. 0x10 - CMD_EXEC_ROM

Description

Commits the KM to continued ROM execution for the lifetime of the current warm-reset epoch. Once this command succeeds, all subsequent `CMD_EXEC_ROM`, `CMD_SRAM_LOAD_EXEC`, and `CMD_SRAM_EXEC` commands return *failure* without taking any action. The inhibit flag is cleared on every warm or cold reset, so the SEP may choose again after reboot.

Payload

This command has no payload.

Response

This command has a `RESP_CMD` response with a return code of *success* on first call, or *failure* if a handover command has already been issued in this epoch (including a prior `CMD_EXEC_ROM`).

6.2.7. 0x11 - CMD_SRAM_LOAD_EXEC

Description

Instructs the KM ROM to accept a mutable firmware image via the mailbox FIFO, load it into SRAM starting at address `0x8000`, verify the image integrity with CRC-32C, and hand control to the loaded firmware.

After sending the command, the SEP waits for the KM to confirm that it is ready to receive the image. The KM sends a `RESP_CMD` success response ("confirmation"), then immediately disables inbound mailbox buffering and waits for the image frame. The SEP must then send exactly one mailbox frame containing the `FW_WORDS` firmware words followed by a CRC-32C of the entire image as the final word. The frame separator must be set on the CRC word.

If the CRC check fails, the KM triggers an unrecoverable fault (`fw_crc` fault code) and halts. Recovery requires a warm or cold reset.

If the CRC check passes, the KM:

1. Shreds every crypto engine sideload key-share register and zeroizes the Adams Bridge ML-KEM shared key. These banks lie outside SRAM, so the scramble below cannot reach them.
2. Locks OTP secret fields and the `rom_persist` SRAM region, and closes the KPV to mutable firmware by read-locking (`lock_use`) every registered slot and shredding every free one.

3. Write-locks the SRAM regions containing the firmware image.
4. Disables all IRQ sources, masks all CPU IRQs, and flushes both mailbox FIFOs to erase residual inbound payload. ROM does not program `IRQ_ENTRY_ADDR`; the loaded firmware sets its own IRQ entry vector before re-enabling interrupts (`IRQ_ENTRY_LOCK` is left unset so it may do so).
5. Scrambles the entire SRAM with a pseudo-random pattern (locked regions are silently skipped by hardware).
6. Clears all CPU general-purpose registers.
7. Jumps to `0x8000` (SRAM base).

The ROM never returns to the main loop. The mutable firmware is responsible for sending its own ready message to the SEP.

Notes

- If the handover-disabled flag is set (from a prior `CMD_EXEC_ROM`), the command returns *failure* immediately and the image is not requested.
- `FW_WORDS` must be non-zero and the loaded image must fit entirely below the static load ceiling `_km_fw_load_limit` (the worst-case ROM stack bound, region-aligned to a 1 KB SRAM write-lock boundary), otherwise `_invalid_arg` is returned and no image is requested. The ceiling is a build-time constant, so a warm-reset `CMD_SRAM_EXEC` restart whose ROM stack runs deeper than at load time can never descend into the loaded image.
- The two execution paths (`CMD_SRAM_LOAD_EXEC` and `CMD_SRAM_EXEC`) both call `rom_handover_finish()`. The stack-less assembly handoff (`rom_handover_jump.S`) runs after all C-level operations complete. This ordering is required because the assembly wipes the C stack.

Payload

[Key Manager SRAM-load command payload](#) shows the image-size word sent with `CMD_SRAM_LOAD_EXEC`.

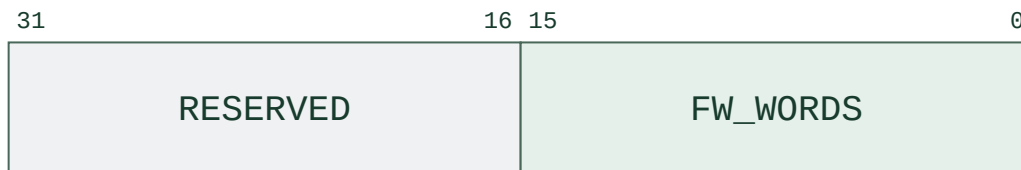


Figure 15. Key Manager SRAM-load command payload

[Key Manager SRAM-load payload fields](#) defines the firmware word count and reserved bits.

Table 34. Key Manager SRAM-load payload fields

Field	Name	Description
[31:16]	RESERVED	Must be written as zero.
[15:0]	FW_WORDS	Firmware image size in 32-bit words. Does <i>not</i> include the trailing CRC-32C word appended by the SEP. Must be non-zero.

Image Frame

After receiving the `RESP_CMD` confirmation, the SEP sends exactly one mailbox frame with the following structure ([Key Manager SRAM image frame](#)).

Table 35. Key Manager SRAM image frame

Word	Content
0 .. FW_WORDS-1	Firmware image words, in order.
FW_WORDS	CRC-32C of the image (words 0 to FW_WORDS-1), in little-endian byte order. The mailbox separator bit must be set on this final word.

Response

If validation succeeds, the KM sends a **RESP_CMD** success response (the "confirmation") directly to the outbound FIFO before reading the image. This response has no return argument. The KM never returns to the dispatcher after sending this response; no second **RESP_CMD** is emitted.

If validation fails (handover disabled, zero FW_WORDS, reserved bits set, or image exceeds the stack guard region), a **RESP_CMD** error response is sent by the normal dispatcher path.

6.2.8. 0x12 - CMD_SRAM_EXEC

Description

Instructs the KM ROM to jump to previously-loaded mutable firmware stored in SRAM. The firmware must have been loaded by a prior **CMD_SRAM_LOAD_EXEC** command in an earlier boot epoch (i.e., `sram_fw_size > 0` in the `rom_persist` region).

The KM sends a **RESP_CMD** success response directly, then executes the same handover finish sequence as **CMD_SRAM_LOAD_EXEC** (lock sensitive data, lock SRAM firmware regions, scramble, clear GPRs, jump). The mutable firmware is responsible for sending its own ready message to the SEP.

Payload

This command has no payload.

Response

If a firmware image is present (`sram_fw_size > 0`), the KM sends a **RESP_CMD** success response directly and then never returns.

If no firmware has been loaded (`sram_fw_size == 0`), the command returns *invalid_arg* via the normal dispatcher.

If the handover-disabled flag is set, the command returns *failure*.

6.2.9. 0x22 - CMD_KEY_GENERATE

Description

Requests a freshly generated random key.

Payload

The size of the requested key in multiples of 32-bit words (equal to `REQ_SIZE+1`), as well as the valid destinations for the key.

[Key-generation command payload](#) shows the requested size and destination-policy words for **CMD_KEY_GENERATE**.

Word 0



Word 1



Figure 16. Key-generation command payload

[Key-generation destination policy](#) defines the permitted-destination bits for a generated key.

Table 36. Key-generation destination policy

Bit	Name	Description
0	<code>hmac_sha2</code>	Key transfer to the HMAC engine is permitted when set to 1.
1	<code>kmac_sha3</code>	Key transfer to the KMAC engine is permitted when set to 1.
2	<code>aes</code>	Key transfer to the AES engine is permitted when set to 1.
3	<code>otbn</code>	Key transfer to the OTBN engine is permitted when set to 1.
4	<code>abr_mldsa_seed</code>	Key transfer to the Adams Bridge ML-DSA seed is permitted when set to 1.
5	<code>abr_mlkem_seed_d</code>	Key transfer to the Adams Bridge ML-KEM seed D is permitted when set to 1.
6	<code>abr_mlkem_seed_z</code>	Key transfer to the Adams Bridge ML-KEM seed Z is permitted when set to 1.
7	<code>abr_mlkem_msg</code>	Key transfer to the Adams Bridge ML-KEM message is permitted when set to 1.

Response

If any arguments are invalid, this command has a `RESP_CMD` response with a return code of `invalid_arg`, and a return argument pointing to the index of the invalid payload word: 0 for `REQ_SIZE`, 1 for `DEST_VALID`.

`REQ_SIZE` is a 7-bit field, so the largest key it can request is 128 words. Any payload word greater than 127 is rejected as `invalid_arg` with `return_arg` = 0, including one whose reserved bits [31:7] are non-zero.

The KM will respond with `failure` if the KPV does not have `REQ_SIZE+1` number of consecutive free slots available to grant.

If the request is granted, this command has a `RESP_CMD` response with a return code of `success`, and a return argument containing the parameters of the generated key, including its handle, its size, and its valid destinations. The granted size must always be equivalent to the requested size.

[Key-generation success response](#) shows the generated key handle, size, and destination policy returned on success.



R = RESERVED [15]

Figure 17. Key-generation success response

6.2.10. 0x23 - CMD_KEY_REVOKE

Description

This command will revoke an existing key from the KPV, given a valid key handle.

Payload

The handle for the key to invalidate.

[Key-revoke command payload](#) shows the key handle sent with CMD_KEY_REVOKE.



Figure 18. Key-revoke command payload

Response

If any arguments are invalid, this command has a **RESP_CMD** response with a return code of *invalid_arg*, and a return argument pointing to the index of the invalid payload word.

If the request cannot be satisfied for any other reason, this command has a **RESP_CMD** response with a return code of *failure*, and no return argument.

If the request is granted, this command has a **RESP_CMD** response with a return code of *success*, and a return argument containing the handle of the revoked key.

[Key-revoke success response](#) shows the revoked key handle returned on success.



Figure 19. Key-revoke success response

6.2.11. 0x24 - CMD_KEY_TRANSFER

Description

This command transfers a key from the KPV to one or more crypto engines, given a valid key handle and a bitfield of requested destinations. Requested destination crypto engines must be permitted by the destination-policy bitmask stored for the key handle in the key registry.

Payload

The handle of the key to transfer in the KPV and the bitmask of the destination crypto engines.

[Key-transfer command payload](#) shows the key handle and requested destination-engine mask.

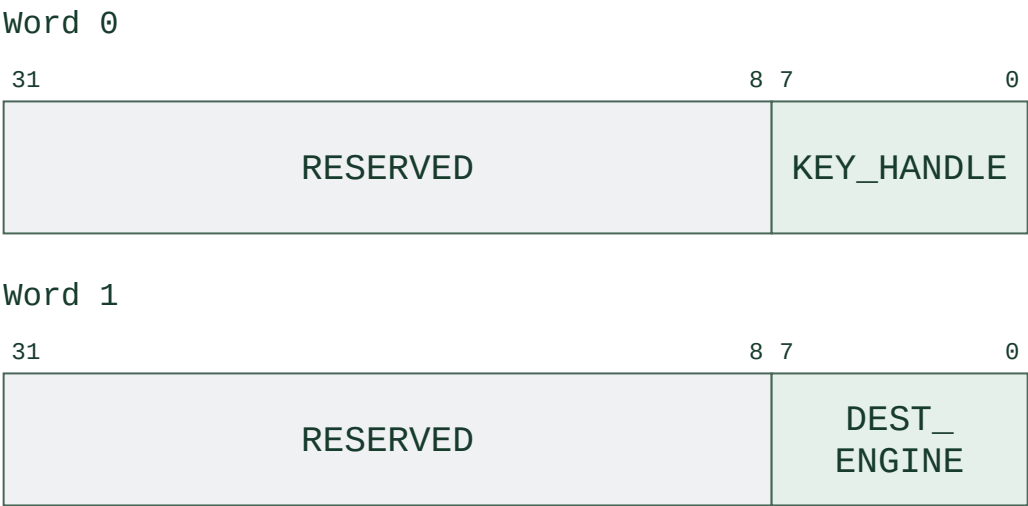


Figure 20. Key-transfer command payload

[Key-transfer requested engines](#) defines the destination-engine bits in the transfer request.

Table 37. Key-transfer requested engines

Bit	Name	Description
0	hmac_sha2	Key transfer to the HMAC engine is requested when set to 1.
1	kmac_sha3	Key transfer to the KMAC engine is requested when set to 1.
2	aes	Key transfer to the AES engine is requested when set to 1.
3	otbn	Key transfer to the OTBN engine is requested when set to 1.
4	abr_mlds_a_seed	Key transfer to the Adams Bridge ML-DSA seed is requested when set to 1.
5	abr_mlkem_seed_d	Key transfer to the Adams Bridge ML-KEM seed D is requested when set to 1.
6	abr_mlkem_seed_z	Key transfer to the Adams Bridge ML-KEM seed Z is requested when set to 1.
7	abr_mlkem_msg	Key transfer to the Adams Bridge ML-KEM message is requested when set to 1.

Response

If any arguments are invalid, this command has a **RESP_CMD** response with a return code of *invalid_arg*, and a return argument pointing to the index of the invalid payload word.

If the request cannot be satisfied for any other reason, this command has a **RESP_CMD** response with a return code of *failure*, and no return argument. If *any* requested crypto engines are disallowed, the request will result in *failure* and no requested key transfers will take place.

If the request is granted, this command has a **RESP_CMD** response with a return code of *success*, and a return argument containing the key handle and a bitfield of the crypto engines to which the key was transferred.

[Key-transfer success response](#) shows the key handle and completed-engine mask returned on success.

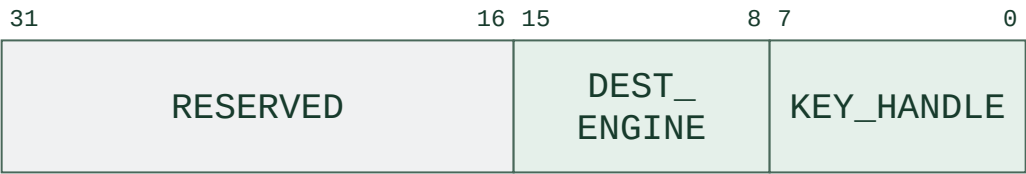


Figure 21. Key-transfer success response

[Key-transfer completed engines](#) defines the destination-engine bits in the transfer response.

Table 38. Key-transfer completed engines

Bit	Name	Description
0	hmac_sha2	Key transfer to the HMAC engine was completed when set to 1.
1	kmac_sha3	Key transfer to the KMAC engine was completed when set to 1.
2	aes	Key transfer to the AES engine was completed when set to 1.
3	otbn	Key transfer to the OTBN engine was completed when set to 1.
4	abr_mldsa_seed	Key transfer to the Adams Bridge ML-DSA seed was completed when set to 1.
5	abr_mlkem_seed_d	Key transfer to the Adams Bridge ML-KEM seed D was completed when set to 1.
6	abr_mlkem_seed_z	Key transfer to the Adams Bridge ML-KEM seed Z was completed when set to 1.
7	abr_mlkem_msg	Key transfer to the Adams Bridge ML-KEM message was completed when set to 1.

6.2.12. 0x25 - CMD_ENGINE_SHRED

Description

This command will purge the key data registers from one or multiple crypto engines. These registers will be scrambled with random data. Any number of engines may be selected.

Payload

A bitfield indicating the crypto engines to purge.

[Key-purge command payload](#) shows the engine mask sent with the key-purge request.

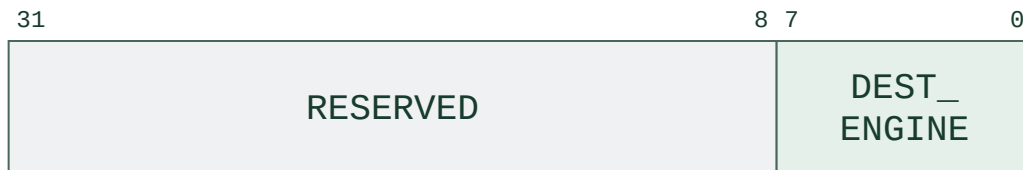


Figure 22. Key-purge command payload

[Key-purge requested engines](#) defines the bits selecting the engines whose key registers are to be purged.

Table 39. Key-purge requested engines

Bit	Name	Description
0	hmac_sha2	Key purging of the HMAC engine is requested when set to 1.
1	kmac_sha3	Key purging of the KMAC engine is requested when set to 1.
2	aes	Key purging of the AES engine is requested when set to 1.
3	otbn	Key purging of the OTBN engine is requested when set to 1.
4	abr_mldsa_seed	Key purging of the Adams Bridge ML-DSA seed is requested when set to 1.
5	abr_mlkem_seed_d	Key purging of the Adams Bridge ML-KEM seed D is requested when set to 1.
6	abr_mlkem_seed_z	Key purging of the Adams Bridge ML-KEM seed Z is requested when set to 1.
7	abr_mlkem_msg	Key purging of the Adams Bridge ML-KEM message is requested when set to 1.

Response

If any arguments are invalid, this command has a **RESP_CMD** response with a return code of *invalid_arg*, and a return argument pointing to the index of the invalid payload word.

If the request cannot be satisfied for any other reason, this command has a **RESP_CMD** response with a return code of *failure*, and no return argument.

If the request is granted, this command has a **RESP_CMD** response with a return code of *success*, and a return argument indicating the crypto engines that were purged.

[Key-purge success response](#) shows the engine mask returned after a successful purge.



Figure 23. Key-purge success response

[Key-purge completed engines](#) defines the bits reporting which engines were purged.

Table 40. Key-purge completed engines

Bit	Name	Description
0	hmac_sha2	Key material of the HMAC engine was purged when set to 1.

Bit	Name	Description
1	kmac_sha3	Key material of the KMAC engine was purged when set to 1.
2	aes	Key material of the AES engine was purged when set to 1.
3	otbn	Key material of the OTBN engine was purged when set to 1.
4	abr_mldsa_seed	Key material of the Adams Bridge ML-DSA seed was purged when set to 1.
5	abr_mlkem_seed_d	Key material of the Adams Bridge ML-KEM seed D was purged when set to 1.
6	abr_mlkem_seed_z	Key material of the Adams Bridge ML-KEM seed Z was purged when set to 1.
7	abr_mlkem_msg	Key material of the Adams Bridge ML-KEM message was purged when set to 1.

6.2.13. 0x26 - CMD_KEY_LOAD

Description

This command loads and write-locks host-provided key material into the KPV.

Payload

The SEP must provide the key size in multiples of 32-bit words (where the key size is `KEY_SIZE+1`) and the valid destinations for the key, followed by the key material itself.

[Key-load command payload](#) shows the size, destination policy, and key words sent with CMD_KEY_LOAD.

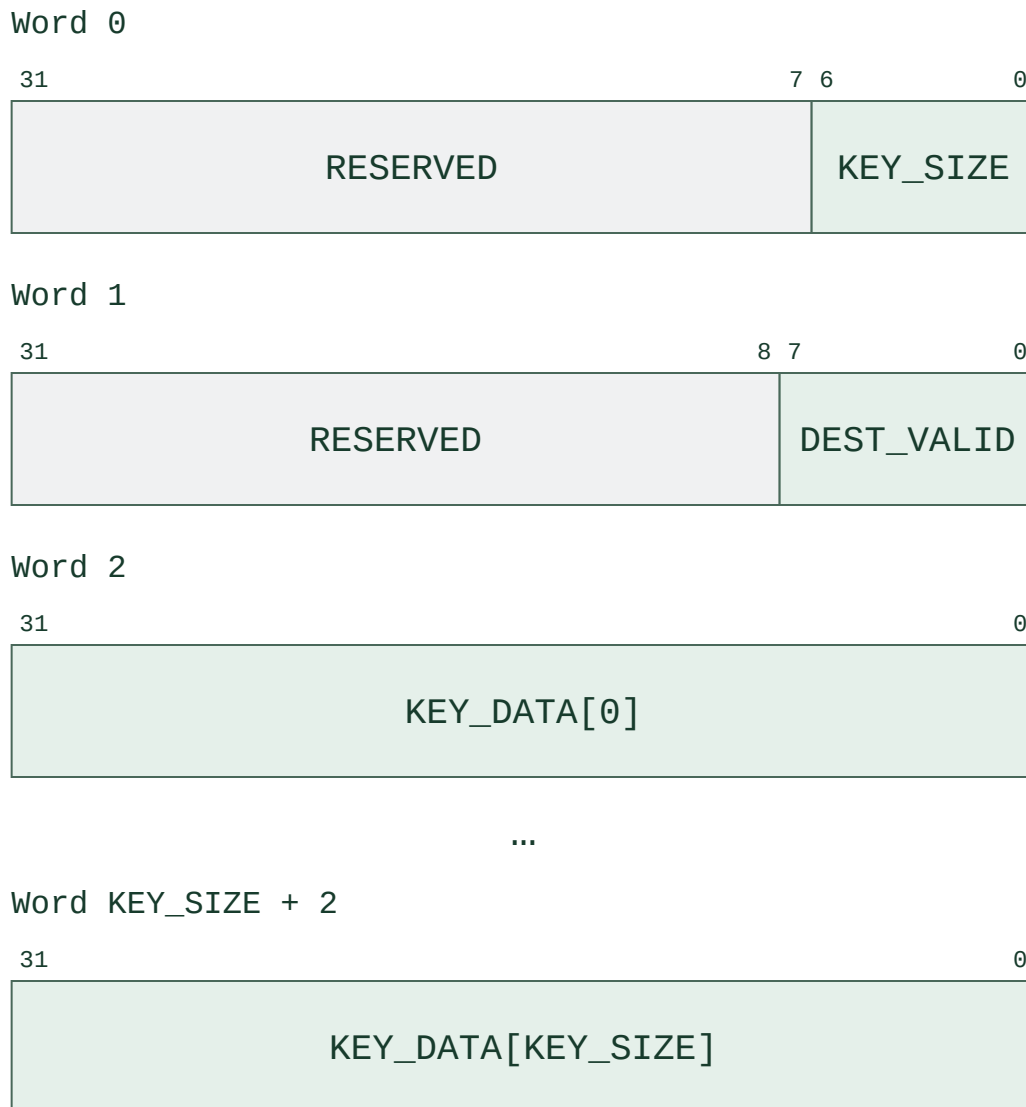


Figure 24. Key-load command payload

[Key-load destination policy](#) defines the permitted-destination bits for a loaded key.

Table 41. Key-load destination policy

Bit	Name	Description
0	<code>hmac_sha2</code>	Key transfer to the HMAC engine is permitted when set to 1.
1	<code>kmac_sha3</code>	Key transfer to the KMAC engine is permitted when set to 1.
2	<code>aes</code>	Key transfer to the AES engine is permitted when set to 1.
3	<code>otbn</code>	Key transfer to the OTBN engine is permitted when set to 1.
4	<code>abr_mlds_a_seed</code>	Key transfer to the Adams Bridge ML-DSA seed is permitted when set to 1.
5	<code>abr_mlkem_seed_d</code>	Key transfer to the Adams Bridge ML-KEM seed D is permitted when set to 1.
6	<code>abr_mlkem_seed_z</code>	Key transfer to the Adams Bridge ML-KEM seed Z is permitted when set to 1.
7	<code>abr_mlkem_msg</code>	Key transfer to the Adams Bridge ML-KEM message is permitted when set to 1.

On-wire payload integrity is provided by the standard mailbox `PAYLOAD_CRC32` trailer; there is no command-specific key CRC word. After the key is accepted, the firmware computes its own CRC-32C over `KEY_DATA` and stores it in the key

registry for later integrity checks (see [Key Registry](#)).

Argument validation

The KM validates the payload header words in a strict, fixed order and returns `RESP_CMD` with return code `invalid_arg` at the first failure. The return argument is the fixed header-word index of the offending field (0 for `KEY_SIZE` / length, 1 for `DEST_VALID`), not an arbitrary `KEY_DATA` position.

[CMD_KEY_LOAD Validation Order](#) lists the validation checks in order and the error returned at each failure.

Table 42. CMD_KEY_LOAD Validation Order

Step	Check	Return Code	Return Arg
1	PAYLOAD_LEN < 3 (cannot parse KEY_SIZE/DEST_VALID)	invalid_arg	0
2	KEY_SIZE reserved bits set ([31:7] != 0)	invalid_arg	0
3	PAYLOAD_LEN != KEY_SIZE + 3 (length inconsistent)	invalid_arg	0
4	DEST_VALID reserved bits set ([31:8])	invalid_arg	1
5	DEST_VALID == 0 (no destination selected)	invalid_arg	1

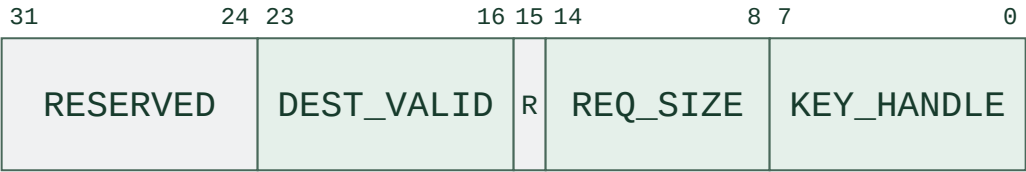
Capacity failure

After argument validation, the KM checks that the KPV has `KEY_SIZE+1` eligible consecutive free slots and that at least one key handle is free before mutating any KPV state. If either resource is exhausted, the command returns `RESP_CMD failure` with no return argument. A *failure* response is also returned, before the handler runs, whenever `RECOVERABLE_ERR` is set in the KMCSR. In every *failure* case the KPV state and key registry are left exactly as they were before the command arrived; the load is atomic.

Success

If the request is granted, the KM shreds the chosen slots, writes the `KEY_DATA`, write-locks the slots, stores the key CRC-32C, and allocates a new non-zero handle. The command responds with `RESP_CMD success` and a return argument containing the parameters of the loaded key: its handle, its size, and its valid destinations. The granted size always equals the requested size, and the returned handle is usable by every other key-management command with no distinction from a `CMD_KEY_GENERATE` handle.

[Key-load success response](#) shows the loaded key handle, size, and destination policy returned on success.



R = RESERVED [15]

Figure 25. Key-load success response

6.2.14. 0x27 - CMD_ABR_SK_TRANSFER

Description

Captures the Adams Bridge ML-KEM shared key into the KPV. Adams Bridge delivers a freshly computed ML-KEM shared key into the KM over the shared-key side-load interface and raises the shared-key interrupt; the KM firmware notifies the SEP with an unsolicited `RESP_ABR_SHARED_KEY_READY` (see [0x56 - RESP_ABR_SHARED_KEY_READY](#)). The SEP then issues this command to direct the KM to consume the pending shared key and store it as a managed KPV key.

The KM reads the 256-bit (8-word) shared key, clears the interface `KEY_VALID` (which zeroizes the source key registers), writes the key into a freshly allocated KPV slot, records the SEP-supplied `DEST_VALID` in the key registry, write-locks the slot, and returns a key handle usable by every other key-management command.

Payload

The valid crypto engine destinations for the captured key.

[Adams Bridge shared-key capture payload](#) shows the destination policy supplied for the captured shared key.



Figure 26. Adams Bridge shared-key capture payload

[Adams Bridge shared-key destination policy](#) defines the permitted destinations for the captured shared key.

Table 43. Adams Bridge shared-key destination policy

Bit	Name	Description
0	hmac_sha2	Key transfer to the HMAC engine is permitted when set to 1.
1	kmac_sha3	Key transfer to the KMAC engine is permitted when set to 1.
2	aes	Key transfer to the AES engine is permitted when set to 1.
3	otbn	Key transfer to the OTBN engine is permitted when set to 1.
4	abr_mldsa_seed	Key transfer to the Adams Bridge ML-DSA seed is permitted when set to 1.
5	abr_mlkem_seed_d	Key transfer to the Adams Bridge ML-KEM seed D is permitted when set to 1.
6	abr_mlkem_seed_z	Key transfer to the Adams Bridge ML-KEM seed Z is permitted when set to 1.
7	abr_mlkem_msg	Key transfer to the Adams Bridge ML-KEM message is permitted when set to 1.

Response

The KM validates the payload in a fixed order and returns `RESP_CMD` with return code `invalid_arg` (return argument 0, the sole payload word index) if `DEST_VALID` has any reserved bit set ([31:8]) or is zero.

If no shared key is currently available (the interface `KEY_VALID` is clear), the command returns `RESP_CMD failure` with no return argument. A `failure` response is also returned if the KPV cannot allocate a slot for the key.

If the request is granted, the command returns `RESP_CMD success` with a return argument containing the handle of the newly stored key.

[Adams Bridge shared-key capture response](#) shows the newly allocated key handle returned on success.



Figure 27. Adams Bridge shared-key capture response

6.2.15. 0x28 - CMD_OTP_READ_LOCK_COLD

Description

Sets one or more cold-reset-domain OTP field read-lock bits in the `OTP_READ_LOCK_COLD` register. The effective read-lock for each OTP field is the OR of the corresponding bit in `OTP_READ_LOCK` (warm reset domain) and `OTP_READ_LOCK_COLD` (cold reset domain). When either lock bit is set, hardware returns zero for that field's value and complement words.

Unlike `OTP_READ_LOCK`, bits set by this command survive warm reset (including soft reset) and are cleared only by cold reset. This allows the SEP host to apply permanent read-locks that persist for the lifetime of a power cycle, regardless of subsequent warm resets.

Payload

A bitmask selecting which OTP fields to lock. Bits `[31:9]` are reserved and must be zero; a non-zero reserved field returns `invalid_arg`. Writing 0 is a no-op; already-set bits are unaffected (write-one-to-set, idempotent).

The command takes exactly one payload word. A frame carrying no payload word returns `invalid_len` rather than `invalid_arg`, because the length check precedes any inspection of the argument.

[OTP cold-read-lock command payload](#) shows the lock bits sent with `CMD_OTP_READ_LOCK_COLD`.



Figure 28. OTP cold-read-lock command payload

[OTP cold-read-lock bit encoding](#) maps each payload bit to the OTP field whose reads it locks.

Table 44. OTP cold-read-lock bit encoding

Bit	Field	Description
0	LC	Lock <code>OTP_LIFE_CYCLE</code> reads.
1	D	Lock <code>OTP_DEMOTION_STATE</code> reads.
2	cU	Lock <code>OTP_CHIPLET_UID_VAL/CPL</code> reads.
3	SiU	Lock <code>OTP_SIP_UID_VAL/CPL</code> reads.
4	SU	Lock <code>OTP_SYS_UID_VAL/CPL</code> reads.
5	CK	Lock <code>OTP_CLASS_KEY_VAL/CPL</code> reads.

Bit	Field	Description
6	cI	Lock <code>OTP_SEP_CHIPLET_ID_VAL/CPL</code> reads.
7	SiI	Lock <code>OTP_SEP_SIP_ID_VAL/CPL</code> reads.
8	SsI	Lock <code>OTP_SEP_SYS_ID_VAL/CPL</code> reads.

The three identity fields (**cI**, **SiI**, **SsI**) hold per-owner public values, so locking them conceals nothing. They carry locks so that the OTP block stays regular and so that firmware has the same means to gate their re-read as it has for the secret UID fields.

Response

If any reserved bit `[31:9]` is set, the command returns `RESP_CMD invalid_arg` with `return_arg` = the raw `lock_bits` word received.

On success the command applies the lock bits (triple-write convention) and returns `RESP_CMD success` with `return_arg` = the resulting `OTP_READ_LOCK_COLD` register value after the write.

[OTP cold-read-lock success response](#) shows the resulting lock-register value returned on success.

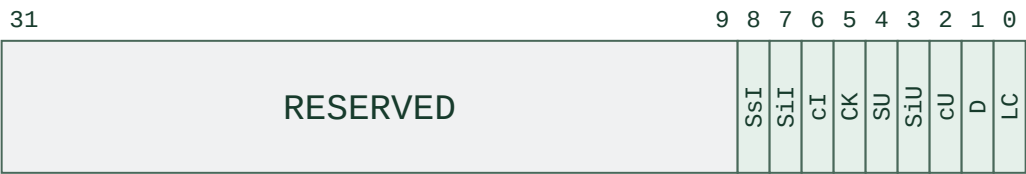


Figure 29. OTP cold-read-lock success response

6.3. Response Message Reference

[Key Manager response-message format](#) shows the response header, payload, and checksum placement.

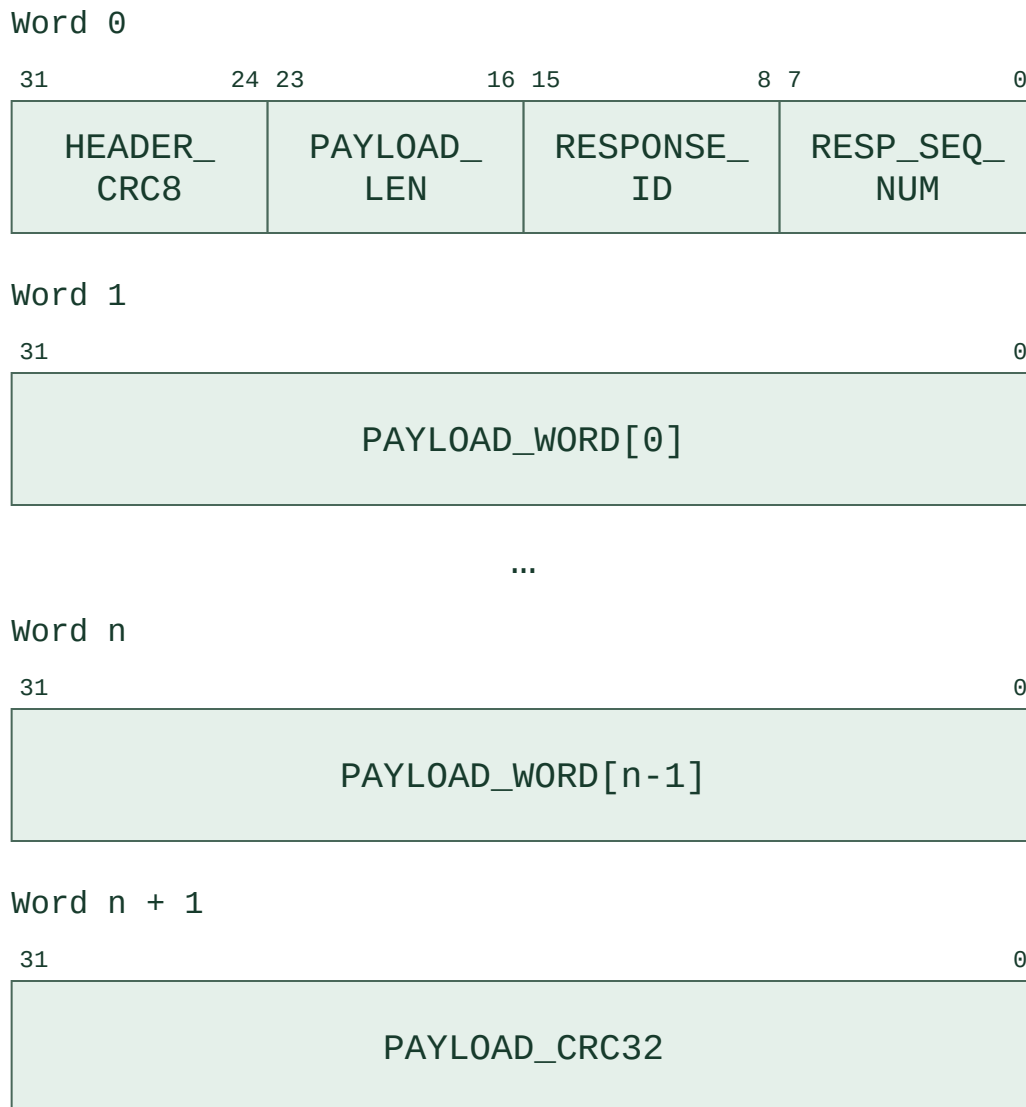


Figure 30. Key Manager response-message format

The KM sends response messages to the SEP using the exact same basic message container format as command messages.

The sequence numbers of response messages do not correspond to the sequence numbers of command messages. There is also no direct relationship between command IDs and response IDs.

[Key Manager response summary](#) lists the response IDs and the events that generate them.

Table 45. Key Manager response summary

ID	Response	Trigger
0x00	RESP_CMD	Reply to every direct SEP command.
0x55	RESP_KM_READY	Unsolicited; KM finished init and entered main loop.
0x56	RESP_ABR_SHARED_KEY_READY	Unsolicited; Adams Bridge ML-KEM shared key is available to capture.
0xFE	RESP_RECOVERABLE_FAULT	Unsolicited; KM hit a recoverable fault.
0xFF	RESP_UNRECOVERABLE_FAULT	Unsolicited; KM hit an unrecoverable fault.

6.3.1. 0x00 - RESP_CMD

Description

A general command response message. The key manager responds to all direct commands from the SEP with this response.

Payload

The sequence number from the source command, the source command ID, a signed 8-bit return code, and an optional argument.

Key Manager command-response payload shows the source command, return code, and optional return argument.

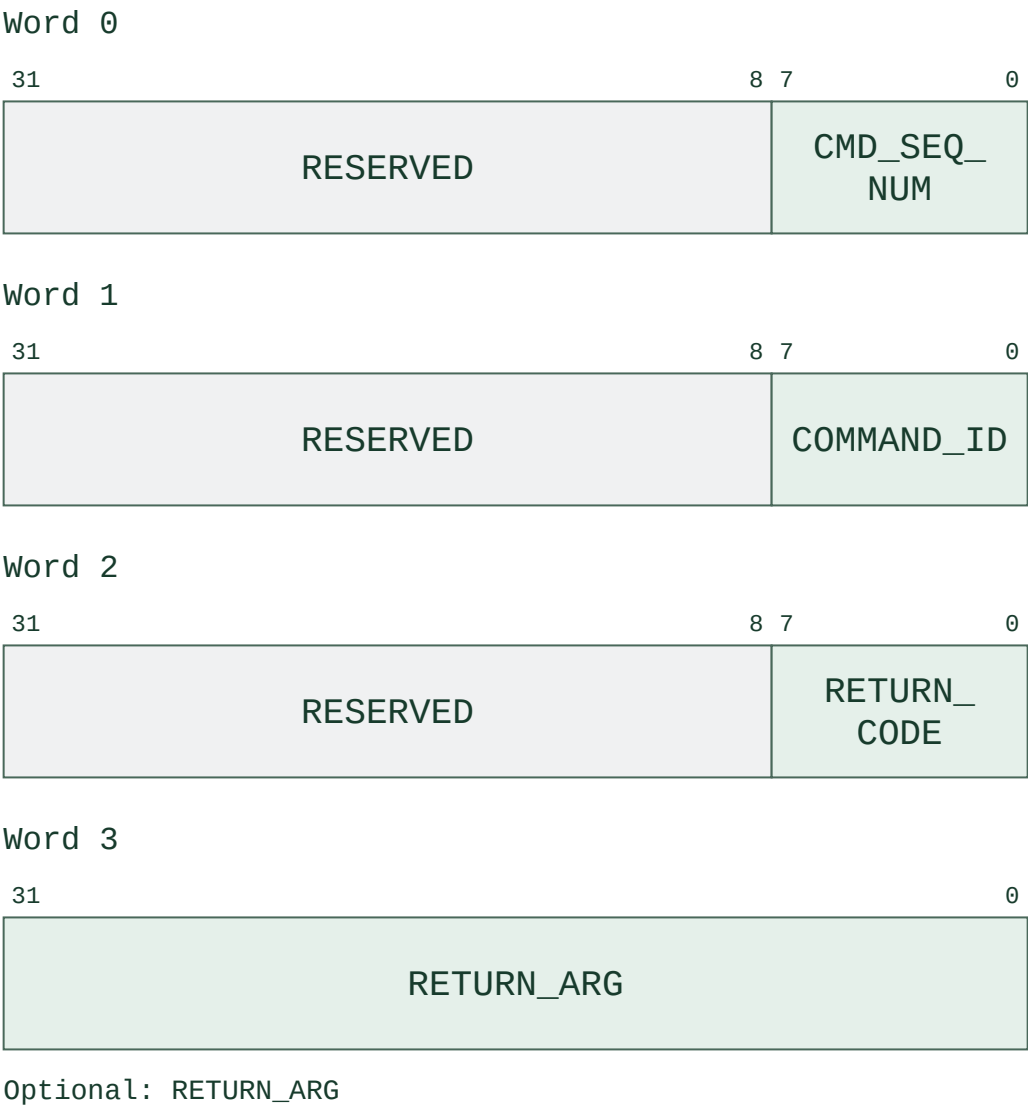


Figure 31. Key Manager command-response payload

Key Manager command return codes defines the command result codes and their return arguments.

Table 46. Key Manager command return codes

Code	Name	Description	Argument Value
0	success	General success code.	Specific to the type of source COMMAND_ID .
-1	failure	General failure code.	Specific to the type of source COMMAND_ID .
-2	header_crc	The header CRC-8/ROHC of the source command was invalid. No action was taken.	The CRC-8/ROHC the KM computed over the received header, i.e. the value the SEP should have sent.
-3	cmd_noseq	The CMD_SEQ_NUM of the source command was not the expected sequence number. No action was taken.	The next expected command sequence number.
-4	invalid_cmd	COMMAND_ID did not map to a valid command. No action was taken.	None.
-5	invalid_len	Either the PAYLOAD_LEN of the source command did not correspond to the size of the mailbox message frame, or the frame was self-consistent but PAYLOAD_LEN was not the length the named command requires. No action was taken.	The actual length of the message payload that was received.
-6	payload_crc	The payload CRC-32C of the source command was invalid. No action was taken.	The CRC-32C the KM computed over the received payload, i.e. the value the SEP should have sent.
-7	invalid_arg	One or more payload arguments of the source command were invalid. No action was taken.	The index for the payload word of the received command with the invalid argument. If the index points to a location outside of the received payload arguments, it indicates an argument was missing.

Both CRC return arguments carry the KM's own computed value rather than the one received. Echoing the received value back would tell the SEP nothing it did not already send; the computed value is what lets it see which side disagrees.

While **RECOVERABLE_ERR** is asserted, only **CMD_HW_VER**, **CMD_ROM_VER**, **CMD_SRAM_VER**, **CMD_STAT** and **CMD_RECOV_ACK** are accepted. Any other command is answered **failure** with no return argument, and its handler is never invoked ([Error Recovery](#)).

6.3.2. 0x55 - RESP_KM_READY

Description

An unsolicited response message that indicates the KM has completed initialization and has entered its main event loop. Serves as a proof of life indicator for the SEP.

Payload

This message has no payload.

6.3.3. 0x56 - RESP_ABR_SHARED_KEY_READY

Description

An unsolicited response message that notifies the SEP that Adams Bridge has delivered an ML-KEM shared key into the KM and it is available to capture. The KM emits this notification from its main loop after the shared-key interrupt fires and its ISR flags the pending event. The SEP responds by issuing **CMD_ABR_SK_TRANSFER** (see [0x27 - CMD_ABR_SK_TRANSFER](#)) to store the key in the KPV and obtain a handle.

Payload

This message has no payload.

6.3.4. 0xFE - RESP_RECOVERABLE_FAULT

Description

An unsolicited response message that indicates the KM has encountered a recoverable fault.

Payload

A signed 8-bit code indicating the cause of the fault, sign-extended to the full payload word. Only [7:0] is meaningful; because every code is negative, [31:8] reads as all ones rather than zero.

[Key Manager recoverable-fault payload](#) shows the sign-extended recoverable-fault code.

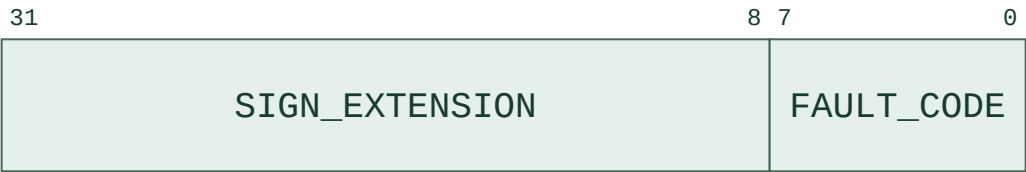


Figure 32. Key Manager recoverable-fault payload

[Key Manager recoverable-fault codes](#) lists the recoverable faults reported to the SEP host.

Table 47. Key Manager recoverable-fault codes

Code	Name	Description
-1	key_slot_crc	The CRC-32C check for a key slot has failed. Whatever operation triggered the fault was aborted.
-2	rx_buff_oflow	The message receive buffer has encountered a message overflow. This is caused by a singular incomplete message filling up the message receive buffer. All message buffers and mailbox FIFOs have been flushed by the KM.
-3	mbox_overflow	The KM outgoing mailbox FIFO has encountered an overflow condition. All message buffers and mailbox FIFOs have been flushed by the KM.
-4	mbox_underflow	The KM incoming mailbox FIFO has encountered an underflow condition. All message buffers and mailbox FIFOs have been flushed by the KM.
-5	flushed_by_sep	The KM mailbox was flushed by the SEP. All message buffers have been flushed by the KM. The mailbox FIFOs are not flushed.

6.3.5. 0xFF - RESP_UNRECOVERABLE_FAULT

Description

An unsolicited response message that indicates the KM has encountered an unrecoverable fault. Core execution has halted.

Payload

A signed 8-bit code indicating the cause of the fault, sign-extended to the full payload word. Only [7:0] is meaningful;

because every code is negative, [31:8] reads as all ones rather than zero.

Key Manager unrecoverable-fault payload shows the sign-extended unrecoverable-fault code.



Figure 33. Key Manager unrecoverable-fault payload

Key Manager unrecoverable-fault codes lists the unrecoverable faults reported to the SEP host.

Table 48. Key Manager unrecoverable-fault codes

Code	Name	Description
-1	wipe_state	The wipe state signal was asserted and the KM has purged all sensitive data.
-2	rom_parity	The ROM encountered a parity error.
-3	sram_parity	The SRAM encountered a parity error.
-4	rom_write	The ROM encountered a write error.
-5	sram_write_lock	The SRAM encountered a write error to a write-locked region.
-6	axi_decerr	The KM system fabric encountered an AXI DECERR.
-7	axi_slvrr	The KM system fabric encountered an unhandled AXI SLVERR with no DRBG error alongside it. A DRBG read timeout also raises SLVERR, and is reported as drbg_err instead.
-8	drbg_err	There was an error reading from the DRBG. A DRBG read timeout answers the CPU with SLVERR in the same cycle as it flags the DRBG error, so both status bits are set; firmware reports this condition as drbg_err rather than the generic axi_slvrr.
-9	illegal_insn	An illegal instruction was encountered.
-10	bus_error	The CPU core encountered an internal bus error or misaligned access.
-11	ebreak	An EBREAK instruction was executed.
-12	spurious_irq	An unrecognised IRQ source was asserted.
-13	otp_sigint	The OTP controller reported a dual-rail integrity violation.
-14	fw_crc	The loaded mutable firmware image failed its CRC-32C integrity check. The KM halts; recovery requires a warm or cold reset.
-15	fw_stack_ovf	The firmware load destination reached the stack guard region boundary, indicating the SEP sent more words than declared in FW_WORDS. The KM halts; recovery requires a warm or cold reset.
-16	shred_range	A shred operation was requested with a word count that exceeded the shred-order buffer. The KM halts; recovery requires a warm or cold reset.
-17	exec	An instruction fetch was attempted from a non-whitelisted (non-executable) memory region other than the ROM (execute-permission whitelist violation). The KM halts; recovery requires a warm or cold reset.
-18	rom_access	A ROM instruction fetch or data read was attempted after the ROM lockout engaged ([km-rom-lockout]). Only reportable as a fault code while the ROM is still live, because the ISR that decodes it is itself inside the locked-out ROM; a real breach after handover halts the KM through the illegal-instruction trap instead. The KM halts; recovery requires a warm or cold reset.

6.4. Common Sequences

6.4.1. Initial Boot

As soon as the KM is taken out of reset, it begins executing out of the ROM. The ROM performs basic setup routines such as initialization of peripherals and their drivers, as well as scrambling sensitive memory locations like the KPV and crypto engine sideload key registers. Once the boot process is complete, the KM sends a liveness message to the SEP via `RESP_KM_READY` and waits for further instruction via mailbox commands.

At this point the SEP may want to query hardware and/or firmware version and status via the `CMD_HW_VER`, `CMD_ROM_VER`, and `CMD_STAT`.

6.4.2. SEP Key Loading

The KM supports loading host-provided key material into the KPV directly over the mailbox using the `CMD_KEY_LOAD` command. The SEP never writes KPV slots directly; the KM firmware performs all slot allocation and key writes on the SEP's behalf. The typical flow is as follows:

1. The SEP issues a `CMD_KEY_LOAD` command containing the key size, the valid crypto engine destinations, and the key material itself, then waits for a response from the KM.
2. The KM allocates a set of free KPV slots, writes the key material into them, write-locks the slots, and registers a key handle.
3. The KM responds with *success* and a return argument containing the key handle, size, and valid destinations.
4. The SEP may now use the provided key handle for any further key management operations.

6.4.3. Key Transfer and Sideload Purge

The SEP requests transfer of a key from the KPV to any crypto engine sideload key register via the `CMD_KEY_TRANSFER` command. The SEP provides a handle for the key and a bitmask indicating the destination sideload key registers. The KM will respond with *success* when the operation is complete. If the key handle is invalid or any requested destination sideload key registers are disallowed, the KM will respond with *failure*.

When the SEP is done with the crypto engine, it issues the `CMD_ENGINE_SHRED` command to purge key material in the key sideload registers. The KM will respond with *success* when the operation is complete.

6.4.4. Adams Bridge Shared-Key Capture

The Adams Bridge ML-KEM shared-key interface flows in the opposite direction from the seed and engine sideloads: Adams Bridge computes a shared key and delivers it *into* the KM. The capture flow is as follows:

1. Adams Bridge loads the 256-bit shared key into the KM shared-key interface and the hardware sets the sticky `IRQ_STATUS.key_valid` bit, which (when `IRQ_ENABLE` is set) raises the shared-key interrupt on PicoRV32 bit 5.
2. The KM ISR acknowledges the interrupt by clearing the sticky `IRQ_STATUS` bit (write-1-to-clear) and flags the main loop. The ISR does not consume the key; `KEY_VALID` remains set.
3. From the main loop, the KM sends the SEP an unsolicited `RESP_ABR_SHARED_KEY_READY` notification.
4. The SEP issues `CMD_ABR_SK_TRANSFER`, supplying the `DEST_VALID` for the captured key.
5. The KM reads the shared key, clears the interface `KEY_VALID` (zeroizing the source registers), writes the key into a

freshly allocated KPV slot, write-locks the slot, and responds with *success* and the new key handle.

6. The SEP may then use the handle with any other key-management command, including transferring the captured key to a permitted engine.

6.4.5. Key Revocation

The SEP requests revocation of a key in the KPV via the `CMD_KEY_REVOKE` command. The KM erases every slot the key occupies and then removes the key from the key registry. It is now no longer accessible or transferable by either the SEP or KM.

The erase overwrites each slot's data through the KPV scrambler and returns the slots to the free pool ([Key Management](#)). A slot sealed before revocation is retired instead of freed, and is neither readable nor reusable until warm reset.

6.4.6. Error Recovery

The KM can encounter a variety of recoverable and unrecoverable errors.

A recoverable error signals a condition that halts critical KM services until acknowledged. It has the following flow:

1. The KM asserts the recoverable error signal.
2. The KM flushes the mailbox *only* for any recoverable error involving the mailbox or message buffers.
3. The KM sends the `RESP_RECOVERABLE_FAULT` response message to the SEP with a code indicating the cause of the fault.
4. The KM will *not* process any command **except**:
 - a. `CMD_HW_VER`
 - b. `CMD_ROM_VER`
 - c. `CMD_SRAM_VER`
 - d. `CMD_STAT`
 - e. `CMD_RECOV_ACK`
5. The SEP takes any action it deems necessary before it clears the fault.
6. The SEP sends a `CMD_RECOV_ACK` command message to clear the fault.
7. The KM deasserts the recoverable error signal.
8. The KM resumes normal operation, handling all valid commands.

When a recoverable fault is triggered by a command handler (for example, a `key_slot_crc` fault raised while processing a command), the KM sends **both** the `RESP_RECOVERABLE_FAULT` and that command's own `RESP_CMD` error response (*failure*). The paired `RESP_CMD` follows immediately after the fault response. The SEP must drain both messages before issuing `CMD_RECOV_ACK`; otherwise a subsequent response read consumes the stale `RESP_CMD` instead of the acknowledgement's reply. Faults raised from within the mailbox interrupt handler (`mbox_overflow`, `mbox_underflow`, `flushed_by_sep`) are not paired with a `RESP_CMD`, because the transmit buffer has been flushed and the fault message is written directly to the outgoing mailbox FIFO.

An unrecoverable error signals a condition that puts the KM into a permanent halt state. It has the following flow:

1. The KM shreds all crypto engine key sideload registers and zeroizes the Adams Bridge ML-KEM shared key.
2. The KM flushes the mailbox.

3. The KM sends the `RESP_UNRECOVERABLE_FAULT` response message to the SEP with a code indicating the cause of the fault.
4. The KM shreds the entire SRAM using one pass of a Weyl sequence followed by two passes of a xoshiro128++ PRNG.
5. The KM CPU permanently halts, which sets the unrecoverable error signal to interrupt the SEP.
6. The SEP must hard-reset the KM to bring it back through the boot sequence.

The two wiping steps — the sideload-register shred and the whole-SRAM shred — are together gated on a single build-time predicate, which is enabled by default. The fault report and the halt are not gated, so a fault is always announced and always fatal whether or not the wipe runs. The predicate is a weak function, so a build can override it; DV tests do exactly that, because a test that needs to observe post-fault state cannot have its SRAM erased underneath it. A production build leaves it enabled.

Note that the fault report is sent **after** the sideload shred but **before** the SRAM shred. Sending it first would leave key material live in the engine registers for the duration of the mailbox write, while deferring it until after the SRAM shred would risk the KM halting with nothing reported.

`shred_range` is the one fault code that does not take this path. It is raised by the shred routine itself on an out-of-range word count, so the KM reports the fault and halts directly, without the sideload shred, the mailbox flush or the SRAM shred; taking the flow above would re-enter the routine that just failed. The CPU halt still asserts the unrecoverable-fault signal to the SEP, so only the wiping is skipped. The condition is reachable only through a firmware bug or memory corruption, since every caller passes a compile-time-bounded count.

6.4.7. SRAM Firmware Handover

The KM ROM supports handing control from immutable ROM to mutable SRAM firmware via the `CMD_SRAM_LOAD_EXEC` (load + run) or `CMD_SRAM_EXEC` (re-run previously loaded firmware) commands.

The high-level steps for `CMD_SRAM_LOAD_EXEC` are:

1. **Validate.** The ROM checks `FW_WORDS > 0` and that the firmware image fits below the static load ceiling `km_fw_load_limit` — a build-time constant computed by the linker as `STACK - rom_max_stack`, rounded down to a 1 KB SRAM write-lock region boundary, where `_rom_max_stack` is the worst-case ROM stack depth validated against the production ELF (see [ROM Stack Budget and the Static Load Ceiling](#)). If either check fails, the command returns `_invalid_arg` and no image is requested.
2. **Confirm.** The ROM clears `sram_fw_size`, waits for the TX buffer to drain, disables inbound mailbox FIFO IRQs, and sends a `RESP_CMD` success ("confirmation") directly to the outbound FIFO. The SEP may then begin streaming the image.
3. **Stream.** The ROM polls the inbound FIFO directly (bypassing the normal message buffer), storing each word into SRAM starting at `0x8000` and accumulating a running CRC-32C. A per-word guard (defense-in-depth on top of the static `__km_fw_load_limit` validation) ensures the destination address never reaches the load ceiling.
4. **Verify.** The trailing CRC-32C word (sent by the SEP as the last word of the frame) is compared against the computed value. A mismatch triggers the `fw_crc` unrecoverable fault.
5. **Persist.** On success, `sram_fw_size` is updated to `FW_WORDS × 4`.
6. **Shred sideload keys.** The HMAC, KMAC, AES, and OTBN sideload key-share registers, together with the four Adams Bridge seed sideload interfaces (ML-DSA seed, ML-KEM seed D, ML-KEM seed Z, ML-KEM message), are overwritten with PRNG data and their `key_valid` bits cleared, and the Adams Bridge ML-KEM shared key is zeroized (`KEY_CTRL` cleared, hardware wipes `KEY[*]`). These register banks lie outside the SRAM address space, so the whole-SRAM scramble below cannot reach them; shredding here denies mutable firmware any key material the ROM previously sideloaded or any shared key delivered but not yet consumed.

7. **Lock sensitive data.** OTP secret fields and the `rom_persist` SRAM region are locked, and the KPV is closed to mutable firmware: using the key registry, every registered slot is read-locked (`lock_use`) and every free slot is shredded through the hardware erase path. Note that this read-lock is what denies mutable firmware the surviving key material; it is not the per-slot `seal` bit (`KPV Driver`), which no ROM flow sets. The erase path is not gated by the slot locks, so even a write-locked free slot — a revoked key, for instance — is wiped and made unreadable.
8. **Write-lock the image.** The SRAM write-lock mask is set to cover all 1 KB regions that contain the loaded image. The load-time bounds check guarantees those regions are disjoint from the ROM stack and data, so write-locking them cannot stop the scramble below from erasing ROM private state.
9. **Reseed the PRNG.** The `xoshiro128++` state is refreshed from the DRBG, so the scramble below runs from fresh entropy. This happens before the IRQs are disabled, so a DRBG fault raised during the reseed is still reported through the KMCSR IRQ.
10. **Disable IRQs and flush the mailbox.** All KMCSR IRQ enables are cleared and the pending sticky status bits with them, so mutable firmware inherits no spurious interrupts and the scramble's writes to write-locked regions raise no `sram_write_lock` fault. The ROM then waits for the outbound FIFO to drain, so the flush cannot discard a `RESP_CMD` confirmation the SEP has not yet read, and flushes both mailbox FIFOs to erase any residual inbound payload (e.g. key material from an earlier `CMD_KEY_LOAD`). Finally the mailbox IRQ enables are cleared and all CPU IRQ sources masked. ROM does not program `IRQ_ENTRY_ADDR`; the vector is left at the ROM default and the loaded firmware sets its own before re-enabling interrupts.
11. **Scramble.** A stack-less assembly routine (`rom_handover_jump.S`) seeds a `xoshiro128++` PRNG from the value passed from C, iterates over the full SRAM range `0x8000-0xFFFF`, and overwrites each address with a pseudo-random word. Hardware silently discards writes to write-locked regions (firmware image and `rom_persist`), so only ROM runtime data (stack, `.data`, `.bss`) is actually overwritten. No stack pointer or memory-resident variable is accessed during or after this step.
12. **Clear GPRs.** The entry address is loaded into `x5` first, and every other general-purpose register is then cleared to zero. `x5` is never cleared: the jump is `jalr zero, 0(x5)`, which writes the link address to `x0` and so discards it rather than writing `x5`. Mutable firmware therefore lands with every register zeroed except `x5`, which still holds its own entry address. Two further registers are used momentarily after the bulk clear, to arm SRAM execution mode below, and are re-zeroed before the jump.
13. **Enable SRAM execution mode.** Immediately before the jump, the assembly performs a triple write to `KMCSR.SRAM_EXEC_MODE.enable` (write-1-only, warm-reset domain). This adds the write-locked SRAM regions to the execute-permission whitelist and **arms** the ROM lockout, which engages on the first committed fetch from write-locked SRAM (`[km-rom-lockout]`). The ROM therefore stays executable across this store, which is what keeps the stack-less handover assembly running after it. The triple write hardens against voltage-glitch attacks that might cause a single store to be skipped. Setting this bit as the very last operation before the jump minimises the window during which SRAM execution is enabled before the first SRAM fetch.
14. **Jump.** `jalr zero, 0(x5)` transfers control to `0x8000`. That fetch is the first from write-locked SRAM, so it is what engages the ROM lockout. The ROM never returns. The mutable firmware is responsible for sending its own ready message to the SEP.

For `CMD_SRAM_EXEC`, the handover begins at step 6 (no image load), provided `sram_fw_size > 0`.

ROM lockout

Handover is an exchange rather than an addition: mutable firmware gains execute permission on its own SRAM regions and loses all access to the ROM it was handed control from.

Arming and engaging are separate events. `SRAM_EXEC_MODE.enable` only arms the exchange; the lockout engages on the first committed instruction fetch from a write-locked SRAM address, which is when the mutable image actually starts running. Splitting the two is what lets the arming store itself, and the rest of the ROM handover assembly that follows it,

keep executing from ROM.

Once engaged, every ROM instruction fetch and every ROM data read is refused and returns zeros rather than ROM content. Zeros are returned on a completion the CPU wrapper synthesizes, because `km_rom_interface` derives its ready from the ROM response and would otherwise stall the CPU forever. ROM **writes** are not part of the lockout; they still reach the ROM interface so that it can raise `rom_write_err` as before.

A refused access sets `IRQ_STATUS.rom_access_violation`, which is reported as the `rom_access` fault (-18) and kept distinct from `exec` (-17) so that a lockout breach is distinguishable from an ordinary non-executable fetch. The status bit is sticky and is set in the cycle of the access, but the ISR that would translate it into a mailbox fault code lives in the ROM the lockout has just revoked. A real breach after handover therefore halts the KM without emitting `rom_access` to the host: containment comes from the illegal-instruction trap on the zeros a blocked fetch returns, and the sticky status bit is read out after the halt. The fault code is observable only while the ROM is still live, or through an interrupt vector the mutable image installed itself.

The lockout is one-way within a boot. `SRAM_EXEC_MODE.enable` is write-1-only and the lockout latch is not software-writable at all. Both are warm-reset-domain state, so mutable firmware cannot clear either and cannot re-enter the ROM by any sequence of writes. Only a warm or cold reset restores ROM access, and that restarts the ROM from its entry point rather than resuming it.

The testbench virtual ROM (VROM, `0x1000_0000`) is exempt, because block-level firmware runs its `.text` from there. VROM does not exist in production silicon and carries no security requirement.

6.4.8. ROM Stack Budget and the Static Load Ceiling

Mutable firmware is loaded from the SRAM base (`0x8000`) and grows upward, while the ROM stack grows downward from `STACK` (the top of the ROM's `.data/.bss`). After a warm reset that follows mutable-firmware execution, the ROM re-runs and a subsequent `CMD_SRAM_EXEC` restart may drive the stack deeper than it was when the image was first loaded. The maximum image size must therefore be bounded by the **worst-case** ROM stack depth, not by the live stack pointer at load time.

The bound is enforced statically:

- The ROM firmware is built with a fully **static** stack — no variable-length arrays or `alloca`. All scratch buffers are fixed-size locals or file-scope `static`.
- `scripts/km_stack_analyze.py` walks the disassembled call graph of the linked production ELF (rooted at `main`, and separately at the dedicated IRQ entry `rom_irq`) and sums per-function stack frames along the deepest path. It reports a failure if it finds a dynamic frame, an unresolved indirect call on a reachable path, or a call-graph cycle.
- The firmware build passes the budget to the linker via `--defsym=__rom_max_stack=0x600`, so the load ceiling below is derived from the same value the analyzer checks the image against.
- The linker (`km_sram_layout.ld`) derives the load ceiling `km_fw_load_limit = (STACK - rom_max_stack)`, rounded down to a 1 KB SRAM write-lock region so the image and ROM stack never share a lock region, and asserts that a non-empty load area remains. `rom_cmd_sram_load_exec()` rejects any image whose top exceeds this symbol.

Run the analyzer on the production image to see the current worst-case main and IRQ stack depths and deepest paths:

```
python3 scripts/km_stack_analyze.py build/tests/rom_main/rom_main.rom.elf \
--objdump riscv64-unknown-elf-objdump --root main --limit 0x600
```

Raise the budget only if the analyzer reports a worst case that exceeds it.

6.5. Bootloader

6.5.1. Boot Sequence

The ROM bootloader will perform the following sequence of steps:

1. Sample whether this is a cold or a warm boot by reading `COLD_BOOT_DONE` before the final step sets it.
2. Clear all KMCSR IRQ status bits, then enable all fault and error related interrupts: ROM and SRAM parity, ROM write, SRAM write lock, AXI SLVERR and DECERR, DRBG error, wipe state, OTP integrity, execute permission violation, and ROM access violation ([\[km-rom-lockout\]](#)).
3. Rewrite `IRQ_ENTRY_ADDR` to the ROM interrupt entry point, then unmask CPU IRQs. `IRQ_ENTRY_ADDR` and `IRQ_ENTRY_LOCK` are warm-reset-domain, so a vector chosen by a previous mutable image cannot persist into ROM execution; the ROM rewrites it anyway rather than relying solely on the hardware reset value. This must precede unmasking.
4. Initialize the DRBG sampler.
5. Seed the firmware PRNG from the DRBG.
6. If the SRAM scrambler is not enabled:
 - a. Write the SRAM scrambler key register `SHRED_ITER+1` number of times with random words sourced from the DRBG.
 - b. Using inline assembly, enable & lock the SRAM scrambler then restart CPU execution (jump to address 0).
7. On a cold boot, initialize the warm-persistent SRAM region ([ROM warm persist region](#)). A warm boot leaves it intact.
8. Initialize the KPV scrambler, enable it, and lock it.
9. If wiping is enabled — a weak predicate, on by default, that block-level tests override to preserve state across a restart — then:
 - a. Shred all KPV slots.
 - b. Shred the HMAC, KMAC, AES and OTBN engine sideload key registers.
 - c. Shred the four Adams Bridge seed sideload interfaces (ML-DSA seed, ML-KEM seed D, ML-KEM seed Z, ML-KEM message).
 - d. Zeroize the Adams Bridge ML-KEM shared key.
10. Initialize the message buffers.
11. Initialize the key registry ([Key Registry](#)).
12. Reset the inbound and outbound message sequence counters.
13. Flush both mailbox FIFOs, then clear all latched mailbox IRQ status.
14. Enable the inbound mailbox data-available and error interrupts, leaving the outgoing data interrupt disabled.
15. Enable the Adams Bridge ML-KEM shared-key interrupt, so the first `KEY_VALID` assertion sends an unsolicited `RESP_ABR_SHARED_KEY_READY` (0x56 - [RESP_ABR_SHARED_KEY_READY](#)).
16. Set `COLD_BOOT_DONE`, then immediately send a `RESP_KM_READY` response message to the SEP. These two steps are adjacent and last by design: `RESP_KM_READY` is the boot-complete barrier, so anything sampling `BOOT_STATUS` on seeing it must already observe `COLD_BOOT_DONE`.
17. Finally, enter the main event loop.

6.6. Message Buffers

The message receive and transmit buffers are fixed size SRAM-based linear buffers that store message frames that have been read from or are ready to be written to the mailbox FIFOs. Since the maximum message payload length is 255 words, each message buffer is 257 words in size (1 header word + 255 payload words + 1 CRC-32C word), which is exactly one maximal frame. Since the KM firmware does not have a heap, the buffers are statically allocated.

Each buffer holds **at most one complete frame at a time**, and every new frame begins at index 0. A buffer therefore needs no head pointer and no per-frame start and end pointers: its state is the word count, a flag for whether a complete frame is present, that frame's length, and a read index. Once a complete frame is present, further writes are rejected until it is consumed, which is what bounds the buffer and gives the receive path its overflow detection.

Holding one frame rather than several is what makes the buffer's occupancy independent of how the SEP paces its commands: a maximal frame always fits, so a legal command can never be refused for want of room, and the KM processes commands strictly in order in any case ([Main Event Loop](#)). The `rx_buff_oflow` fault therefore signals a single frame longer than a legal message, not an accumulation of pending ones.

All access to the message buffers must go through a well defined interface. The interface must define functions for initializing, reading, writing, and getting status from a message buffer. The interface must support writing a frame word-by-word as the mailbox ISR receives it and reading it back word-by-word as the main loop parses it, and it must prevent buffer overflow and underflow.

Each buffer has exactly one writer and one reader: the mailbox ISR is the sole writer of the receive buffer and sole reader of the transmit buffer, and the main loop is the sole writer of the transmit buffer and sole reader of the receive buffer. Main-loop code that mutates a buffer shared with the ISR must mask the mailbox IRQ for the duration of the mutation and then restore it to its previous state.

6.7. Pseudorandom Generators

6.7.1. Firmware PRNG

The KM firmware shall use the xoshiro128++ algorithm as its general-purpose pseudorandom number generator. The PRNG is used for generating shred data and seeding pseudorandom-order operations. It must not be used for generating cryptographic key material; all key material must be derived from a master key or sourced directly from the DRBG.

State

The PRNG maintains a 128-bit internal state consisting of four 32-bit words, denoted `s[0]`, `s[1]`, `s[2]`, and `s[3]`.

Seed

The PRNG is seeded by reading four consecutive words from the DRBG and assigning them to `s[0]` through `s[3]`. An all-zero state is an absorbing fixed point of xoshiro128++ and must be avoided. If all four seed words are zero, `s[0]` must be set to 1 before use. The PRNG is reseeded on-demand from the DRBG via a function call.

Implementation Constraints

The SRAM shred loop must hold the PRNG state, write pointer, loop bound and temporaries in CPU registers for the duration of the loop, and must perform no SRAM read or write of PRNG state while it runs. The loop overwrites the memory the state would otherwise occupy, so state held in SRAM would be corrupted by the loop's own output.

The constraint is confined to that loop. Elsewhere the PRNG state is a static object in SRAM and the PRNG routines are called from drivers throughout the ROM. The state object is overwritten along with the rest of SRAM when the shred runs.

6.7.2. Register Access Shuffling

All operations requiring pseudorandom-order register access must use the Fisher-Yates shuffle to generate the access order. This covers the shred and key write operations on the KPV slots and on the crypto engine key-share registers. The shuffle operates on an index array initialized to `[0, 1, ..., n-1]` and produces a uniformly random permutation.

The SRAM and KPV scrambler key registers are filled with successive DRBG words in index order. Each word written is itself fresh entropy, and the register holds no key whose words an observer could correlate, so the write order carries no information to conceal.

Index Generation

The random index `j` in the range `[0, bound)` is generated using bit-masked rejection sampling from a PRNG-fed bit pool. This avoids division and eliminates modulo bias.

The bit pool is a 32-bit register refilled from the PRNG whenever insufficient bits remain. Bits are consumed from LSB to MSB.

6.8. Interrupt Handlers

The PicoRV32 has a very basic interrupt controller that does not support hardware interrupt priority or nested interrupts. Due to these limitations, interrupt priority must be managed by firmware and all interrupt handlers must be kept as short as possible to minimize interrupt response latency. The main ISR must process all currently active interrupts within a single call.

6.8.1. Interrupt Priority

Interrupt priority must be handled by firmware within the PicoRV32's main interrupt service routine (ISR). Any new interrupts that arrive while the PicoRV32 ISR is executing will trigger a pending interrupt signal that will then re-execute the main ISR after it returns. Since higher priority interrupts cannot preempt lower priority interrupts, short and deterministic ISR execution time is important.

The ISR should prioritize interrupts in the following order, from highest priority to lowest priority:

1. EBREAK / Illegal instruction
2. Bus error or misaligned access
3. Wipe state
4. ROM parity error
5. SRAM parity error
6. ROM write error
7. SRAM write lock error
8. AXI DECERR
9. DRBG error

10. AXI SLVERR
11. OTP integrity violation
12. Execute permission violation
13. ROM access violation
14. OTP change status
15. Mailbox
16. Adams Bridge ML-KEM shared key
17. Spurious IRQ (catch-all for unrecognised sources)

DRBG error is ranked above AXI SLVERR deliberately. A DRBG read timeout answers the CPU with SLVERR in the same cycle as it flags the DRBG error, so both status bits are always set together; testing SLVERR first would attribute every entropy failure to the fabric and send debugging toward the memory map. An unrelated fabric error sets SLVERR on its own and is unaffected by the ordering.

Each KMCSR source is tested against `IRQ_STATUS` and its `IRQ_ENABLE` bit, so a sticky status bit whose enable is clear is passed over rather than handled. The boot sequence ([Boot Sequence](#)) determines which sources are enabled in ROM.

6.8.2. EBREAK / Illegal Instruction

The EBREAK / illegal instruction interrupt handler distinguishes the faulting instruction by reading the opcode at the saved PC. An EBREAK instruction triggers an unrecoverable fault with the `ebreak` fault code. Any other illegal instruction triggers an unrecoverable fault with the `illegal_insn` fault code. The interrupt status bit is not cleared.

6.8.3. Bus Error or Misaligned Access

The bus error or misaligned access interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.4. Wipe State

The wipe state interrupt handler will purge keys from all crypto engines, trigger an unrecoverable fault, and finally shred the entire SRAM. Hardware will automatically clear the KPV, so firmware can skip it. The interrupt status bit is not cleared.

SRAM shredding is done last, and must be implemented entirely within assembly code that does not perform any SRAM read operations. Any operations after SRAM shredding must also be implemented in assembly since all program state (stack, etc.) has been destroyed.

NOTE

To ensure the wipe state process does not hang, the PRNG is *never* reseeded from the DRBG during any shred operation in the wipe state flow. This diverges from typical shred operations that normally reseed the PRNG from the DRBG.

6.8.5. ROM Parity Error

The ROM parity error interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.6. SRAM Parity Error

The SRAM parity error interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.7. ROM Write Error

The ROM write error interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.8. SRAM Write Lock Error

The SRAM write lock error interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.9. AXI DECERR

The AXI DECERR interrupt handler will trigger an unrecoverable fault. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.10. AXI SLVERR

The AXI SLVERR interrupt handler will trigger an unrecoverable fault with the `axi_slvrr` fault code. It is reached only when the DRBG error status bit is clear, since the DRBG handler is checked first ([Interrupt Priority](#)). The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.11. DRBG Error

The DRBG error interrupt handler will trigger an unrecoverable fault with the `drbg_err` fault code. It is checked before the AXI SLVERR handler, because a DRBG read timeout sets both status bits and the specific code is the useful one ([Interrupt Priority](#)). A stale DRBG error cannot misattribute a later fabric fault, because the handler halts the CPU rather than returning. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.12. OTP Integrity Violation

The OTP integrity violation interrupt handler will trigger an unrecoverable fault with the `otp_sigint` fault code. This interrupt is asserted when the OTP controller detects a dual-rail integrity violation on a redundantly-encoded OTP field. The interrupt status bit is not cleared. This behavior may change in future firmware revisions.

6.8.13. Execute Permission Violation

The execute permission violation interrupt handler will trigger an unrecoverable fault with the `exec` fault code. This interrupt is asserted when a committed instruction fetch targets a memory region outside the execute-permission whitelist, other than the ROM. The interrupt status bit is not cleared.

6.8.14. ROM Access Violation

The ROM access violation interrupt handler will trigger an unrecoverable fault with the `rom_access` fault code. This interrupt is asserted when a ROM instruction fetch or data read is attempted after the ROM lockout has engaged ([\[km-](#)

[rom-lockout](#)]). The interrupt status bit is not cleared.

This handler is only reachable while the ROM is still live, since it is itself ROM code and the lockout revokes the ROM that contains it. It therefore reports a breach detected before handover completes, or one raised through [IRQ_SET](#); after the lockout engages, containment is the illegal-instruction trap on the zeros a blocked fetch returns.

6.8.15. OTP Change Status

The OTP change status interrupt is an informational (non-fault) source that indicates one or more monitored OTP fields have changed. Unlike the fault handlers, this handler does not halt the KM. It has the following execution flow:

1. Read the OTP change status to determine which fields changed.
2. Invoke the OTP change hook to take any firmware-defined action. In the ROM this is a weak no-op hook intended to be overridden by later firmware.
3. Clear (write-1-to-clear) the change status bits that were handled.

The boot sequence does not enable this source, so the handler is dormant in ROM; leaving an informational interrupt masked keeps it from adding ISR entries during boot. Firmware that overrides the hook enables the source itself.

6.8.16. Mailbox

The mailbox interrupt handler transfers message frames between the mailbox FIFOs and the message buffers. It also handles any other interrupts raised by the mailbox due to error or status conditions. It does not check the properties of incoming messages or construct the headers of outgoing messages, as this is the job of the message handlers.

The mailbox interrupt handler has the following execution flow:

1. Check for and handle any mailbox error interrupts.
 - a. If the [OUTBOUND_OVERFLOW](#) interrupt is enabled and asserted:
 - i. Set the [RECOVERABLE_ERR](#) register of the KMCSR.
 - ii. Flush *all* message buffers and mailbox FIFOs.
 - iii. Message sequence numbers are reset to zero.
 - iv. A [RESP_RECOVERABLE_FAULT](#) response is sent to the SEP with a *mbx_overflow* fault code through the mailbox FIFOs directly. The message transmit buffer is bypassed.
 - v. Clear the [OUTBOUND_OVERFLOW](#) interrupt status bit.
 - vi. Enable the incoming mailbox FIFO interrupt if necessary.
 - vii. Return immediately.
 - b. If the [INBOUND_UNDERFLOW](#) interrupt status is enabled and asserted:
 - i. Set the [RECOVERABLE_ERR](#) register of the KMCSR.
 - ii. Flush *all* message buffers and mailbox FIFOs.
 - iii. Message sequence numbers are reset to zero.
 - iv. A [RESP_RECOVERABLE_FAULT](#) response is sent to the SEP with a *mbx_underflow* fault code through the mailbox FIFOs directly. The message transmit buffer is bypassed.
 - v. Clear the [INBOUND_UNDERFLOW](#) interrupt status bit.

- vi. Enable the incoming mailbox FIFO interrupt if necessary.
 - vii. Return immediately.
- c. If the `FLUSHED_BY_SEP` interrupt status is enabled and asserted:
- i. Set the `RECOVERABLE_ERR` register of the KMCSR.
 - ii. Flush *all* message buffers, but do not flush the mailbox FIFOs.
 - iii. Message sequence numbers are reset to zero.
 - iv. A `RESP_RECOVERABLE_FAULT` response is sent to the SEP with a *flushed_by_sep* fault code through the mailbox FIFOs directly. The message transmit buffer is bypassed.
 - v. Clear the `FLUSHED_BY_SEP` interrupt status bit.
 - vi. Enable the incoming mailbox FIFO interrupt if necessary.
 - vii. Return immediately.
2. Check for any pending words in the message transmit buffer.
- a. Calculate the minimum between the number of words currently in the message transmit buffer, and the available capacity of the outgoing mailbox FIFO.
 - b. Transfer this determined count of words from the message transmit buffer into the outgoing mailbox FIFO.
 - c. Set the separator bit in the FIFO before writing the final message frame word.
 - d. After words are transferred, if the message transmit buffer is empty, disable the outgoing mailbox FIFO interrupt.
3. Check for any pending words in the incoming mailbox FIFO.
- a. Calculate the minimum between the number of words currently in the incoming mailbox FIFO, and the available capacity of the message receive buffer.
 - b. Transfer this determined count of words from the incoming mailbox FIFO into the message receive buffer.
 - c. Check the separator bit after each word is moved from the FIFO to the buffer. The separator bit is aligned with the last word of each message frame.
 - d. Record pointers for the start/end words of each message frame.
 - e. After the words are transferred, if the message receive buffer is full, disable the incoming mailbox FIFO interrupt.

6.8.17. Adams Bridge ML-KEM Shared Key

The Adams Bridge shared-key interrupt is an informational (non-fault) source delivered on its own PicoRV32 interrupt (bit 5), separate from the KMCSR-aggregated and mailbox interrupts. It signals that Adams Bridge has delivered a new ML-KEM shared key into the KM. The handler is intentionally minimal:

1. Acknowledge the interrupt by clearing the sticky `IRQ_STATUS.key_valid` bit (write-1-to-clear) in the shared-key sideload register block.
2. Set a pending-notify flag for the main loop.

The handler does not read or consume the shared key — `KEY_VALID` stays set so the key remains available for the subsequent SEP-issued `CMD_ABR_SK_TRANSFER`. Because the interrupt status is sticky and cleared via W1C (rather than by disabling `IRQ_ENABLE`), a later shared-key delivery re-arms the interrupt automatically. The main loop sends `RESP_ABR_SHARED_KEY_READY` to the SEP when it observes the pending-notify flag.

6.8.18. Spurious IRQ

After all known IRQ sources have been serviced, the ISR checks for any unrecognised bits in the IRQ mask. If any are set, the ISR triggers an unrecoverable fault with the *spurious_irq* fault code. This ensures security-critical firmware never silently ignores unexpected interrupt sources.

6.9. Incoming Message Handler

The incoming message handler parses incoming message frames from the message receive buffer in SRAM, checks their properties, and calls the associated command handler. The incoming message handler is called as part of the main task loop of the KM system firmware, and it processes a single message frame per-call to the function. It must ensure that invalid or malformed messages are not processed by the KM. It must use functions provided by the message buffer interface and may not manipulate the message buffer directly. Partial message frames, i.e. message frames that have not been fully received with a message separator indicator, are skipped.

The incoming message handler has no arguments. It has the following execution flow:

1. If the message receive buffer is full AND it only contains a single partial message frame:
 - a. Flush *all* message buffers and mailbox FIFOs.
 - b. Reset both the command and response sequence counters to zero. The flush discards whatever the SEP had in flight, so neither side can agree on where the count stands; restarting both from zero is what lets the link resynchronize after the SEP acknowledges the fault.
 - c. Set the `RECOVERABLE_ERR` register of the KMCSR.
 - d. A `RESP_RECOVERABLE_FAULT` response is sent to the SEP with an `rx_buff_overflow` fault code.
 - e. Enable the incoming mailbox FIFO interrupt if necessary.
 - f. Return immediately.
2. If the message receive buffer is empty OR it only contains a single partial message frame:
 - a. Enable the incoming mailbox FIFO interrupt if necessary.
 - b. Return immediately.
3. Acquire the next fully valid message frame from the message receive buffer.
4. If the message header CRC-8/ROHC is invalid:
 - a. A `RESP_CMD` response is sent back to the SEP with a `header_crc` return code and the handler returns.
5. If the sequence number is not the next expected value:
 - a. A `RESP_CMD` response is sent back to the SEP with a `cmd_noseq` return code and the handler returns.
6. If the command ID is not a valid command:
 - a. A `RESP_CMD` response is sent back to the SEP with an `invalid_cmd` return code and the handler returns.
7. If the payload length is not consistent with the message frame length counted during transfer from the incoming mailbox FIFO to the message receive buffer:
 - a. A `RESP_CMD` response is sent back to the SEP with an `invalid_len` return code and the handler returns.
8. If the message has a payload and the payload CRC-32C is invalid:
 - a. A `RESP_CMD` response is sent back to the SEP with a `payload_crc` return code and the handler returns.
9. If the `RECOVERABLE_ERR` bit is set in the KMCSR:

- a. Only allow the following commands:
 - i. `CMD_HW_VER`
 - ii. `CMD_ROM_VER`
 - iii. `CMD_SRAM_VER`
 - iv. `CMD_STAT`
 - v. `CMD_RECOV_ACK`
 - b. A `RESP_CMD` response is sent back to the SEP with the return code and return argument from the command handler.
 - c. Respond to all other commands with a `RESP_CMD` response of *failure* and do not execute the command.
10. If the `RECOVERABLE_ERR` bit is *not* set in the KMCSR:
- a. Call the command handler corresponding to the command ID and include the following arguments:
 - i. The length of the message payload.
 - ii. A structure containing the message payload.
 - b. A `RESP_CMD` response is sent back to the SEP with the return code and return argument from the command handler.
11. Enable the incoming mailbox FIFO interrupt if necessary.
12. Return.

6.10. Outgoing Message Handler

The outgoing message handler constructs a message from a given response payload and writes it into the message transmit buffer in SRAM. The outgoing message handler is called from command handlers and other system functions to send responses to the SEP. It must use functions provided by the message buffer interface and may not manipulate the message buffer directly.

The outgoing message handler takes a response ID and a payload as arguments. It has the following execution flow:

1. The next message sequence number is inserted into the message header.
2. The response ID is inserted into the message header.
3. The payload length is calculated and inserted into the message header.
4. The message header CRC-8/ROHC is calculated and inserted into the header.
5. If the payload length is nonzero, the payload CRC-32C is calculated and appended to the message.
6. The entire message frame is written into the message transmit buffer. If the buffer is full, enable the outgoing mailbox FIFO interrupt and block (spin-wait) until enough free space is available to write the entire frame. Further execution depends on the SEP draining the outgoing mailbox FIFO.
7. Enable the outgoing mailbox FIFO interrupt if necessary.
8. Return.

The mailbox ISR is what moves those frames onward, draining the transmit buffer into the outbound FIFO as space appears; the buffer does not wait on SEP activity alone.

Some messages are framed identically but written straight to the outbound FIFO, bypassing the transmit buffer: the

unrecoverable fault report, the mailbox-error recoverable faults raised from the ISR itself, and the `RESP_CMD` confirmation sent at handover. Each of these is followed immediately by a CPU halt or by interrupt masking, so a buffered frame would never be drained.

6.11. Command Handlers

Each supported KM command has an associated command handler function. They are called by the incoming message handler, validate the arguments provided in the command message, and then call the appropriate system or key management function. If a command handler encounters an error it will return with an error code and an optional return argument. This includes errors encountered during payload parsing. Otherwise, if there are no errors it will return a *success* code and provide an optional return argument.

6.12. DRBG Driver

Initialize DRBG

Configures the DRBG and waits for it to be ready. It has the following execution flow:

1. Wait for `DRBG_READY` in the STATUS register.

Get Status

Returns the value of the STATUS register.

Get Word

Returns a single random word from the DRBG, read from the `DATA` register.

Get Block

Gets a block of random words from the DRBG, given a number of words to read. It has the following execution flow:

1. Enable DRBG prefetch.
2. Read the given number of random words, each from the `DATA` register.
3. Disable DRBG prefetch.
4. Return the random words.

Every word must be read from `DATA`, including in the prefetching block path. Only a `DATA` read consumes the prefetched word and starts the next fetch. `PREFETCH_DATA` is a read-only debug mirror of the same holding register, so reading it in a loop returns one value repeatedly instead of a stream — which would silently yield a key made of a single repeated word.

6.13. KPV Driver

These functions provide firmware abstractions for the KPV. The vault holds 64 slots of 16 words each, and each slot carries a `CTRL` register holding the `lock_write`, `lock_use` and `seal` bits and the `erase` trigger, in bits 0 through 3. All four are write-one-to-set, and a warm reset clears every one of them so a slot cannot carry stale control state into a later erase.

Every entry point operates on exactly one slot, addressed by slot index (0–63), and knows nothing of keys. The KPV

stores key data and lock state only: it has no field describing how many slots a key spans or how many words of its final slot are meaningful. That extent is software state held in the key registry ([Key Registry](#)), so a caller operating on a whole key loops over its slots itself.

Because hardware masks nothing, a read of a slot returns all the words asked for, and words past the end of a key read back as whatever the preceding shred left there rather than as zero. Only `lock_use` forces read data to zero.

Initialize Scrambler

If unlocked, writes the KPV scrambler key register `SHRED_ITER+1` number of times with random words sourced from the DRBG.

Scrambler Enable

Enables the KPV scrambler.

Scrambler Disable

Disables the KPV scrambler, unless it is already locked. Used by tests to read back raw stored data and confirm that scrambling took effect; production firmware locks the scrambler instead.

Scrambler Lock

Locks the KPV scrambler.

Erase Slot

Hardware-erases one slot:

1. Asserts the slot's `CTRL.erase` trigger. The write-1 trigger is issued three times so a single skipped store (e.g. from a fault-injection glitch) cannot prevent the erase from starting.
2. Polls the slot's `erase` bit until the hardware self-clears it.

The hardware overwrites every key word of the slot through the KPV scrambler. Nothing gates an erase — not `lock_write`, not `lock_use`, not a seal — but the slot's seal state decides what the slot looks like afterwards. An unsealed slot comes back with its control register cleared and is reusable. A sealed slot is **retired** instead: `lock_write` stays set, `lock_use` is set, and the slot can be neither read, rewritten nor reused until the next warm reset. Erase being ungated is what keeps revocation available on a sealed slot; retirement is what stops the freed slot being refilled with a value of the caller's choosing.

Shred Slot

Shreds a single slot by invoking **Erase Slot** on it `SHRED_ITER+1` times. Because it uses the hardware erase path it also wipes a write-locked slot, and a sealed slot ends up retired rather than free.

Shred KPV

Shreds every KPV slot by invoking **Shred Slot** on each slot, so each slot is hardware-erased `SHRED_ITER+1` times. Write-locked slots are wiped too.

Write Slot

Writes up to a slot's worth of key data, given a slot index, the data, and a word count of `1–16`. The words land starting at word 0 of the slot; any remaining words keep whatever the preceding shred put there. The function returns an error

and writes nothing if the word count is outside 1–16 or the slot is write-locked, which includes any sealed or retired slot. The count is bounded because slots are contiguous, so an over-long write would run into the neighbouring slot.

Read Slot

Reads key data out of one slot, given a slot index, an output buffer, and a word count of 1–16. The words are read starting at word 0 of the slot. How many of them are meaningful is the caller's business — the extent lives in the key registry. The function returns an error and reads nothing if the word count is outside 1–16 or the slot is read-locked, which includes any retired slot.

Write-Lock Slot

Sets `lock_write` on one slot. The register write is issued three times so a single skipped store cannot leave the slot writable. The value is built from zero rather than read-modify-written, because every control field is write-one-to-set and a written-back read value could only ever set bits — which would re-trigger an in-flight erase.

Read-Lock Slot

Sets `lock_use` on one slot, by the same triple-write convention, preventing further reads of its key data.

Seal Slot

Seals one slot, leaving its data readable but not overwritable until the next warm reset. Only the seal bit is written, by the same triple-write convention: hardware sets `lock_write` alongside the seal, and reading both back is the fault check the function reports on.

Sealing commits the slot as well as its contents. The material can still be revoked, because erase is never gated, but the erase retires the slot, so the slot never carries anything else within the boot cycle. That is what the seal exists for: key material can cross a trust boundary revocable but not replaceable.

No ROM boot or handover flow in this revision seals a slot. The primitive and its two predicates are the driver surface a future flow — or mutable firmware — uses to take and to check a seal.

Slot Sealed

Reports whether a slot is sealed, which is the check a consumer of surviving key material makes before trusting it: a sealed slot cannot have been altered since the seal was taken, because a seal holds `lock_write` set. It is true of a retired slot as well, so pair it with **Slot Retired** to tell a live sealed slot from one whose material is gone.

Slot Retired

Reports whether a slot is retired, i.e. a sealed slot that has since been erased. A sealed slot gains `lock_use` only from an erase completing, so a set `lock_use` is what separates a retired slot from a sealed live one.

6.14. Crypto Engine Drivers (HMAC / KMAC / AES / OTBN / Adams Bridge)

Each crypto engine (HMAC, KMAC, AES, OTBN) and each of the four Adams Bridge seed inputs (ML-DSA seed, ML-KEM seed D, ML-KEM seed Z, ML-KEM message) has a driver that manages key material for that destination. All of these drivers expose the same interface and share the same execution flow; they differ only in the engine they target.

Shred Key

Purges the sideload key share registers with random data from the PRNG. It has the following execution flow:

1. Clears the key valid bit.
2. Writes random data sourced from the PRNG to both key share registers.
 - a. Reseeds the PRNG from the DRBG.
 - b. Writes the registers with random data from the PRNG across both key share registers in a pseudorandom order that is seeded from the PRNG. Write order should be randomly interleaved across both key share registers to enhance obfuscation.
 - c. Performs this entire sequence a total of `SHRED_ITER+1` times.

Write Key

Writes key data to the sideload key share registers in a pseudorandom order that is seeded from the PRNG. The two key share registers XOR together within the accelerator to produce the real key. It has the following execution flow:

1. Pads the given key, `KEY`, with random data from the DRBG up to the full length of one of the sideload key share registers.
2. Collect a random array of data, `RAND`, from the DRBG the same size as the padded key.
3. For each word, `i`: `KEY_RAND[i] = XOR(KEY[i], RAND[i])`.
4. Writes `RAND` to `KEY_SHARE0` and `KEY_RAND` to `KEY_SHARE1` in a pseudorandom order that is seeded from the PRNG. Write order should be randomly interleaved across both key share registers to enhance obfuscation.
5. Sets the key valid bit.

Adams Bridge ML-KEM shared-key driver

The Adams Bridge ML-KEM shared-key interface is the one inbound path (Adams Bridge → KM); it is read-only to firmware and consumed rather than written:

- **Read (consume) shared key.** Reads the eight 256-bit key words into a caller buffer, then writes `KEY_CTRL = 0` to clear `KEY_VALID`, which triggers a hardware zeroize of the source key registers. Invoked by the `CMD_ABR_SK_TRANSFER` handler.
- **Acknowledge interrupt.** Clears the sticky `IRQ_STATUS.key_valid` bit by writing 1 (W1C). Called by the shared-key ISR to acknowledge the interrupt without consuming the key.
- **Interrupt enable / disable.** Set or clear the `IRQ_ENABLE.key_valid_en` bit. The masked CPU interrupt is `IRQ_STATUS & IRQ_ENABLE`; firmware enables it during boot and leaves it enabled, relying on the sticky-status W1C acknowledgement rather than toggling the enable per event.

6.15. Key Derivation Function Library

The `rom_kdf` library implements the NIST SP 800-108r1 counter-mode key derivation function (KDF) with HMAC-SHA-256 as its pseudorandom function. The KDF is defined by the following construction, where `i` and `L` are unsigned 16-bit big-endian values and `L` is the requested output length in bits:

```
K(i) = HMAC-SHA-256(key, BE16(i) || input[192] || BE16(L))
out  = leftmost L bits of K(1) || K(2) || ...
```

The counter starts at 1. The library computes `ceil(L / 256)` HMAC iterations and copies `ceil(L / 8)` bytes to the caller's output buffer. `L` is in the range 1 through 8192 bits. If `L` is not byte-aligned, the final output byte is copied in full and the caller ignores the bits beyond `L`. Because `L` is included in every HMAC message, outputs requested at different lengths are cryptographically separated; a shorter output is not a prefix of a longer output.

The input is a fixed 192-byte `km_kdf_input_t`. The library copies this structure into the HMAC message verbatim. Structure fields use the PicoRV32 in-memory byte representation; only the surrounding counter and output-length fields are big-endian.

[KDF Fixed Input Layout](#) defines the byte layout of the fixed input mixed into key derivation.

Table 49. KDF Fixed Input Layout

Byte Offset	Field	Description
0-1	<code>version</code>	KM ABI version mixed into the derivation.
2	<code>out_class</code>	Class of the derived key.
3	<code>out_type</code>	Type of the derived key.
4	<code>out_owner</code>	Owner of the derived key.
5	<code>out_domain</code>	Parent-inherited domain, forged by the KM.
6-7	<code>flags</code>	Provenance and operational flags, forged by the KM.
8-9	<code>purpose</code>	Public purpose metadata.
10-11	<code>out_bits</code>	Requested derived-key size recorded in the input.
12-15	<code>caps</code>	Capabilities requested for the derived key.
16-19	<code>device_state</code>	Live life-cycle and demotion state sampled by the KM.
20-31	<code>rsvd</code>	Reserved; all bytes are zero.
32-63	<code>label</code>	Fixed operation label providing domain separation.
64-127	<code>context</code>	Caller- or ROM-provided derivation context.
128-191	<code>entropy</code>	DRBG input for ephemeral operations; otherwise all zero.

`rom_kdf_800_108()` uses `rom_hmac_sha256()` for each counter iteration. An HMAC key of at most 64 bytes is passed directly to HMAC. A longer key is first reduced to a 32-byte SHA-256 digest as specified by RFC 2104. This normalization accepts a parent key of any length — comfortably including the 512-byte KM symmetric-key limit of 128 words — while retaining the HMAC library's 64-byte key limit. The KDF itself does not enforce the KM key-size limit; a caller that must respect it enforces it before calling.

The requested output length is expressed in bits, and the supported range is 1–8192 bits (1024 bytes). A single derivation can therefore produce more material than one KM key can hold. The KDF rejects null required pointers, a null key with non-zero length, and an output length of zero or above that ceiling. Invalid arguments leave the output buffer unchanged. If SHA-256 or HMAC fails after validation, the KDF zeroes the complete requested output region before returning an error. The implementation also securely clears its normalized key, HMAC message, and digest scratch buffers before returning.

`rom_kdf_selftest()` provides the KDF power-on known-answer test. It derives a 512-bit output from an embedded 32-byte key and 192-byte input, thereby exercising two counter iterations, and compares the result with the embedded expected value.

The shipped ROM does not call this self-test, and neither its boot sequence ([Bootloader](#)) nor any command handler is gated on it. It is available for a caller that wants the check; a boot flow that derives real key material should run it and fail closed, but nothing in the current ROM enforces that.

6.16. CRC Function Library

The CRC function library contains routines for CRC computations. The high-level functions (`rom_crc8_rohc`, `rom_crc32c`) preserve their public signatures and golden-vector results, but are implemented on top of the hardware CRC accelerator rather than software lookup loops: each consumes one accelerator instruction per input unit (one byte per CRC-8/ROHC call, one 32-bit word per CRC-32C word call, and one byte per CRC-32C tail byte). The accelerator returns the CRC-32C running remainder before the final XOR, so `rom_crc32c` seeds `0xFFFF_FFFF`, chains word and tail-byte updates, and applies the final XOR once at the end. See the CRC Accelerator section of the architecture documentation for the instruction encodings and engine behavior.

6.16.1. CRC-8/ROHC

Compute the CRC-8/ROHC for a given block of data.

[CRC-8/ROHC Parameters](#) specifies the CRC-8 parameters used for message headers.

Table 50. CRC-8/ROHC Parameters

Parameter	Value
Width	8
Poly	<code>0x07</code>
Init	<code>0xFF</code>
RefIn	true
RefOut	true
XorOut	<code>0x00</code>
Check ("123456789")	<code>0xD0</code>

The CRC-8/ROHC is computed over the input bytes in little endian order, from least significant byte to most significant byte.

6.16.2. CRC-32C

Compute the CRC-32C (Castagnoli) for a given block of data.

[CRC-32C Parameters](#) specifies the CRC-32C parameters used for payload integrity.

Table 51. CRC-32C Parameters

Parameter	Value
Width	32
Poly (normal)	<code>0x1EDC6F41</code>
Poly (reflected)	<code>0x82F63B78</code>
Init	<code>0xFFFFFFFF</code>
RefIn	true
RefOut	true
XorOut	<code>0xFFFFFFFF</code>
Check ("123456789")	<code>0xE3069283</code>

The CRC-32C is computed over the input bytes in little endian order, from least significant byte to most significant byte.

6.17. Key Registry

KM firmware maintains a mapping between key handles and key slots in the KPV, called the key registry. Key handles are unique 8-bit numbers that act as aliases to key slots in the KPV. Handle `0x00` is reserved for "null", i.e. no handle is assigned. Key handles are simply incremented as they are allocated in the registry. Key handles cannot be reused, and are permanently invalidated when they are destroyed.

Because handles are never reused, the handle space is exhaustible independently of the vault: the 64 KPV slots bound how many keys can be **live** at once, not how many handles have ever been issued. A host that repeatedly generates and revokes a key frees the slot each time but consumes a fresh handle, so 255 such cycles exhaust the space with the vault almost empty throughout. The registry therefore checks for exhaustion on every allocation and fails the request rather than wrapping to `0`, which would alias the null handle. Recovering the handle space requires a reset.

There are two data structures in the key registry:

- A mapping from key handles to a base slot index, the key's length in 32-bit words, a CRC-32C value for the associated key, and an 8-bit `DEST_VALID` destination-policy bitmask.
- A mapping from slot indexes to a key handle. A key handle value of NULL (`0x00`) indicates the slot does not have an associated key handle.

Each handle entry also carries a validity flag. Destroying a handle clears that flag and clears the slots' entries in the reverse map, but leaves the entry's other metadata in place; every lookup below tests the flag, so a destroyed handle is indistinguishable from one never issued.

The registry is the **only** record of a key's extent. The KPV has no field describing how many slots a key spans or how many words of its final slot are meaningful (*KPV Driver*), so a key's length is recorded once here at generation time and the slot span is derived from it. That keeps the recorded length and the set of slots claimed for the key from ever disagreeing.

All access to the key registry must go through a well defined interface. The following functions manipulate the key registry.

Initialize Registry

Clears every handle entry and every slot mapping, and sets the next allocatable handle to `1`. Called once from the boot sequence (*Bootloader*).

Get Key Handle

Returns the registered key handle, given a key slot index. The given key slot index does not need to be the base slot for the key. Unlike the lookups below, this function does not signal an error: a slot with no associated handle, or a slot index past the end of the vault, returns NULL (`0x00`). Allocation uses it as the free-slot predicate on that basis.

Get Key Slot

Returns the registered base key slot index, given a key handle. Returns an error if the key handle does not exist or has been invalidated.

Get Key CRC

Returns the registered CRC-32C for a key, given a key handle. Returns an error if the key handle does not exist or has been invalidated.

Get Key Destinations

Returns the registered `DEST_VALID` destination-policy bitmask for a key, given a key handle. Returns an error if the key handle does not exist or has been invalidated.

Get Key Words

Returns the registered length of a key in 32-bit words, given a key handle. Returns an error if the key handle does not exist or has been invalidated. Every caller that has to walk a key's slots — CRC computation, key transfer, revocation — takes its extent from here.

Generate Handle

Returns and registers a unique 8-bit key handle, given the following:

- The base slot index for a key
- The key length in 32-bit words
- A CRC-32C of the key data
- The valid crypto engine destinations for the key

It has the following execution flow:

1. Returns with an error if the key length is outside `1-128` words, or if the key's slot span derived from that length would run past the last KPV slot. Bounding the span here is what keeps the reverse slot-to-handle map from being written past its end.
2. Assigns the next handle. Handles are allocated monotonically from `1` rather than by searching for a free value, so a handle is never reused once invalidated.
 - a. Returns with an error if the next handle would wrap to `0`, i.e. handles are exhausted.
3. Updates the registry with the generated key handle, base slot, key length, CRC-32C, and destination-policy bitmask, and points every slot the key occupies at that handle in the reverse map.

This function does not check whether the requested slots are already registered. Allocating a free run of slots is the caller's responsibility ([Key Management](#)).

Destroy Handle

Invalidates a given key handle in the registry. The key handle may not be used again. Returns with an error if the given key handle does not exist.

6.18. Key Management

These functions provide high-level key management operations. In general, they are called by command handler functions.

Check Key

Checks the integrity of a stored key, given a key handle. Returns an error if the key handle does not exist or the CRC-32C is invalid. Triggers a recoverable fault if the CRC-32C is invalid. It has the following execution flow:

1. Get the key slot, key length and CRC-32C for the given key handle, returning with an error if invalid.

2. Calculate the CRC-32C over the key data, reading one slot at a time across the key's slot span and taking only the remaining words from the final slot. The span comes from the registered key length, since the KPV records no extent of its own.
3. If the calculated CRC-32C does not match the CRC-32C stored in the key registry:
 - a. Set the `RECOVERABLE_ERR` register of the KMCSR.
 - b. A `RESP_RECOVERABLE_FAULT` response is sent to the SEP with a `key_slot_crc` fault code.
 - c. Return with an error.

Generate Key

Generates a random key, given the following:

- The size of the key in multiples of 32-bit words
- Valid crypto engine destinations for the key

It collects the key material from the DRBG and then stores it by the same path as **Load Key** below, zeroizing its scratch buffer afterwards.

Load Key

Stores key material in the KPV, given the following:

- The size of the key in multiples of 32-bit words
- Valid crypto engine destinations for the key
- The key data

This is the single store path: **Generate Key** supplies DRBG output as the key data, and the SEP-facing load and Adams Bridge capture commands supply theirs. It has the following execution flow:

1. Returns with an error if the encoded size exceeds `127`, the destination mask is zero, or the resulting slot span exceeds the span of the largest supported key.
2. Finds a set of unoccupied consecutive key slots, returning with an error if none are available.
 - a. The search starts at a slot index selected randomly using data from the DRBG and wraps once around the vault.
 - b. A run that would extend past the last slot is skipped rather than wrapped, so a key never straddles the end of the vault.
 - c. Selected slots must not be write-locked or read-locked, which also excludes any sealed or retired slot.
 - d. Selected slots must not be associated with an existing key handle.
3. Computes the CRC-32C over the key data, before anything is written to the KPV.
4. Registers the handle, recording the base slot, key length, CRC-32C and destination mask. This happens **before** any KPV mutation, because the registry holds the key's only record of its own length; if a subsequent slot write fails, the handle is destroyed again so no registry entry survives without the key material it describes.
5. Shreds all slots associated with the key.
6. Writes the key material one slot at a time, the final slot taking only the words that remain.
7. Write-locks all slots associated with the key.
8. Returns the handle for the key.

Revoke Key

Revokes a key, given a key handle. It has the following execution flow:

1. Gets the base slot index and key length associated with the key handle, returning with an error if invalid.
2. Hardware-erases every slot across the key's span: each slot's `CTRL.erase` trigger is asserted, and firmware waits until the hardware self-clears it. The erase overwrites the key data through the KPV scrambler. An unsealed slot also has its CTRL register cleared, including the lock bits, so it is wiped and reusable; a sealed slot is retired instead and cannot be reused until warm reset ([KPV Driver](#)).
3. Destroys the given key handle.
4. Returns success.

Transfer Key

Transfers a key from the KPV to one or more crypto engines, given the following:

- The handle for the key
- A bitfield of crypto engine destinations for the key

It has the following execution flow:

1. Gets the base slot index associated with the key handle, returning with an error if invalid.
2. Checks the integrity of the key associated with the given handle, returning with an error if invalid.
3. Gets the destination-policy bitmask from the key registry and checks that **all** requested crypto engine destinations are allowed, returning with an error if any are invalid.
4. Transfers the key to all requested crypto engines.
5. Returns success.

6.19. Main Event Loop

After the bootloader initializes the KM, the main loop of the KM continuously monitors the incoming message buffer for commands. When the incoming message buffer is empty, the main loop should wait for an interrupt. All other tasks are handled by the main ISR.

The KM processes commands strictly in the order that they are received.

The main loop also dispatches the Adams Bridge shared-key notification. The shared-key ISR sets a pending-notify flag; each iteration the main loop samples and clears that flag and, if set, sends the SEP an unsolicited `RESP_ABR_SHARED_KEY_READY`. The mailbox and shared-key interrupts are masked around the flag sample/clear and the wait-for-interrupt decision so that a notification set by the ISR can neither be dropped (lost update) nor leave the CPU sleeping with work pending (lost wakeup).

6.19.1. Fault Handling

When the KM encounters an unrecoverable fault it purges key material, flushes the mailbox FIFOs, and then writes a `RESP_UNRECOVERABLE_FAULT` message directly to the outgoing mailbox FIFO. The message transmit buffer is entirely bypassed for this operation. Finally the PicoRV32 traps so that it halts execution. At this point the SEP must hard-reset the KM. [Error Recovery](#) gives the full ordered flow and states which of its steps are gated on the build-time wipe

predicate.

When the KM encounters a recoverable fault, it should set the `RECOVERABLE_ERR` register bit in the KMCSR. The intent is that the KM should halt critical services until the SEP has had a chance to acknowledge the fault and take any desired action. While this bit is set the KM answers every command with *failure* except the four query commands (`CMD_HW_VER`, `CMD_ROM_VER`, `CMD_SRAM_VER`, `CMD_STAT`) and `CMD_RECOV_ACK` itself, rejecting the command before its handler runs. The `RESP_RECOVERABLE_FAULT` message is always sent back to the SEP in addition to any error messages returned from any command that may have triggered the fault.